CLINICAL STUDY PROTOCOL

Study Title:	A Phase 1, Dose Escalation, and Cohort Expansion Study Evaluating NX-5948, a Bruton's Tyrosine Kinase (BTK) Degrader, in Adults with Relapsed/Refractory B-cell Malignancies
Protocol Number:	NX-5948-301
Active Investigational Product:	NX-5948 (BTK degrader)
Indication:	Relapsed/Refractory B-cell Malignancies
Sponsor:	Nurix Therapeutics, Inc.
Development Phase:	Phase 1a/1b
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Date of Protocol Amendment 8.1:	14 Mar 2025
IND Number	163508
EU Clinical Trial Number	2023-510541-25-00

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SPONSOR SIGNATURE PAGE

A Phase 1, Dose Escalation, and Cohort Expansion Study Evaluating NX-5948, a Bruton's Tyrosine Kinase (BTK) Degrader, in Adults with Relapsed/Refractory B-cell Malignancies

Protocol Number: NX-5948-301

Protocol Amendment 8.1: 14 Mar 2025

This protocol has been approved by Nurix Therapeutics, Inc. The following signature documents this approval.

Signed by:
Sarah Injac
Signer Name: Sarah Injac
Signing Reason: I approve this document
Signing Time: 14-Mar-2025 | 12:29:15 PM PDT
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14-Mar-2025 | 12:30:06 PM PDT

Sarah Injac, MD, PhD

Date

Senior Director, Clinical Development

INVESTIGATOR STATEMENT

- I confirm agreement to conduct the study in compliance with NX-5948-301 Protocol Amendment 8.1 dated 14 Mar 2025.
- I acknowledge that I am responsible for overall study conduct. I agree to personally conduct or supervise the described study.
- I agree to ensure that all associates, colleagues, and employees assisting in the conduct of the study understand their obligations and will comply with the study protocol.
- Mechanisms are in place to ensure that site staff receive the appropriate training and information throughout the study.

Principal Investigator Name (Printed)	Signature
Date	Site

1. SYNOPSIS

TITLE OF STUDY:

A Phase 1, Dose Escalation, and Cohort Expansion Study Evaluating NX-5948, a Bruton's Tyrosine Kinase (BTK) Degrader, in Adults with Relapsed/Refractory B-cell Malignancies

PROTOCOL NUMBER:

NX-5948-301

STUDY SITES:

Multicenter study conducted at approximately 60 sites in the European Union, Switzerland, United Kingdom, and the United States.

PHASE OF DEVELOPMENT:

Phase 1a/1b

STUDY RATIONALE:

NX-5948 is a chimeric targeting molecule that harnesses normal cellular protein degradation mechanisms (ie, E3 ligase-mediated ubiquitin-proteasome pathway) to degrade BTK, both wild type and mutated forms resistant to commercially available BTK inhibitor (BTKi) and has been shown to inhibit tumor growth in mouse models.

The therapeutic hypothesis is that BTK degradation will be effective in treating patients with chronic lymphocytic leukemia (CLL) and other B-cell malignancies where inhibition of BTK has demonstrated clinical effectiveness.

BTK degradation may be effective in patients who have developed resistance to BTKi, or in patients with BTKi intolerance, and potentially in B-cell indications where treatment with BTKi has been less effective.

NX-5948 has also demonstrated ability to cross the blood-brain barrier in animal studies and may be useful in treating B-cell malignancies present in the central nervous system (CNS).

OBJECTIVES AND ENDPOINTS:

Objectives and endpoints for Phase 1a and Phase 1b Parts 1 and 2 are summarized below:

Phase 1a (Dose Escalation)

Objectives	Endpoints
Primary • To evaluate the safety and tolerability of BTK degrader NX-5948, when taken orally, in adult patients with	• Incidence of treatment-emergent adverse events (TEAEs); Grade 3, 4, 5 TEAEs; serious adverse events (SAEs); TEAEs leading to study drug

relapsed/refractory (R/R) B-cell malignancies. • To establish the maximum tolerated dose (MTD) and/or recommended Phase 1b dose(s) of NX-5948 in adult patients with R/R B-cell malignancies.	discontinuation; deaths due to TEAEs, and all deaths. • Changes from baseline in safety parameters. • Incidence of dose-limiting toxicities (DLTs).
Secondary • To characterize the pharmacokinetic (PK) profile of NX-5948. • To characterize the pharmacodynamic (PD) profile of NX-5948. • To assess preliminary anti-tumor activity of NX-5948.	• NX-5948 PK parameters in plasma (maximum concentration [C_{max}], time at which C_{max} is observed [T_{max}], half-life, area under the concentration time curve up to the last measurable concentration [$AUC_{0\text{-last}}$], area under the concentration time curve to the end of a dosing period [$AUC_{0\text{-tau}}$], minimum concentration [C_{min}], accumulation ratio). • Changes from baseline of BTK levels in B-cells. • Overall response rate (ORR). • Complete response (CR) rate. o For patients with CLL, CR/complete response with incomplete recovery (CRi). • Time to first response. • Duration of response (DOR). • Progression-free survival (PFS). • Time to next therapy.
Exploratory • To assess overall survival (OS). • To explore mechanisms of response/resistance. • To characterize any PK/PD relationships with anti-tumor activity, biomarkers, and/or safety. • To evaluate NX-5948 metabolism. • To assess the impact of NX-5948 on cognitive function.	• OS. • Biomarkers of the BTK signaling and resistance pathways and target engagement, eg, including and not limited to: o Evaluation of immune cell phenotypes and B-cell receptor (BCR) signaling in peripheral blood mononuclear cells (PBMCs).

	○ Circulating cytokine and chemokine levels. ○ Next generation sequencing (NGS): retrospective evaluation of gene expression, mutation status, and allele frequency. • Identification of potential NX-5948 metabolites. • Mini Mental State Exam, 2nd Edition: Brief Version (MMSE-2: BV).

Phase 1b Part 1 (Safety Expansion)

Objectives	Endpoints
Primary • To evaluate the anti-tumor activity of NX-5948 in least 2 dose levels in patients with CLL/small lymphocytic lymphoma (SLL) selected for Phase 1b safety expansion. • To evaluate the anti-tumor activity of NX-5948 in patients with non-Hodgkin lymphomas (NHLs; mantle cell lymphoma [MCL], marginal zone lymphoma [MZL], diffuse large B-cell lymphoma [DLBCL]/high-grade B-cell lymphoma [HGBL], follicular lymphoma [FL], primary central nervous system lymphoma [PCNSL]/secondary central nervous system lymphoma [SCNSL]) or Waldenström macroglobulinemia (WM) based on the dose levels selected for Phase 1b safety expansion. • To further characterize and compare the safety and tolerability of at least 2 dose levels of NX-5948 in patients with CLL/SLL based on the dose(s) selected in Phase 1a for Phase 1b safety expansion. • To further characterize the safety and tolerability of NX-5948 in patients with NHLs (MCL, MZL, DLBCL/HGBL, FL, PCNSL/SCNSL) or WM based on the dose levels selected in Phase 1a for Phase 1b safety expansion.	• ORR as assessed by Investigator. • Incidence of TEAEs; Grade 3, 4, 5 TEAEs; SAEs; TEAEs leading to discontinuation; deaths due to TEAEs; all deaths; and changes in safety parameters.

• To identify the optimized dose for patients with CLL/SLL based on safety, tolerability, anti-tumor activity, and PK/PD profile for further evaluation of dose expansion cohorts.	
Secondary • To further characterize the PK profile of at least 2 dose levels of NX-5948. • To further characterize the PD profile of at least 2 dose levels of NX-5948. • To further assess anti-tumor activity of NX-5948 in at least 2 dose levels selected for Phase 1b safety expansion.	• NX-5948 PK parameters (C_{max}, T_{max}, half-life, $AUC_{0\text{-}last}$, $AUC_{0\text{-}tau}$, C_{min}, accumulation ratio, cerebrospinal fluid [CSF]:plasma concentration ratio) in plasma and CNS (for patients with CNS disease only). • Changes from baseline of BTK levels in B-cells. • CR rate. o For patients with CLL, CR/CRi. • Time to first response. • DOR. • PFS. • Time to next therapy.

Exploratory
• To assess OS.
• To evaluate NX-5948 metabolism.
• To explore mechanisms of response/resistance.
• To characterize any PK/PD relationships with anti-tumor activity, biomarkers, and/or safety.
• To assess the impact of NX-5948 on cognitive function.
• To further assess efficacy of NX-5948 in CLL/SLL subsets defined by mutations associated with clinical risk and/ or treatment resistance.
• To assess the efficacy of NX-5948 in CLL with secondary warm autoimmune hemolytic anemia (wAIHA) in Cohort 16
	• OS.
	• Biomarkers of the BTK signaling and resistance pathways and target engagement, eg, including and not limited to:
	  • Evaluation of immune cell phenotypes and BCR signaling in PBMCs.
	  • Circulating cytokine and chemokine levels.
	  • NGS: retrospective evaluation of gene expression, mutation status, and allele frequency.
	  • Minimal residual disease (MRD): evaluation of disease burden to understand depth and durability of response by NX-5948.

	• Explore patient-reported outcome (PRO) measure for overall impact of treatment toxicity (using Functional Assessment of Chronic Illness Therapy [FACIT-GP5] single item) for each cohort in Phase 1b Part 1.
	• Identification of potential NX-5948 metabolites.
	• MMSE-2: BV.
	• ORR and DOR for CLL/SLL mutation subsets including but not limited to BTK, PLCG2, TP53.
	• In patients with CLL with secondary wAIHA:
	  • Percentage of patients with autoimmune hemolytic anemia (AIHA) complete response (CR^{AIHA}), AIHA partial response (PR^{AIHA}), and overall AIHA response as assessed by Investigator
	  • Probability of maintained response
	  • DOR
	  • Time to response
	  • Relapse-free survival: Time from start of treatment until death or relapse of AIHA
	  • Frequency of packed red blood cell (PRBC) transfusion while receiving NX-5948

Phase 1b Part 2 (Cohort Expansion)

Objectives	Endpoints
Primary • To further evaluate the efficacy of NX-5948 in patients with CLL/SLL at the optimized dose identified in Phase 1b Part 1.	• ORR as assessed by Investigator.
Secondary	

• To further assess efficacy of NX-5948 at the optimized dose identified in Phase 1b Part 1. • To further characterize the safety and tolerability of NX-5948 in patients with CLL. • To further characterize the PK profile of NX-5948 at the optimized dose identified in Phase 1b Part 1. • To further characterize the PD profile of NX-5948 at the optimized dose identified in Phase 1b Part 1.	• Rate of CR/CRi. • Time to first response. • DOR. • PFS. • Time to next therapy. • Incidence of TEAEs; Grade 3, 4, 5 TEAEs; SAEs; TEAEs leading to discontinuation; deaths due to TEAEs; all deaths; and changes in safety parameters. • NX-5948 PK parameters (C_{max}, T_{max}, half-life, $AUC_{0\text{-last}}$, $AUC_{0\text{-tau}}$, C_{min}, accumulation ratio, CSF:plasma concentration ratio) in plasma and CNS (for patients with CNS disease only). • Changes from baseline of BTK levels in B cells.

Exploratory
• To assess OS.
• To further assess efficacy of NX-5948 in CLL/SLL subsets defined by mutations associated with clinical risk and/ or treatment resistance.
• To evaluate NX-5948 metabolism.
• To explore mechanisms of response/resistance.
• To characterize any PK/PD relationships with anti-tumor activity, biomarkers, and/or safety.
• To assess the impact of NX-5948 on cognitive function.
	• OS.
	• ORR and DOR for CLL/SLL mutation subsets including but not limited to BTK, PLCG2, TP53.
	• Biomarkers of the BTK signaling and resistance pathways and target engagement, eg, including and not limited to:
	  o Evaluation of immune cell phenotypes and BCR signaling in PBMCs.
	  o Circulating cytokine and chemokine levels.
	  o NGS: retrospective evaluation of gene expression, mutation status, and allele frequency.
	  o MRD: evaluation of disease burden to understand depth and durability of response by NX-5948.
	  o Explore PRO measure for overall impact of treatment toxicity (using FACIT-GP5 single item) for each cohort in Phase 1b Part 2.
	• Identification of potential NX-5948 metabolites.

	• MMSE-2: BV.

STUDY DESIGN AND PLAN:

Design:

The study will be divided into: Phase 1a (Dose Escalation) followed by Phase 1b Part 1 (Safety Expansion) and Phase 1b Part 2 (Cohort Expansion).

Phase 1a will evaluate the safety and tolerability of NX-5948 in patients with R/R B-cell malignancies who have received at least 2 prior lines of therapies (or at least 1 prior line of therapy for patients with PCNSL), and for whom no other therapies are known to provide clinical benefit.

Phase 1b Part 1 (Safety Expansion) will further investigate the safety, tolerability, and anti-tumor activity of NX-5948 in patients with B-cell malignancies at the recommended dose(s) selected in Phase 1a.

Phase 1b Part 2 (Cohort Expansion) will further investigate the anti-tumor activity of NX-5948 in patients with R/R CLL/SLL, with prior exposure to a BTKi and a B-cell lymphoma-2 inhibitor (BCL-2i) at the optimized dose identified in Phase 1b Part 1.

Phase 1a – Dose Escalation

Phase 1a of the study will utilize a standard 3+3 dose escalation design of 3 patients per dose level in the absence of DLTs. The study will commence with accrual to Dose Level (DL) 1 (50 mg/day). Patients at DL 1 will be treated for a minimum of 28 days (1 cycle of treatment) before enrolling any additional patients and/or subsequent dose levels.

Adverse events (AEs) will be assessed per the National Cancer Institute's (NCI) Common Terminology Criteria for Adverse Events (CTCAE) v5.0 (Section 15.2). The scheduled assessments at each clinic visit are shown in Section 15.3.

Assuming no (0) DLTs at any given dose level, accrual may advance to the next dose level. A 3-patient cohort with 1 DLT during Cycle 1 will be expanded to 6 patients.

If ≥ 33% of patients have a DLT in any dose level cohort during Cycle 1, then accrual at that dose will stop and a lower dose level cohort will be expanded up to 6 patients. This lower dose level cohort may be at the previous lower dose (if 6 patients have not already been accrued at that lower dose) or at an intermediate dose level subject to Safety Review Committee (SRC) approval. If ≥ 33% of patients have a DLT in DL 1, a lower dose level and/or alternative dosing design may be explored following SRC approval. To accommodate logistics of enrollment at multiple centers, over-enrollment of a 3-patient dose level cohort by an additional patient may be allowed provided no DLT has been observed in that dose level cohort. If a DLT occurs, a total of 6 evaluable patients will be enrolled. All patients will be considered for DLT assessments.

During dose escalation, up to a maximum of 12 backfill patients per previously cleared dose level may be enrolled in order to further explore safety and tolerability, with approval of the Sponsor. Any AEs that may occur in these patients during Cycle 1 will be considered as DLT evaluable, and the SRC will consider these AEs when determining the recommended Phase 1b dose(s).

DLTs are defined as TEAEs or laboratory abnormalities occurring during the first cycle (28 days) of study drug administration during Phase 1a Dose Escalation that meet 1 of the NCI CTCAE v5.0 criteria (see Section 4.1.3.2) unless clearly and incontrovertibly due to the underlying disease or an extraneous cause, and assessed as not related to study drug by the Investigator. Toxicities occurring in treatment Cycles 2 or later (Phase 1a) or in Phase 1b will be reviewed and their impact on dose escalation and dosing frequency assessed.

DLT Evaluation Criteria:

The following criteria apply in order to be evaluable for dose level cohort DLT/MTD decisions:

If a patient in Phase 1a misses more than 25% of doses for Cycle 1 (eg, > 7 doses of NX-5948 in 28 days) during the DLT period for reasons other than an AE, additional patient(s) may be enrolled to provide adequate data for dose escalation decision making.

Patients withdrawn from dosing during the DLT period in any dose level cohort for reasons unrelated to NX-5948 safety may be replaced in order to have 3 or 6 patients evaluable for dose escalation (dependent on dose level cohort size).

Dose reductions during the DLT period are not advised. Patients with dose interruption may be discontinued from treatment if not able to resume treatment within the timelines listed in Table 7 and Table 8. After Day 28, patients will continue receiving daily doses of NX-5948 in 28-day treatment cycles. In Cycle 2 and beyond for patients enrolled in Phase 1a and throughout the study for patients enrolled in Phase 1b, dose modifications for safety purposes are shown in Table 7 and Table 8 in Section 6.7. At the time of dose modification, safety and clinical laboratory assessments are to be performed. If required in Cycle 2+, dose reductions from the starting dose of 50 mg daily to DL -1 (25 mg daily) or DL -2 (25 mg every other day) will be decided by the Investigator who may consult the Medical Monitor. Sponsor should be notified in advance of dose modifications. Please see Section 6.7 for additional information on dose modifications after the DLT period.

All NX-5948-related Grade 4 non-hematologic events will be cause for study drug discontinuation.

Note: After completing 2 cycles of study drug treatment, intrapatient dose escalation may occur at the discretion of the Principal Investigator (PI) in consultation with the Medical Monitor once a higher dose level has been cleared and reviewed by the SRC. Any related AEs from prior NX-5948 administration must have improved to Grade ≤ 1 or baseline before intrapatient dose escalation. Intrapatient dose escalation may only occur up to a maximum of 2 times, always after completing 2 cycles of study drug treatment, up to the highest dose that has been cleared at the discretion of the PI in consultation with the Medical Monitor.

MTD is defined as the highest dose level with < 33% DLTs during Phase 1a of the study. This will be determined by the SRC after review of all available data. The recommended Phase 1b dose(s) will be determined using all available PK, PD, safety, anti-tumor activity, and tolerability data. The recommended Phase 1b dose(s) will not be higher than the MTD. Indication-specific MTDs/recommended Phase 1b dose(s) may be observed during dose escalation. Once an MTD/recommended Phase 1b dose(s) has been identified for 1 indication, the Sponsor may continue to concurrently enroll patients into the Phase 1a portion of the study for other indications.

The SRC will be comprised of representatives of the Sponsor, the Sponsor's Medical Monitor, the Contract Research Organization's (CRO's) Medical Monitor (if applicable), and at a minimum, a quorum of Investigators with patients under review. One or more relevant external medical experts may be added at the committee's discretion.

The SRC's responsibilities will be detailed in a separate Charter, and at a minimum will entail:

- Meet to evaluate the DLT period for each dose-escalation cohort to review any significant safety event (such as drug-related death or the equivalent of ≥ 33% of patients experiencing DLT events) as they occur.
- Review all relevant safety findings and confirm all DLTs.
- Determine all cohort dose escalation and de-escalation decisions during Phase 1a (Dose Escalation), including any cohort dose modifications based on toxicity.
- Recommend the dose(s) or dosing schedule(s), to extend into Phase 1b (safety and cohort expansion; ie, the recommended Phase 1b dose[s]).

Phase 1b Part 1 – Safety Expansion

Once the recommended Phase 1b dose(s) has been determined, the anti-tumor activity of NX-5948 will be investigated in up to 17 safety expansion cohorts of patients with B-cell malignancies. Eligibility for these cohorts and associated doses will be determined based on indication-specific criteria.

Safety expansion cohorts of up to 20 patients each per cohort (or per dose level for Cohort 1) will be recruited to further assess safety, tolerability, PK, PD, and evaluate preliminary anti-tumor activity of study drug. The data derived from the CLL cohorts will be used to determine the optimized dose for further study (recommended Phase 1b Part 2 dose).

Cumulative data from Phase 1a and available data from Phase 1b Part 1 Cohort 1 will be reviewed prior to initiating subsequent safety expansion cohorts. Cohorts 2 to 5, and 15 to 17 may be opened or filled based on emerging data.

Phase 1b Part 2 – Cohort Expansion

Following the review of maturing data from Phase 1a and available data from Phase1b Cohort 1 after the completion of Cohort 1 enrollment, and the identification of a recommended Phase 1b Part 2 dose, up to an additional 80 patients with CLL/SLL (post-BTKi, post-BCL-2i unless deemed ineligible for BCL-2i by Investigator at time of study enrollment) may be enrolled. ORR as assessed by Investigator will be utilized to further assess the anti-tumor activity of NX-5948 in this population in addition to its safety, tolerability, and PK/PD properties.

NUMBER OF PATIENTS PLANNED:

A total of up to approximately 572 patients are planned to be enrolled.

Approximately 30 to 66 evaluable patients in <u>each</u> of the parallel dose escalations for CLL/SLL and NHL/WM (ie, meeting DLT evaluability criteria and up to a maximum of 12 patients per previously cleared dose level) are planned to be enrolled in Phase 1a (dose escalation). The total is dependent on the number of dose levels investigated and will include at least 6 patients at the dose(s) selected for Phase 1b for additional analysis of safety, PK and PD, and clinical activity.

Up to 40 evaluable patients (ie, meeting efficacy evaluable criteria) in CLL/SLL Cohort 1 (20 patients per dose level) and up to 20 patients in all other cohorts in up to 17 B-cell malignancy safety expansion cohorts in Phase 1b Part 1 (up to 360 patients in total). Following dose optimization, up to an additional 80 patients with CLL/SLL may be enrolled in Phase 1b Part 2.

CRITERIA FOR INCLUSION AND EXCLUSION:

Individuals meeting all the following inclusion criteria and none of the exclusion criteria at Screening will be eligible for participation in the study.

Key Inclusion Criteria (refer to protocol Section 5.1 for detailed criteria and exceptional cases):

1. Each patient must sign an informed consent form (ICF) indicating that he or she understands the purpose of procedures required for the study and is willing to participate in the study.
2. Patients must be ≥ 18 years of age.
3. Patients in **Phase 1a (Dose Escalation)** must have 1 of the following histologically confirmed R/R B-cell malignancies:
   - R/R CLL, SLL
   - DLBCL of the following subgroups:
     - DLBCL not otherwise specified (NOS), germinal center B-cell type, activated B-cell type (includes transformed indolent lymphoma [eg, Grade 3b/transformed FL] and Richter-transformed DLBCL)
     - HGBL with *MYC* and *BCL*-2 and/or *BCL*-6 rearrangements, HGBL NOS
       - (note: other subgroups of DLBCL or large B-cell lymphoma (LBCL) from the 2016 World Health Organization [WHO] classification of lymphoid malignancies [Swerdlow 2016] are excluded)
   - FL (Grade 1–3a; eligibility for systemic treatment as determined by the Groupe d'Etude des Lymphomes Folliculaires [GELF] criteria; Section 15.8)
   - MCL
   - MZL (eligible subtypes include extranodal marginal zone lymphoma [EMZL], mucosa-associated lymphoid tissue [MALT], nodal marginal zone lymphoma [NMZL], and splenic marginal zone lymphoma [SMZL])
   - WM
   - PCNSL
4. Patients in **Phase 1a (Dose Escalation)** must meet the following:
   a. For non-PCNSL indications, received at least 2 prior lines of therapy and have no other therapies known to provide clinical benefit.
   b. For PCNSL, received at least 1 prior line of therapy.
5. Patients in **Phase 1b (Safety Expansion and Cohort Expansion)** must have 1 of the following histologically documented B-cell malignancies, meet criteria for systemic treatment, and have prior treatments and/or molecular features described for their assigned cohort:

CLL Cohorts:

Cohort 1 and Phase 1b Part 2:

- CLL or SLL with prior exposure to both a BTKi and a BCL-2i (unless deemed ineligible for a BCL-2i by Investigator at time of study enrollment).

Cohort 2:

- CLL or SLL patients who meet eligibility criteria for Cohort 1 and Phase 1b Part 2 (described above) **and** have documentation of a mutation in BTK other than C481S (eg, T474 or L528) within 6 months prior to study enrollment.

Cohort 3:

- CLL or SLL following treatment with a noncovalent BTKi and are BCL-2i naïve.

Cohort 4:

- CLL or SLL with documented TP53 mutation or 17p deletion following upfront therapy that included a BTKi and are BCL-2i naïve.

Cohort 5:

- CLL or SLL following at least 1 line of therapy that included a BTKi and BCL-2i naïve.

  (Note: For Cohorts 3 to 5, if patients meet multiple cohort criteria, refer to Section 4.1.4 for decision-making schema).

Cohort 15:

- First-line (1L)+ CLL or SLL requiring treatment with no prior exposure to a BTKi
  - A minimum of 6 (33%) and no more than 10 (50%) treatment-naïve patients may enroll in this cohort.
  - All other patients in this cohort: Second-line (2L)+ with prior exposure to non-BTKi therapy, including BCL-2i.
- Are intolerant to standard of care (SoC) treatment, or
- Refused SoC treatment, or
- Have been offered SoC and investigational medicinal product (IMP) but refused SoC in favor of IMP (as per consenting requirements), or
- The patient's treating physician considers that the lack of SoC treatment is not detrimental for the patient.

Cohort 16:

- BTKi-exposed R/R CLL or SLL requiring treatment, with secondary wAIHA
- Patients must have:
  - Received at least 1 line of therapy for CLL/SLL and at least 1 line of therapy for wAIHA

    (Note: In the event a therapeutic modality was intended to treat **both** CLL/SLL requiring treatment per International Workshop on CLL (iwCLL) **and** wAIHA requiring treatment, it may be utilized to meet both treatment requirements)
  - Anemia with hemoglobin ≤ 10 g/dL
  - Active wAIHA requiring treatment demonstrated by laboratory evidence of immune mediated hemolysis defined as:

- Positive direct antiglobulin test (DAT)
- Haptoglobin below the lower limit of normal (LLN)
- Minimum of one of the following:
  - Lactate dehydrogenase above the upper limit of normal (ULN)
  - Increased indirect bilirubin
  - Reticulocytosis appropriate for degree of anemia

- Uncontrolled AIHA defined as one of the following:
  - Relapsed after previous treatment with corticosteroids (with or without rituximab), BTKi, or splenectomy
  - Steroid resistant as evidenced by failure to obtain a hematologic response within 3 weeks on at least 1 mg/kg prednisone or prednisolone
  - Steroid dependent (on a dose of ≤ 20 mg/day of prednisone or prednisolone) and on stable doses for ≥ 2 weeks
- Note: Patients with multiple co-occurring autoimmune cytopenias (eg, Evans syndrome) are not eligible to enroll in Cohort 16

Cohort 17:

- BTKi-exposed R/R CLL or SLL requiring treatment due to one of the following:
  - Synchronous progression of systemic CLL/SLL and progression or new onset of CNS involvement **OR**
  - Progression of systemic disease in the presence of measurable ongoing CNS involvement **OR**
  - New or worsening CNS disease in the presence of systemic disease that is stable or demonstrating evidence of clinical response to therapy

NHL/WM Cohorts:

Cohort 6 (MCL):

- Non-blastoid MCL with prior exposure to a BTKi and an anti-CD20 monoclonal antibody (mAb)-based chemoimmunotherapy regimen.

Cohort 7 (MZL):

- MZL (EMZL, MALT, NMZL, SMZL) with prior exposure to an anti-CD20 mAb-based chemoimmunotherapy regimen and an additional line of therapy.

Cohort 8 (3L+ WM):

- WM with prior exposure to a BTKi and an additional line of therapy.

Cohort 9 (2L WM):

- WM following upfront therapy with a BTKi.

Cohort 10 (DLBCL):

- DLBCL of the following subgroups with prior exposure to an anthracycline (unless previously deemed ineligible to receive), an anti-CD20 mAb-based chemoimmunotherapy regimen, and an additional line of therapy:
  - Transformed indolent lymphoma (eg, Grade 3b/transformed FL)
  - Richter-transformed DLBCL

  (Note: other subgroups of DLBCL or LBCL from the 2016 WHO classification of lymphoid malignancies [Swerdlow 2016] are excluded).

Cohort 11 (FL):

- FL (Grade 1–3a; eligibility for systemic treatment as determined by the GELF criteria; Section 15.8) with prior exposure to an anti-CD20 mAb-based chemoimmunotherapy regimen and an additional line of therapy.

Cohort 12 (PCNSL/SCNSL):

- PCNSL following at least 1 prior line of therapy that included a BTKi (2L+) or following 2 or more prior lines of therapy (3L+).
- SCNSL meeting criteria for any indication described (with the exception of CLL/SLL) with secondary CNS involvement of their disease.

  (Note: CLL/SLL patients with CNS involvement will enroll in Cohort 17)

Cohort 13 (PCSNL):

- PCNSL following upfront therapy and with no prior exposure to a BTKi (2L).

Cohort 14 (1L WM):

- Patients with treatment-naïve WM deemed unfit for chemoimmunotherapy by Investigator at time of study entry.
- Are intolerant to SoC treatment, or
- Refused SoC treatment, or
- Have been offered SoC and IMP but refused SoC in favor of IMP (as per consenting requirements), or
- The patient's treating physician considers that the lack of SoC treatment is not detrimental for the patient.

6. Patients must have measurable disease per response criteria specific to the malignancy.
   a. For patients with NHL, must have either at least 1 metabolically active lesion defined as fluorodeoxyglucose (FDG)-avid (5PS score of 4 or 5) as assessed by positron emission tomography (PET)-computed tomography (CT) or at least 1 target lesion that is > 1.5 cm for lymph nodes or > 1.0 cm for extranodal lesion(s) in the longest diameter as assessed by CT.
   b. For patients with WM, serum monoclonal immunoglobulin M (IgM) paraprotein > 0.5 g/dL.
   c. Patients with SCNSL (Cohort 12) must have measurable disease per International PCNSL Collaboration Group criteria, as evidenced by at least one of the following: CNS disease on magnetic resonance imaging (MRI), detectable in CSF, or on ophthalmologic exam. The presence or absence of systemic

disease should be assessed and documented during Screening, but systemic disease is not required for enrollment.

7. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 (0 to 2 for patients with PCNSL and SCNSL).
8. Anticipated life expectancy of at least 12 weeks from the time of Screening.
9. Adequate organ and bone marrow function as follows:
   - Hemoglobin > 90 g/dL (or > 70 g/dL if anemia is due to bone marrow involvement or AIHA per Investigator assessment); a minimum 5-day transfusion free interval is required.
   - Absolute neutrophil count > 1.0×10^9/L unless neutropenia is due to bone marrow involvement per Investigator assessment, in which case there is no minimum ANC requirement. Granulocyte-colony stimulating factor (G-CSF) allowed for bone marrow involvement of disease.
   - Platelet counts:
     - Platelets > 50×10^9/L (or > 30×10^9/L if thrombocytopenia is due to bone marrow or splenic involvement).
     - For patients with PCNSL or SCNSL: platelets > 50×10^9/L regardless of etiology.
   - For all patients, a minimum 14-day transfusion-free interval is required prior to first dose.
   - Coagulation (international normalized ratio [INR] and partial thromboplastin time [PTT] or activated partial thromboplastin time [aPTT]) ≤ 1.5 × ULN; patients undergoing lumbar puncture must have PTT or aPTT < ULN.
   - Serum amylase and lipase ≤ 1.5 × ULN.
   - Aspartate aminotransferase (AST) and alanine aminotransferase (ALT) ≤ 2.5 × ULN; total bilirubin ≤ 1.5 × ULN (or ≤ 5 × ULN for Gilbert's syndrome) except for patients with indirect hyperbilirubinemia and evidence of hemolysis (haptoglobin below LLN) who may enroll in the absence of laboratory or clinical evidence of liver dysfunction.
   - Calculated glomerular filtration rate (GFR) > 45 mL/min/1.73 m^2.
   - QT interval corrected, Fredericia formula (QTcF) < 450 msec (triplicate electrocardiogram [ECG] average); for patients with pacemaker or bundle branch block QTcF < 475 msec.
   - Oxygen saturation (SpO2) > 92% on room air.
10. Women of childbearing potential (WOCBP) must agree to use highly effective contraception (failure rate of < 1% per year) during the study from Screening through 6 months after the last dose of study drug. WOCBP must agree to not donate eggs (ova, oocytes) during the same timeframe.
11. Male patients with WOCBP partners must agree to use highly effective contraception and barrier contraception during the study from Screening through 3 months after the last dose of study drug. Men must agree to not donate sperm during the same timeframe.

12. Patients must be willing and able to adhere to the prohibitions and restrictions specified in this protocol.

Key Exclusion Criteria (refer to protocol Section 5.1 for detailed criteria and exceptional cases):

Individuals who meet any of the following exclusion criteria at Screening will not be eligible to participate in the study:

1. Known or suspected active prolymphocytic leukemia or Richter's transformation to Hodgkin's lymphoma prior to enrollment.
2. Prior treatment for the indication under study for anti-cancer intent that includes:
    a. Radiotherapy within 2 weeks of planned start of study drug (excluding limited palliative radiation).
    b. Prior systemic chemotherapy within 2 weeks of planned start of study drug.
    c. Prior mAb therapy within 4 weeks of planned start of study drug, except for patients enrolling in Cohort 16 (CLL with secondary wAIHA) for whom a 16-week washout period is required.
    d. Prior small molecule therapy within 5 half-lives or 2 weeks (whichever is shorter) of planned start of study drug.
    e. Autologous or allogeneic stem cell transplant within 100 days prior to planned start of study drug.
    f. Chimeric antigen receptor (CAR) T-cell therapy within 100 days prior to start of study drug (within 60 days prior to start of study drug for Phase 1b).
    g. Use of systemic corticosteroids outside of dosing limits described below and within 7 days prior to initiation of study treatment excepting those used as prophylaxis for radio diagnostic contrast. Patients with CNSL (including both primary and secondary CNSL): no greater than 40 mg/day prednisone, or equivalent; CNSL patients using > 20 mg/day prednisone, or equivalent, must be clinically stable at that dose for 7 days. All other diagnoses: no greater than 20 mg/day prednisone or equivalent.
    h. Use of systemic immunosuppressive drugs other than systemic corticosteroids for any medical condition within 60 days prior to first dose of study drug.
    i. Previously treated with a BTK degrader.
3. Active, uncontrolled autoimmune hemolytic anemia (except for patients enrolling in Cohort 16) or active, uncontrolled autoimmune thrombocytopenia.
4. Unable to swallow capsules or malabsorption syndrome, disease or procedure significantly affecting gastrointestinal function, likely to interfere with the delivery, absorption, or metabolism of study drug.
5. Patients with active graft-versus-host disease (GVHD) or who have required anti-GVHD treatment or prophylaxis within 60 days of planned start of study drug.
6. History of known or suspected uncontrolled seizure disorder not related to CNS involvement of underlying malignancy.
7. Patient has any of the following:

a. Myocardial infarction, unstable angina, unstable symptomatic ischemic heart disease, or placement of a coronary arterial stent within 6 months of planned start of study drug.

b. Uncontrolled atrial fibrillation or other clinically significant arrhythmias, conduction abnormalities, or New York Heart Association (NYHA) class III or IV heart failure within 6 months of planned start of study drug.

c. Thromboembolic events (eg, deep vein thrombosis, pulmonary embolism, or symptomatic cerebrovascular events), stroke, or intracranial hemorrhage within 6 months of planned start of study drug.

d. Any other significant cardiac condition (eg, pericardial effusion, restrictive cardiomyopathy, severe untreated valvular stenosis, severe congenital heart disease, or persistent uncontrolled hypertension defined as systolic blood pressure > 160 mmHg or diastolic blood pressure > 100 mmHg despite optimal medical management) within 6 months of planned start of study drug.

8. Bleeding diathesis, or other known risk for acute blood loss.
9. History of Grade ≥ 2 hemorrhage within 28 days of planned start of study drug.
10. Patients requiring ongoing treatment with warfarin or an equivalent vitamin K antagonist within 168 hours prior to first dose of study drug.
11. Toxicities from previous anti-cancer therapies must have resolved to baseline levels or to Grade 1 (see exceptions in protocol).
12. Active known concurrent malignancy or malignancy other than the one under study within the past 3 years (see exceptions in protocol).
13. Patient who has known allergies, hypersensitivity, or intolerance to components of study drug.
14. Patient who requires tumor lysis syndrome (TLS) prophylaxis and has known allergy to both xanthine oxidase inhibitors and rasburicase.
15. Pregnant, breastfeeding, or planning to become pregnant while enrolled in this study or within 6 months after the last dose of study drug.
16. Patient is a man who plans to father a child while enrolled in this study or within 3 months after the last dose of study drug.
17. Patient has had major surgery (eg, requiring general anesthesia) within 4 weeks before the planned first dose of study drug.
18. Active, significant bacterial, fungal, parasitic, or viral infection (see details in protocol)
19. Active, uncontrolled diabetes (eg, hemoglobin A1c > 9%, symptomatic, with or without end organ complications, etc.) as assessed by the Investigator (note: poor glycemic control in the context of disease modifying corticosteroid treatment is not exclusionary).
20. Any other medical or psychiatric condition or social situation that, in the opinion of the Investigator, would compromise patient safety, or interfere with the objectives, the protocol, or completion of treatment per protocol.
21. Vaccinated with a live vaccine within 28 days prior to the first dose of study drug or completion of any vaccinations within 14 days prior to first dose of study drug.

Live vaccines are prohibited throughout study treatment and vaccinations of any kind are prohibited during the DLT period (Phase 1a) or Cycle 1 (Phase 1b).

22. Administration of any strong cytochrome P450 3A (CYP3A) inhibitors or inducers within 14 days or 5 half-lives, whichever is longer, and moderate CYP3A inhibitors or inducers within 168 hours, or 5 half-lives, whichever is longer, prior to first dose of study drug (see Section 6.8.2.2).
23. Administration of any inhibitors of P-glycoprotein or breast cancer resistance protein (BCRP) transporters within 48 hours prior to first dose of study drug (see Section 6.8.2.3).
24. Administration of a proton pump inhibitor (prescription or over-the-counter) within 168 hours prior to first dose of study drug. See Section 6.8.2.4 for allowed use of alternative agents.
25. Use of biotin (ie, vitamin B7) or supplements containing biotin higher than the daily adequate intake of 30 µg within 72 hours prior to Screening and during study treatment.
26. Dual treatment with an anticoagulant and antiplatelet agent is prohibited within 2 weeks of treatment initiation and while on study.

INVESTIGATIONAL PRODUCT(S), DOSE, ROUTE, AND REGIMEN:

Study drug (NX-5948) will be supplied as 25 mg, 50 mg, 100 mg, and 150 mg capsules to be taken orally once daily (QD) under fasting conditions. Prior to any dose modifications, PK trough levels of NX-5948 should be collected. As the effect of food on absorption is currently unknown, all patients should fast for at least 2 hours before and 2 hours after taking NX-5948. In addition, on intensive PK collection days (Cycle 1 Day 1 and Cycle 2 Day 1), patients should fast for a minimum of 8 hours before and for 2 hours after taking NX-5948. Patients will be treated with NX-5948 in 28-day cycles. The starting dose of study drug will be 50 mg/day.

PROHIBITED CONCOMITANT MEDICATIONS:

All systemic immunosuppressive agents with the exception of corticosteroids are prohibited within 60 days of first dose of study drug. Any corticosteroids (at dosages equivalent to > 20 mg/day prednisone for patients without CNSL or > 40 mg/day prednisone for patients with CNSL [except for prophylaxis for radiodiagnostic contrast reactions] within 7 days prior to the first dose of study drug) are prohibited. Live vaccines within 28 days prior to the first dose of study drug and at any time on study, and any vaccine within 14 days prior to the first dose of study drug are prohibited until completion of the first cycle. **Vaccinations of any kind are prohibited during the DLT period (Phase 1a) and Cycle 1 (Phase 1b).**

Any systemic anti-cancer therapy that is directed at the tumor under study, bone marrow transplant, investigational therapy, and radiotherapy are prohibited (focal palliative radiation is permitted following discussion with the Medical Monitor).

The use of biotin (ie, vitamin B7) or supplements containing biotin higher than the daily adequate intake of 30 µg is prohibited within 72 hours prior to Screening and during the study. A total daily dose of ≤ 30 µg is allowed.

Anticoagulants/Antiplatelet Medications:

Treatment with warfarin or an equivalent vitamin K antagonist is prohibited within 168 hours of starting study drug and while on study. Dual treatment with an anticoagulant and antiplatelet agent is prohibited within 2 weeks of treatment initiation and while on study.

CYP3A Strong or Moderate Inhibitors and Inducers:

Patients may not be treated with any substances that are known strong or moderate inducers or strong or moderate inhibitors of CYP3A, including drugs, herbs, nutritional supplements, or grapefruit- or Seville orange-containing products as they may alter the PK of NX-5948. If the patient is taking any of these restricted substances at the time of Screening, then the time between the last dose or consumption of that substance and study drug administration must be at least 14 days or 5 half-lives, whichever is longer, for strong inhibitors or inducers; and at least 168 hours or 5 half-lives, whichever is longer, for moderate inhibitors or inducers. In the event that a patient must receive time-limited therapy with a moderate or strong CYP3A inhibitor or inducer for treatment of an emergent condition, please notify Sponsor and manage as per protocol Section 6.8.2.2.

Inhibitors of P-glycoprotein and BCRP:

Patients may not be treated with any substances at doses at which they are clinically relevant inhibitors of P-glycoprotein or BCRP, including drugs, herbs, or nutritional supplements as they may alter the PK of NX-5948. If the patient is taking any of these restricted substances at the time of Screening or administered any of these restricted substances while on study treatment, the time between the last dose of that substance and study drug administration must be at least 48 hours.

Proton Pump Inhibitors and Other Acid-reducing Agents:

Patients may not be treated with any proton pump inhibitors (prescription or over-the-counter) as NX-5948 exhibits pH dependent solubility. If the patient is taking any of these restricted substances at the time of Screening, the time between the last dose of that substance and study drug administration must be at least 168 hours. Alternative acid reducing agents may be used: H2 antagonists (eg, famotidine) are permitted at least 12 hours prior to study drug administration and at least 4 hours after study drug administration; antacids (eg, calcium carbonate) are permitted at least 4 hours prior to or 4 hours following study drug administration; and alginates are prohibited at least 4 hours prior to or 4 hours following study drug administration.

RISK INFORMATION FOR NX-5948:

Possible clinical risks for NX-5948 are informed from nonclinical toxicity studies as well as selected known risks associated with BTKi that may be related to the proposed mechanism of action; these possible clinical risks include hemorrhage, mucosal effects, serious and opportunistic infections (including hepatitis B virus and other viral reactivations), cytopenias, cardiac arrhythmias, hypertension, and TLS.

Please refer to Section 6.6 and the NX-5948 Investigator's Brochure for details on risk information for NX-5948.

DURATION OF TREATMENT:

Patients will receive treatment with study drug until they experience documented disease progression, unacceptable toxicity, until they meet any of the withdrawal criteria, or until the study is terminated by the Sponsor.

DURATION OF STUDY:

The end of study is defined as when the last patient has either been in Long-term Follow-up for 1 year or has progressed, died, withdrawn from study, or been lost to follow-up, whichever occurs first. Anticipated duration of the study is approximately 6 years.

REFERENCE THERAPIES, DOSE, ROUTE, AND REGIMEN:

No reference therapy will be evaluated in this study.

STATISTICAL METHODS:

Safety Analysis Set:

Safety analysis set (SAS) consists of all patients who receive at least 1 dose of NX-5948. All safety and tolerability evaluations will be based on this analysis set.

DLT/MTD Evaluable Analysis Set:

The DLT/MTD evaluable analysis set will include all patients in Phase 1a (Dose Escalation) who complete 1 cycle of treatment (receiving ≥ 21 doses of NX-5948) and assessment, or who discontinue before completing the first cycle because of toxicity. This will be the analysis set for MTD/recommended Phase 1b dose(s) evaluations.

PK Analysis Set:

PK analysis set (PKAS) includes all patients without protocol deviations affecting interpretability of PK. For noncompartmental assessments, patients must have sufficient plasma concentration data to allow for estimation of at least 1 PK parameter. PK analyses will be based on this analysis set.

PD Analysis Set:

PD analysis set (PDAS) includes all patients without protocol deviations affecting the interpretability of the PD. All PD analyses are based on this analysis set.

Efficacy Analysis Set:

Efficacy analysis set (EAS) includes all patients who received study treatment, have a baseline disease assessment, and at least 1 post-baseline indication-specific disease assessment, died, or discontinued due to clinical progression or toxicity. Unless otherwise specified, all efficacy analyses will be based on this analysis set.

Per Protocol Set:

The per protocol set (PPS) is a subset of the EAS that may also be used to analyze select efficacy endpoints and will exclude patients with certain major protocol deviations. The criteria for inclusion in PPS will be finalized and documented prior to database lock.

SAMPLE SIZE JUSTIFICATION:

Phase 1a:

Approximately 30 to 66 evaluable patients in <u>each</u> of the parallel dose escalations for CLL/SLL and NHL/WM are planned to be enrolled in Phase 1a Dose Escalation. The Phase 1a Dose Escalation stage of the study is a traditional 3+3 dose escalation design with no formal statistical hypothesis. During dose escalation, up to a maximum of 12 patients per previously cleared dose levels may be enrolled in order to further explore safety and tolerability, with approval of the

Sponsor. Under this design, there is a 71% probability of escalation (ie, observing 0 DLTs in 3 patients or 1 DLT in 6 patients) if the true but unknown rate of DLT is 20%.

Given the exploratory nature of the Phase 1a Dose Escalation stage, the sample size is not based on power calculations. The number of patients to be included is typical for this kind of study, and the results give sufficient safety and tolerability data as well as for the description of PK and PD parameters without exposing too many patients.

Phase 1b Part 1 (Safety Expansion):

For the Phase 1b Part 1 portion of the study, up to approximately 360 additional response-evaluable patients (total) in 17 B-cell malignancy safety expansion cohorts are planned to be enrolled. A Simon 2-stage approach (Simon 1989) will be used for each dose level in Cohort 1, with a 1-sided type I error rate of 5% and power of 80%. Up to 20 evaluable patients will be included in each dose level.

For all remaining cohorts, analyses will be descriptive in nature, and the sample size is determined by practical, rather than power, considerations. With up to 20 evaluable patients anticipated per cohort, the precision of the point estimate for the ORR, based on the width of its 95% Clopper-Pearson confidence interval (CI), ranges from a width of 0.247 (95% CI = [0.001, 0.249]) for a response rate of 1/20 to a maximum width of 0.456 (95% CI = [0.272, 0.728]) when the response rate is 10/20.

Phase 1b Part 2 (Cohort Expansion):

For the Phase 1b Part 2 portion of the study, Stage 2 of the Simon 2-stage design for Cohort 1 will be initiated at the dose(s) established in Phase 1b Part 1, at the Sponsor's discretion upon establishing lack of futility. Up to an additional 80 response-evaluable patients with CLL/SLL following prior exposure to a BTKi and a BCL-2i may be enrolled.

STATISTICAL ANALYSIS:

Analyses of Phase 1a and Phase 1b portions of the study will be reported separately. The data will be summarized by dose level cohort (Phase 1a), safety expansion cohort (Phase 1b Part 1), cohort expansion cohorts (Phase 1b Part 2), and all patients.

The Simon 2-stage design will be formally implemented as a rule-based futility stopping rule for CLL/SLL patients prior to initiating Phase 1b Part 2 enrollment. The decision criteria at the end of Phase 1b Part 1, Cohort 1 will adhere to the operating characteristics detailed in Section 4.2.

Safety expansion cohorts in Phase 1b will be summarized separately. Summary statistics for continuous variables will include the number of patients (n), mean, standard deviation, median, minimum, and maximum values. Summary statistics for categorical variables will include the frequency and percentage of patients in each category. All assessments before the first dose of study drug will be considered as pre-dose assessments. If there are multiple pre-dose assessments, the most recent one will be used as "baseline assessment" for statistical analysis. No formal hypothesis testing is planned for efficacy, safety, PK, or PD parameters.

In Phase 1b, 2-sided 95% exact CIs will be presented for the ORR for EAS. Analyses of ORR in Phase 1b may be repeated and presented for the PPS. CR rate (and CR/CRi rate for patients with CLL) will be analyzed using a similar method. Best overall response will be summarized with frequencies and percentages. PFS will also be calculated using a Kaplan-Meier method overall and for appropriate subgroups and cohorts, using B-cell malignancy indication-specific criteria, as appropriate. The 6-, 12-, etc., month time points for PFS may also be reported. All time-to-event type endpoints (such as time to first response, DOR, time to next therapy) will be analyzed using a similar statistical method.

COVID-19 PUBLIC HEALTH EMERGENCY IMPACT ON PROTOCOL ASSESSMENTS:

During study conduct, challenges may arise, for example, from quarantines, site closures, travel limitations, interruptions to the supply chain for the investigational product, or other considerations if site personnel or trial patients become infected with the SARS-Cov-2 virus and develop coronavirus disease 2019 (COVID-19). These challenges may lead to difficulties in meeting protocol-specified procedures, including administration of the study drug or adhering to protocol-mandated visits and laboratory/diagnostic testing. During the course of study conduct, protocol modifications may be required, and there may be unavoidable protocol deviations due to COVID-19 illness and/or COVID-19 public health control measures. In such cases, these incidences should be documented.

Since trial participants may potentially not be able to come to the investigational site for protocol-specified visits, alternative methods for safety assessments (eg, phone contact, virtual visit, telemedicine, alternative location for assessment, including local labs or imaging centers) may be implemented when necessary and feasible, and would be sufficient to assure the safety of trial participants.

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LIST OF ABBREVIATIONS AND DEFINITIONS OF TERMS

Abbreviation	Definition
1L	first line
2L	second line
3L	third line
ABC	activated B-cell type
ADL	activities of daily living
AE	adverse event
AIDS	acquired immunodeficiency syndrome
AIHA	autoimmune hemolytic anemia
ALCOA	Attributable, Legible, Contemporaneous, Original, and Accurate
ALT	alanine aminotransferase
ANC	absolute neutrophil count
aPTT	activated partial thromboplastin time
AST	aspartate aminotransferase
$AUC_{0\text{-last}}$	area under the concentration time curve up to the last measurable concentration
$AUC_{0\text{-tau}}$	area under the concentration time curve to the end of a dosing period
BCL-2	B-cell lymphoma-2
BCL-2i	B-cell lymphoma-2 inhibitor
BCR	B-cell receptor
BCRP	breast cancer resistance protein
BTK	Bruton's tyrosine kinase
BTKi	Bruton's tyrosine kinase inhibitor
BUN	blood urea nitrogen
C481	cysteine 481
CAR	chimeric antigen receptor
CD20	B lymphocyte antigen CD20
CD4	cluster of differentiation 4
CFR	Code of Federal Regulations
CI	confidence interval
CKD-EPI	Chronic Kidney Disease Epidemiology Collaboration
CLL	chronic lymphocytic leukemia
C_{max}	maximum concentration
C_{min}	minimum concentration
CMV	cytomegalovirus
CNS	central nervous system

Abbreviation	Definition
CNSL	central nervous system lymphoma
COVID-19	coronavirus disease 2019
CR	complete response
CRA	clinical research associate
CR^{AIHA}	autoimmune hemolytic anemia complete response
CRBN	cereblon
CRF	case report form
CRi	complete response with incomplete recovery
CRO	Contract Research Organization
CSF	cerebrospinal fluid
CT	computed tomography
CTCAE	Common Terminology Criteria for Adverse Events
ctDNA	circulating tumor deoxyribonucleic acid
CTFG	Clinical Trial Facilitation Group
CTM	chimeric targeting molecule
CYP3A	cytochrome P450 3A
CYP450	cytochrome P450
DAT	direct antiglobulin test
DC_{50}	concentration resulting in 50% degradation of target
DHL	double-hit lymphoma
DL	dose level
DLBCL	diffuse large B-cell lymphoma
DLT	dose-limiting toxicity
DNA	deoxyribonucleic acid
DOR	duration of response
EAS	efficacy analysis set
EBV	Epstein-Barr virus
EC	ethics committee
ECG	electrocardiogram
ECOG	Eastern Cooperative Oncology Group
EDC	electronic data capture
eGFR	estimated glomerular filtration rate
EMZL	extranodal marginal zone lymphoma
EOT	end of treatment
ERP	Eligibility Review Packet
EU	European Union
FACIT-GP5	Functional Assessment of Chronic Illness Therapy
FaSSIF	fasted state stimulated intestinal fluid
FDA	Food and Drug Administration
FDG	fluorodeoxyglucose
FeSSIF	fed state stimulated intestinal fluid

Abbreviation	Definition
FIH	first-in-human
FL	follicular lymphoma
G-CSF	granulocyte-colony stimulating factor
GCB	germinal center B-cell subtype
GCP	Good Clinical Practice
GDPR	General Data Protection Regulation
GELF	Groupe d'Etude des Lymphomes Folliculaires
GFR	glomerular filtration rate
GLP	Good Laboratory Practice
GMP	Good Manufacturing Practice
GVHD	graft-versus-host disease
HbcAb	hepatitis B core antibody
HbsAb	hepatitis B virus surface antibody
HbsAg	hepatitis B virus surface antigen
HBV	hepatitis B virus
HCV	hepatitis C virus
HGBL	high-grade B-cell lymphoma
HIPAA	Health Insurance Portability and Accountability Act
HIV	human immunodeficiency virus
HNSTD	highest non-severely toxic dose
ICF	informed consent form
ICH	International Council for Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use
ID	identification
IDMC	Independent Data Monitoring Committee
Ig	immunoglobulin
IMP	investigational medicinal product
INR	international normalized ratio
IRB	Institutional Review Board
iwCLL	International Workshop on CLL
LBCL	large B-cell lymphoma
LDH	lactate dehydrogenase
LDT	Laboratory Developed Test
LFT	liver function test
LLN	lower limit of normal
LPL	lymphoplasmacytic lymphoma
LTFU	long-term follow-up
mAb	monoclonal antibody
MAD	maximum administered dose
MALT	mucosa-associated lymphoid tissue
MATE1	multidrug and toxin extrusion transporter 1
MCL	mantle cell lymphoma
MCV	mean corpuscular volume
MEC	molar extinction coefficient

Abbreviation	Definition
MedDRA	Medical Dictionary for Regulatory Activities
MMSE-2: BV	Mini Mental State Exam, 2nd Edition: Brief Version
MRD	minimal residual disease
MRI	magnetic resonance imaging
MTD	maximum tolerated dose
MUGA	multiple gated acquisition scan
MZL	marginal zone lymphoma
ncBTKi	noncovalent BTK inhibitor
NCI	National Cancer Institute
NGS	next generation sequencing
NHL	non-Hodgkin lymphoma
NMZL	nodal marginal zone lymphoma
nPR	nodular partial response
NOAEL	no observed-adverse-effect level
NOS	not otherwise specified
NYHA	New York Heart Association
ORR	overall response rate
OS	overall survival
PBMC	peripheral blood mononuclear cell
PCNSL	primary central nervous system lymphoma
PCR	polymerase chain reaction
PD	pharmacodynamic
PDAS	pharmacodynamic analysis set
PET	positron emission tomography
PI	Principal Investigator
PFS	progression-free survival
PK	pharmacokinetic
PKAS	pharmacokinetic analysis set
PPS	per protocol set
PR	partial response
PR^{AIHA}	autoimmune hemolytic anemia partial response
PRBC	packed red blood cell
PR-L	partial response with lymphocytosis
PRO	patient reported outcome
PTT	partial thromboplastin time
QD	once daily
QTc	QT interval corrected
QTcF	QT interval corrected, Fredericia formula
RBC	red blood cell (count)
R-CHOP	rituximab, cyclophosphamide, doxorubicin hydrochloride, vincristine sulfate, and prednisone
R/R	relapsed/refractory
RS	Richter's syndrome

Abbreviation	Definition
SAE	serious adverse event
SAP	statistical analysis plan
SAS	safety analysis set
SCNSL	secondary central nervous system lymphoma
SD	stable disease
SFU	safety follow-up
SLL	small lymphocytic lymphoma
SMZL	splenic marginal zone lymphoma
SoC	standard of care
SpO_2	oxygen saturation
SRC	Safety Review Committee
STD10	severely toxic dose to 10% of the animals
T3	triiodothyronine
T4	Thyroxine
TEAE	treatment-emergent adverse events
THL	triple-hit lymphoma
TLS	tumor lysis syndrome
T_{max}	time at which C_{max} is observed
TSH	thyroid stimulating hormone
US	United States
UK	United Kingdom
ULN	upper limit of normal
wAIHA	warm autoimmune hemolytic anemia
WBC	white blood cell (count)
WHO	World Health Organization
WOCBP	women of childbearing potential
WM	Waldenström macroglobulinemia
WT	wild type

2. INTRODUCTION

A comprehensive review of study drug, NX-5948, is contained in the Investigator's Brochure supplied by Nurix Therapeutics, Inc. Investigators are to review this document prior to initiating this study.

2.1. Study Background and Clinical Rationale

2.1.1. Disease Background

Cancer is a leading cause of morbidity and mortality worldwide, responsible for 18 million diagnoses and nearly 10 million deaths per year (Bray 2018). It is now considered 1 of the major public health challenges of the developed world and has become the leading cause of death in several Western countries. In the United Kingdom (UK), there are approximately 367,000 new cancer cases diagnosed annually and incidence rates have increased by approximately 5% over the past decade. The lifetime risk for developing some form of cancer in the UK is 1:2 in people born after 1960. The 10-year survival rate across indications in the UK is approximately 50% (Cancer Research UK 2020). In the United States (US) in 2018, over 1.7 million people were expected to be diagnosed and over 600,000 people were expected to die of cancer (American Cancer Society 2018). The lifetime risk for both men and women to develop cancer is about 1:3 and 1:5 to die from it.

Lymphoid neoplasms of mature B cells are classified in part based upon a comparison of the immunophenotype and genotype of the tumor cells to normal stages of B-cell development as well as other characteristic immunophenotypic and genetic features.

These tumors can be derived from any stage of mature B-cell development, including I B cells, germinal center B cells (GCBs), post-germinal center memory B cells, or plasma cells. The most common B-cell neoplasms derive from cells that have experienced a germinal center reaction, which is initiated when antigen-stimulated B cells migrate into the germinal centers (or follicles) of secondary lymphoid organs (eg, lymph nodes, spleen, and mucosa associated lymphoid tissues). GCBs proliferate and undergo 2 events that permit the diversification of immunoglobulin (Ig) genes, somatic hypermutation, and heavy chain class switching. Most tumors of mature B cells, including plasma cell neoplasms, show evidence of somatic hypermutation, and it is thought that "mistakes" that occur during somatic hypermutation and class switching are responsible for many of the acquired mutations that lead to B-cell transformation. The World Health Organization (WHO) classification, updated in 2016, identifies more than 40 major B-cell malignancy subtypes with distinct genetic, morphologic, and clinical features (Swerdlow 2016). Among these, in decreasing frequency of occurrence, are:

- Diffuse large B-cell lymphoma (DLBCL);
- Chronic lymphocytic leukemia (CLL) and small lymphocytic lymphoma (SLL);

- Follicular lymphoma (FL);
- Marginal zone lymphoma (MZL);
- Mantle cell lymphoma (MCL);
- Waldenström macroglobulinemia (WM); and
- Primary and secondary central nervous system (CNS) lymphoma (PCNSL and SCNSL).

2.1.2. DLBCL and High-grade B-cell Lymphomas

DLBCL is the most common type of non-Hodgkin lymphoma (NHL), accounting for approximately 30% of all cases (Jaffe 2008). Age-standardized incidence rates range from 4–7 per 100,000 person-years in Western countries (Sant 2010). In the US, the rate of new cases of DLBCL is 5.6 per 100,000 per year, and the death rate is 1.8 per 100,000 per year (SEER 2013–2017 age adjusted rates). The median age at presentation is 70; however, DLBCL can occur at any age, with a slightly higher incidence in men (Smith 2011). Five-year relative survival rate is 63.8% (SEER 2010–2016).

The crude incidence of DLBCL in the European Union (EU) is estimated at 3–4 per 100,000 per year (Tilly 2015). The annual incidence rate of DLBCL in the UK is 7.3 cases per 100,000. Incidence increases with age from 2.5 per 100,000 per year in those aged 35–39 years to 32 per 100,000 per year in those aged 70–74 years, with an expected 4820 new UK cases per year (HMRN 2010-2016).

DLBCL is clinically, morphologically, and genetically a highly heterogeneous lymphoid malignancy that can arise from mature B cells during multiple stages of differentiation. DLBCL may arise as primary (de novo) or from the progression or transformation of other B-cell neoplasms such as CLL/SLL, lymphoplasmacytic lymphoma (LPL), MZL, and FL (Martelli 2013). Gene mutations in B cells alter their gene expression thereby promoting neoplastic transformation. This results in malignant lymphoid cells with a characteristic nuclear size equal to or exceeding normal macrophage nuclei and more than twice the size of a normal lymphocyte (Gouveia 2012).

Gene expression profiling separates DLBCL into 2 major groups:

- GCB, and
- Activated B-cell type (ABC).

GCB tumor cells retain a gene expression profile that resembles a normal GCB and demonstrate t(14;18) translocations in approximately 30–40% of cases. Conversely, ABC tumors have a gene expression profile that resembles an ABC. These tumors are thought to develop from post-GCBs that have commenced early stages of plasmacytic differentiation. When compared with GCB tumors, ABC tumors are associated with inferior outcomes following standard rituximab, cyclophosphamide, doxorubicin hydrochloride, vincristine

sulfate, and prednisone (R-CHOP) therapy; 5-year overall survival (OS) rates of 56% compared to 78%, respectively (Scott 2015).

High-grade B-cell lymphomas (HGBL) is a unique category of NHLs that harbor *MYC* and *BCL-2* and/or *BCL-6* translocations or double-hit/triple-hit lymphomas (DHL/THL), as updated in the 2017 WHO classification. HGBL constitutes approximately 2–11% of all newly diagnosed DLBCL and is associated with inferior outcomes. Additionally, retrospective studies suggest that more intense chemoimmunotherapy regimens may result in better outcomes in patients with DHL/THL among standard therapies used in DLBCL (NCCN Guidelines for B-cell Lymphomas 2023).

2.1.3. CLL/SLL

CLL/SLL is the most common leukemia in Western countries. The annual incidence rate of CLL/SLL in the UK is 7.1 cases per 100,000 with an expected 4680 new UK cases per year (HMRN 2010-16). CLL/SLL accounts for 25–30% of all leukemias in the US (Siegel 2020). The incidence of CLL/SLL is extremely low in Asian countries such as China and Japan, occurring at a frequency approximately 10% of that seen in Western countries (Miranda-Filho 2018). The rate of new US cases of CLL/SLL is 4.6 per 100,000 per year. The death rate is 1.0 per 100,000 per year (SEER 2013–2017 age adjusted rates). Five-year relative survival rate is 85.7% (SEER 2010-2016). The disorder is more common in men, with a male to female ratio of approximately 1.3:1 to 1.7:1. CLL/SLL is considered to be mainly a disease of older adults, with a median age at diagnosis of approximately 70 years.

CLL/SLL is a malignancy of small, mature-appearing lymphocytes, presenting primarily either as a leukemia (CLL) or a lymphoma (SLL). In roughly 60% of cases, the Ig genes are somatically hypermutated, while the remaining cases appear to derive from naïve B cells. CLL/SLL is an extremely heterogeneous disease. In approximately 30% of cases, it can be indolent associated with a prolonged (ie, 10–20 years) clinical course with the eventual cause of death unrelated to CLL. On the other hand, some patients suffer from an aggressive form of the disease succumbing within 2–3 years from diagnosis to their disease or treatment-related complications. Other patients live for 5–10 years with an initial indolent and relatively benign course followed by a more aggressive terminal phase lasting 1–2 years. During this terminal phase, there is considerable morbidity, both from the disease itself and from complications of therapy. In recent years, small molecule therapies targeting Bruton's tyrosine kinase (BTK) activity and B-cell lymphoma-2 (BCL-2) activity have markedly changed approaches to therapy and improved outcomes in patients with CLL. For patients whose disease fails to respond to or progresses following these therapies, options are limited (Roeker 2022). CNS involvement of CLL is exceedingly rare in terms of reported cases with an approximate prevalence of 0.8% based on a single center study over a period of 15 years; however, there is no standard initial regimen for CNS CLL (or Richter's syndrome; RS) due to the rarity of these conditions. Despite use of systemic fludarabine-based therapies, which have CNS penetrance, the prognosis for patients with significant CNS involvement by CLL or RS is poor, with a median OS of 12 and 11 months, respectively (Strati 2016).

2.1.4. FL

FL accounts for approximately 15–20% of adult NHL in Western countries and is the second most common lymphoma in the US and Western Europe (Swerdlow 2016). FL is characterized by an indolent clinical course with a survival range of 8–15 years (Horning 1984). The age-standardized incidence rate ranges from 2–3 per 100,000 person years in Western countries with a median age at diagnosis in the 60s (Morton 2006; Freedman 2020). The annual incidence rate of FL in the UK is 3.3 cases per 100,000, with an estimated 2150 patients diagnosed each year with FL (HMRN 2010-16). The rate of new US cases of FL was 2.7 per 100,000 per year. The death rate was 0.5 per 100,000 per year (SEER 2013–2017 age adjusted rates). Five-year relative survival rate is 89% (SEER 2010- 2016).

Patients usually present with painless peripheral adenopathy. FL is comprised of neoplastic GCBs and recapitulates the appearance of normal germinal centers of secondary lymphoid follicles. The vast majority (> 85%) of FL express BCL-2.

2.1.5. MZL

MZL is an indolent lymphoma subtype that accounts for about 5–10% of NHL. The age-standardized incidence rate ranges from 0.4–1.0 per 100,000 person-years in Western countries. The annual incidence rate of MZL in the UK is 4.1 cases per 100,000. In the UK, an estimated 2690 patients are diagnosed every year with MZL (HMRN 2010-2016). In the current WHO classification, MZL is divided into 3 distinct diseases:

- Extranodal MZL (EMZL) of mucosa-associated lymphoid tissue (MALT);
- Splenic MZL (SMZL); and
- Nodal MZL (NMZL).

In the US during 2001-2009, 4081 (incidence rate: 5.7/1 million person-years) and 8821 (incidence rate: 12.3/1 million person-years) individuals were diagnosed with NMZL and EMZL, respectively (Khalil 2014).

Surveillance, Epidemiology, and End Results Program database reports 5-year relative survival rates of 84% for MZL as a whole, 77% for NMZL, 89% for MALT, and 80% for SMZL. Several inflammatory pathological states are associated with specific subtypes of EMZL, most commonly *H. pylori* infection in gastric MALT lymphoma.

2.1.6. MCL

MCL is an aggressive B-cell lymphoma with poor prognosis (median survival 3–5 years) and relatively rare, accounting for only 2–4% of NHL cases (Swerdlow 2016). The age-standardized incidence rate is approximately 0.5 per 100,000 person-years in Western countries. The annual incidence rate of MCL in the UK is 0.9 cases per 100,000, with an estimated 560 new cases diagnosed each year (HMRN 2010-16). In the US, MCL accounted for over 4000 new cases in 2022 (American Cancer Society 2019; Siegel 2022). Approximately three-quarters of patients are male, and Caucasians are affected almost twice as

frequently as Blacks. The median age at diagnosis is 68 years. In about 80% of cases, the cell of origin is a naïve B cell, while the remaining cases are somatically hypermutated and likely derived from antigen-stimulated B cells.

2.1.7. WM

WM is a distinct clinicopathologic entity demonstrating LPL in the bone marrow with an IgM monoclonal gammopathy in the blood. The MYD88 L265P gene mutation has been identified in over 90% of patients with WM and can be of help in differentiating WM from other conditions (Treon 2012). The estimated annual UK incidence of WM is 0.55 per 100,000 (Owen 2014), with approximately 400 new cases diagnosed each year (WMUK 2020). Approximately 1400 new cases are diagnosed in the US each year (Fonseca 2007; Groves 1998). The median age at diagnosis is 70 years; less than 10% of patients are under 50 years of age, and approximately 60% are males (Castillo 2015; Kyle 2018). Median survival is reported as approximately 8 years (Castillo 2015).

2.1.8. CNS Lymphoma

Lymphoma of the CNS can occur both as *de novo* (primary) disease or in the context of metastatic spread or relapse of extra-CNS disease (secondary). PCNSL is a highly aggressive lymphoma that accounts for 1% of adult NHL (Hoang-Xuan 2015). The disease is limited to the brain, eyes, leptomeninges, and spinal cord, without systemic spread. PCNSL is relatively rare with an estimated annual UK incidence of 3 per 1 million people per year (Hoffman 2006) and approximately 170 cases diagnosed each year (HMRN 2010-2016). The US incidence is approximately 5 per 1 million people per year, and there has been an increase in incidence, especially in patients older than 60 years of age (Grommes 2017).

PCNSL develops in both immunocompetent and immunosuppressed patients, including those with human immunodeficiency virus (HIV)/acquired immunodeficiency syndrome (AIDS) and patients on chronic immune suppression following solid organ transplant. Five-year OS is low in both groups at 30% and 10%, respectively (Shiels 2016). Patients with PCNSL usually present with neurologic symptoms that progress over several weeks, including focal neurological deficits, mental status changes, headache, nausea, vomiting, and seizures (Grommes 2017).

Aggressive NHL subtypes that involve the CNS as a site of relapse (SCNSL) pose a clinical challenge in terms of effective therapy. Approximately 2–10% of patients with aggressive subtypes of systemic NHL (eg, DLBCL) will experience direct involvement of the CNS at some time during their disease course (Bernstein 2009). The incidence is much higher in highly aggressive NHL (eg, Burkitt lymphoma/leukemia, lymphoblastic lymphoma) and lower in indolent NHL (eg, FL).

2.2. Scientific Rationale for NX-5948 in B-cell Malignancies

BTK is a key component of the B-cell receptor (BCR) signaling pathway and has been clinically validated as a target in the treatment of B-cell malignancies. BCR signaling is

indispensable for the adhesion, survival, and growth of human B cells. Chronic activation of the BTK-mediated BCR signaling is a hallmark of many B-cell lymphoid malignancies making it an attractive therapeutic target (Woyach 2012; Pal Singh 2018). Therapies targeting BTK are approved for the treatment of various B-cell lymphomas, including MCL, CLL, SLL, WM, and MZL (Miklos 2017). First generation BTK inhibitors (BTKi) bind covalently to cysteine 481 (C481) of the BTK protein and profoundly impair BCR signaling; however, acquired mutations of C481 are associated with clinical resistance to these agents and subsequent relapse (Woyach 2014; Woyach 2017). Several noncovalent BTKi (ncBTKi) that do not require binding to C481 are currently being investigated in clinical trials as potential therapies for patients with relapsed/refractory (R/R) disease. While ncBTKi have shown promising activity, many patients treated with these compounds ultimately develop resistance mutations in BTK at sites other than C481. These mutations include BTK-V416L, BTK-T474I, and BTK-L528W and have been shown to decrease the *in vitro* activity of noncovalent inhibitors (Wang 2022; Naeem 2023). Interestingly, a number of these newly identified mutations reduce or abolish the enzymatic activity of BTK while continuing to drive tumor cell growth (Montoya 2022). These findings suggest a non-enzymatic scaffolding function for BTK that is not addressed by kinase inhibitors.

BTK inhibition has been shown to be safe and effective for several B-cell malignancies, but BTKi-resistant disease remains a clinical challenge with limited options for salvage therapy and overall poor patient outcomes. BTK degradation may provide a promising mechanism of overcoming such resistance.

By targeting BTK protein levels rather than BTK activity, small molecule-induced protein degradation of BTK offers a unique approach to overcome such resistant mutations. NX-5948 is a chimeric targeting molecule (CTM) that induces the degradation of BTK in cells through recruitment of cereblon (CRBN), an adaptor protein of the E3 ubiquitin ligase complex, and promotes the formation of a ternary complex of CRBN, NX-5948, and BTK. Within this complex, ubiquitin is transferred to BTK directing it for degradation by the proteasome (Figure 1).

Nurix has generated data demonstrating that NX-5948 degrades both wild type (WT; degradation concentration [DC_{50}] of 1.81 nM in primary human B cells) and ibrutinib-resistant forms of the BTK protein in a lymphoma cell line, representing a potential approach to targeting both WT and mutant proteins (refer to the Investigator's Brochure). In addition, NX-5948 demonstrates potent tumor growth inhibition across orally delivered doses in BTK-dependent mouse xenograft tumor models expressing either WT or ibrutinib-resistant C481S BTK-mutant protein. This therapeutic modality (ie, targeted degradation) may also be beneficial in BTK-driven malignancies bearing non-C481 mutations arising in patients treated with noncovalent second-generation BTKi (Wang 2019).

Figure 1 NX-5948 Mechanism of Action

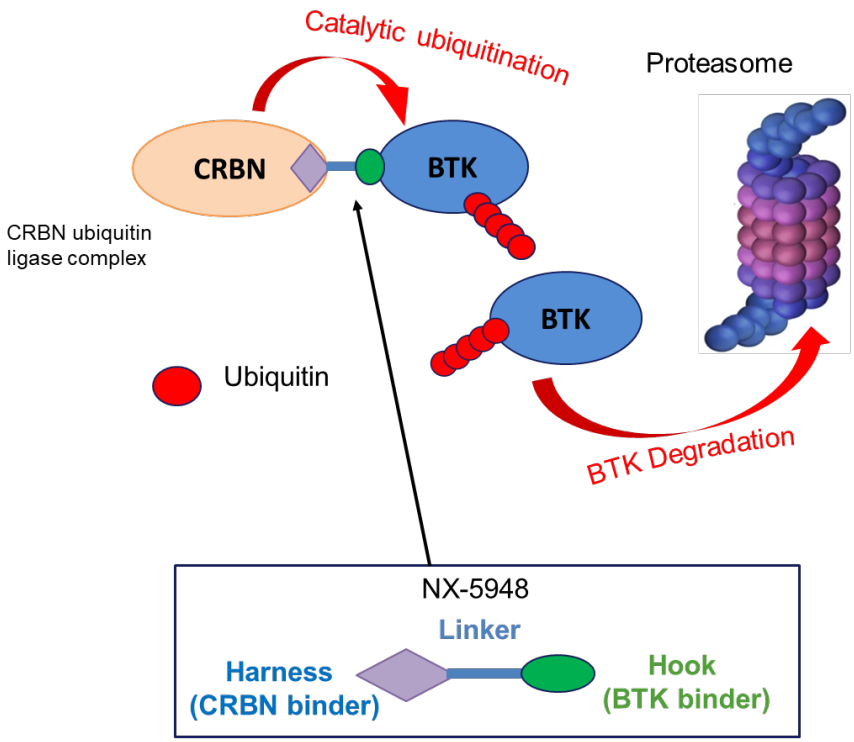


Figure 1: NX-5948 contains 3 moieties: a harness, a linker, and a hook. The harness binds to CRBN, which is an adaptor of a ubiquitin ligase complex. The hook binds to BTK and brings BTK in proximity to the ligase complex triggering its ubiquitination and subsequent proteasomal degradation.
Abbreviations: BTK, Bruton's tyrosine kinase; CRBN, cereblon.

2.3. Nonclinical Study Findings for NX-5948

In the definitive toxicity study conducted in mice, once-daily (QD) oral administration of 30, 100, or 300 mg/kg/day NX-5948 for 28 consecutive days was well-tolerated with no test article-related mortality/moribundity. There were no NX-5948-related clinical observations or effect on body weight or food consumption for either sex at any dose level evaluated. Ophthalmology examinations conducted at the end of the treatment period were unremarkable. NX-5948-related clinical pathology findings at the end of the treatment period were limited to minimal decreases in red cell mass in females at all dose levels, with no dose dependency; this effect was partially reversed in high-dose females at the end of the 28-day treatment-free recovery period. NX-5948-related post-mortem findings at the terminal necropsy were noted in the mandibular lymph node (scattered pigmented macrophages, most likely representing hemosiderin) in a single male at 100 mg/kg/day and in females at all dose levels, and in the spleen (increased extramedullary hematopoiesis in the red pulp) of high-dose females. Following the 28-day treatment-free period, microscopic findings persisted in the mandibular lymph nodes of high-dose females. Under the conditions of this study, the severely toxic dose to 10% of the animals (STD10) for NX-5948 was considered to be > 300 mg/kg/day, and the no-observed-adverse-effect level (NOAEL) for NX-5948 was 300 mg/kg/day. Steady-state (Day 28) systemic exposure (maximum concentration [C_{max}] and

area under the concentration time curve up to the last measurable concentration [$AUC_{0\text{-last}}$]) to NX-5948 at the NOAEL on Day 28 was 2900 ng/mL and 40,900 hr*ng/mL, respectively, in males and 4010 ng/mL and 59,300 hr*ng/mL, respectively, in females.

In the definitive toxicity study conducted in monkeys, QD oral administration of NX-5948 at dosages of 1, 10, 30, or 100 mg/kg/day was tolerated by both sexes with no mortality/moribundity or effects on body weight or food consumption. NX-5948-related clinical observations were limited to a dose-responsive occurrence of discolored (purple or red) focal areas of the skin in males at dosages ≥ 10 mg/kg/day and in females at all dose levels. These observations were noted throughout the treatment period, beginning as early as Week 1. The discoloration occurred in multiple areas over the body surface and was slightly more prominent in females than in males. Discoloration persisted in 2 high-dose males at the end of the 28-day treatment-free recovery period. Ophthalmologic and electrocardiographic examinations were unremarkable. NX-5948-related effects on clinical pathology parameters at the end of the treatment period were limited to minimal decreases in red cell mass with an appropriate reticulocyte response in females at 30 and 100 mg/kg/day, with no dose-dependency. At the end of the recovery period, red cell mass remained decreased in one 30 mg/kg/day female, with a robust reticulocytosis.

At the terminal necropsy, NX-5948-related macroscopic observations of discolored (purple or red) focal areas in the stomach, dark mandibular lymph nodes, and abnormal color (mostly purple) in various skin locations were noted at a higher incidence in both sexes at all NX-5948 dose levels as compared to the vehicle-treated control group. These findings correlated microscopically with submucosal hemorrhage and gastric erosion/ulceration in the stomach, sinusoidal erythrocytosis in the mandibular lymph node, and hemorrhage in the skin. Hemorrhage in the stomach was also rarely associated with submucosal vascular necrosis or thrombosis at ≥ 10 mg/kg/day. NX-5948-related pancreatic islet angiectasis/hemorrhage was noted in some males at all NX-5948 dose levels and in some females at ≥ 10 mg/kg/day. In addition, NX-5948-related increased adrenal gland weight parameters were noted in females at 100 mg/kg/day, with no macroscopic or microscopic correlates. All NX-5948-related macroscopic (except for red focus in the stomach of 1 female at 30 mg/kg/day), microscopic, and organ weight findings were fully resolved at the end of the 28-day treatment-free period. Under the conditions of this study, the highest non-severely toxic dose (HNSTD) of NX-5948 was > 100 mg/kg/day, and the NOAEL was 100 mg/kg/day. Systemic exposure (C_{max} and $AUC_{0\text{-last}}$) to NX-5948 at the NOAEL on Day 28 was 21.0 ng/mL and 243 ng*h/mL, respectively, in males, and 19.7 ng/mL and 210 ng*h/mL, respectively, in females.

Details of nonclinical study findings for NX-5948 are provided in the Investigator's Brochure.

2.4. Rationale for Starting Dose of NX-5948

The results of the 28-day definitive toxicity studies were used to calculate a reasonably safe clinical starting dose for NX-5948 in the planned first-in-human (FIH) trial. The approach used is consistent with guidance provided in the International Council for Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH) S9

guideline and involves setting the starting dose at 1/10 of the STD10 in rodents. If the nonrodent is the most appropriate species, then 1/6 of the HNSTD is considered an appropriate starting dose.

Step #1: The NOAEL for NX-5948 in mice was considered to be 300 mg/kg/day, and the STD10 was identified as being > 300 mg/kg/day, the highest dose level evaluated. In monkeys, the NOAEL for NX-5948 was considered to be 100 mg/kg/day, and the HNSTD was identified as being > 100 mg/kg/day, the highest dose level evaluated. For the purpose of starting dose calculation, the NOAEL values in both species were used in lieu of experimentally determined STD10 and HNSTD values.

- STD10 = 300 mg/kg/day.
- HNSTD = 100 mg/kg/day.

Step #2: Converting the STD10 and HNSTD dosages from units of mg/kg/day to units of mg/m^2:

- Mouse STD10 = 300 mg/kg/day × 3 (mouse k_m value) = 900 mg/m^2.
- Monkey HNSTD = 100 mg/kg/day × 12 (monkey k_m value) = 1200 mg/m^2.

Step #3: The safe starting dose in humans is 1/10 of the mouse STD10 or 1/6 of the monkey HNSTD:

- Mouse STD10 = 900 mg/m^2 ÷ 10 = 90 mg/m^2.
- Monkey HNSTD = 1200 mg/m^2 ÷ 6 = 200 mg/m^2.

Step #4: The lower of the 2 dosages (in mg/m^2) is identified as the starting dose in humans:

- 90 mg/m^2 < 200 mg/m^2.

For an average 60-kg individual with a body surface area of 1.62 m^2, this would be a dose of 145.8 mg (90 mg/m^2 × 1.62 m^2) per day.

Further consideration of the starting dose and expected efficacious exposures were evaluated based on animal efficacy models and pharmacokinetic (PK)/pharmacodynamic (PD) relationships between NX-5948 plasma exposures and BTK degradation. A PK/PD model in animals was developed, which links drug concentration to BTK degradation using a direct response turnover model. Assuming similar PK/PD in humans, the PK component of this mathematical model was scaled using allometry to project PK, associated BTK degradation, and efficacious doses in humans. It is anticipated that > 85% BTK degradation will be achieved with a dose of 100–200 mg daily. The starting dose for oncology patients was selected based on weighing both the pharmacological and safety profile of NX-5948; a dose of 50 mg is less than 1/10 the STD10 in mice based on body surface area per ICH S9 and is projected to have approximately 80–90% BTK degradation based on a PK/PD model and observed in *in vivo* pharmacology.

In vitro studies were conducted to evaluate NX-5948 as a substrate, inducer, and inhibitor of cytochrome P450 (CYP450) enzymes (1A2, 2B6, 2D6, 3A4, 2C9, and 2C19) and as a substrate and inhibitor of common drug transporters. The results indicate that NX-5948 is not an inducer or inhibitor of the CYP450 enzymes tested. NX-5948 is a substrate of CYP450 3A (CYP3A), and therefore NX-5948 plasma exposures may be altered by co-administration with other medications that induce or inhibit CYP3A. Potent or moderate inhibitors and inducers of CYP3A are excluded from the protocol.

NX-5948 has low solubility at pH ranges from 1–8, with greater solubility at pH of 1 (1.1 mg/mL) compared to pH of 5 and above (< 0.1 ug/mL). In biorelevant media, the solubility in fasted state simulated intestinal fluid, pH 6.5 (FaSSIFv-2) is less than in fed state simulated intestinal fluid, pH 5.0 (FeSSIF). Therefore, the risk of a drug-drug interaction with acid-reducing agents is anticipated and will be controlled for. As food is anticipated to increase exposure based on solubility in FeSSIF, NX-5948 will be administered in the fasted state in this study, pending an evaluation of the effect of food on NX-5948 PK. Prohibited concomitant medications are described in Section 6.8.2 and dietary restrictions in Section 6.8.4.

In summary, several non-Good Laboratory Practice (GLP) studies and 2 GLP toxicology studies indicate that NX-5948 does not have any known safety liabilities outside what is already established for marketed BTKi.

NX-5948 was found to absorb photons in the range of natural sunlight (290-700 nm), with 2 absorption maxima: (1) λ_{max} at 305 nm, with a calculated molar extinction coefficient (MEC) of 46,613 L mol^{-1} cm^{-1}, and (2) λ_{max} at 371 nm, with a MEC of 13,465 L mol^{-1} cm^{-1}. As both of these MEC values exceed the 1000 L mol^{-1} cm^{-1} value stipulated in the ICH S10 guideline, a 3T3 neural red uptake assay was conducted to further characterize the phototoxic potential of NX-5948. In this assay, NX-5948 did not demonstrate phototoxic potential.

2.5. Preliminary Clinical Study Findings for NX-5948

NX-5948-301 is an FIH trial. There are currently no completed clinical trials for this compound.

Please refer to the NX-5948 Investigator's Brochure for preliminary data from this ongoing trial. In addition, clinical data have been presented at scientific and medical meetings (Hansen 2024; Linton 2024; Searle 2023; Shah 2024).

These early results indicate the potential benefit of the treatment with NX-5948 and support further evaluation in the additional proposed cohorts as outlined in Section 4.1.4. See Section 2.6.1 for specific population rationale.

2.6. Study Rationale

NX-5948 is a CTM that harnesses normal cellular protein degradation mechanisms (ie, E3 ligase-mediated ubiquitin-proteasome pathway) to degrade BTK, both WT and mutated forms resistant to commercially available BTKi, and has been shown to inhibit tumor growth in mouse models. The therapeutic hypothesis is that BTK degradation will be effective in patients

with CLL and other B-cell malignancies where inhibition of BTK has demonstrated clinical effectiveness. In addition, BTK degradation may be effective in patients who have developed resistance to BTKi, or in patients with BTKi intolerance, and potentially in B-cell indications where treatment with BTKi has been less effective. NX-5948 also crosses the blood-brain-barrier, thus positioning it as a therapeutic agent in patients with PCNSL and other lymphomas that have CNS involvement.

The intended indication for NX-5948 is the treatment of R/R B-cell malignancies, including PCNSL. Preliminary clinical data supports the therapeutic hypothesis with NX-5948 demonstrating anti-tumor activity in multiple indications, including patients harboring BTK resistance mutations and patients with CNS disease (see Section 2.5). These and subsequent data from Phase 1a (Dose Escalation) will be used to determine which indications warrant further investigation in Phase 1b.

2.6.1. Rationale for Inclusion of Specific Sub-populations in Phase 1b

In addition to indication determined cohorts, specific sub-populations have been identified based on a combination of unmet need, preliminary anti-tumor activity, and prevalence that warrant investigation in the form of dedicated cohorts.

These include the following populations (the rationales for which follow):

CLL/SLL cohorts: CLL/SLL with prior exposure to a BTKi and a BCL-2 inhibitor (BCL-2i) harboring BTK mutations other than C481S **(Cohort 2),** BCL-2i-naïve CLL/SLL with prior exposure to a ncBTKi **(Cohort 3),** second-line (2L) BCL-2i-naïve CLL/SLL with prior exposure to a BTKi harboring TP53 and/or 17p deletions **(Cohort 4**), 2L+ BCL-2i-naïve CLL/SLL with prior exposure to a BTKi **(Cohort 5)**, BTKi-naïve CLL/SLL **(Cohort 15)**, CLL with secondary warm autoimmune hemolytic anemia (wAIHA) **(Cohort 16**), and CLL/SLL with CNS involvement **(Cohort 17)**.

NHL/WM cohorts: 2L WM with prior exposure to a BTKi **(Cohort 9)**, 2L BTKi-naïve PCNSL **(Cohort 13),** and treatment-naïve WM unfit for chemoimmunotherapy **(Cohort 14)**.

2.6.1.1. Rationale for Additional CLL Cohorts 2 to 5 and 15 to 17

As of 10 October 2024, 60 patients with R/R CLL had been treated with NX-5948, the majority of whom (49 of 60; 81.7%) had prior exposure to both a BTKi and a BCL-2i. Of the 49 patients who were efficacy-evaluable at this timepoint, 37 had demonstrated a clinical response (36 partial responses [PRs] and 1 partial response with lymphocytosis [PR-L] for an overall response rate [ORR] of 75.5%).

Among 38 efficacy-evaluable patients who had 2 disease assessments (or an event leading to discontinuation), 32 of 38 (84.2%) demonstrated a clinical response, all of which were PRs (Shah 2024).

This preliminary activity in R/R CLL, as well as available safety data (Linton 2024; Shah 2024), is in keeping with data recently submitted in support of a successful application for

FDA accelerated approval of a ncBTKi (Pirtobrutinib PI) in the same population, and presents a risk/benefit profile supportive of enrolling multiple CLL populations.

Briefly, following full enrollment of an initial Phase 1b cohort consisting of patients with prior BTKi and BCL-2i exposure (Cohort 1), and no significant new safety signals, 7 additional CLL cohorts may be opened to enrollment.

Refer to Section 4.1.4 for full cohort descriptions and design.

The goal of expanding the number of CLL patients enrolled in Phase 1b is to enrich for the enrollment of specific CLL patient sub-populations that are either ineligible to enroll or unlikely to be enrolled in sufficient numbers in Cohort 1 but whose outcomes may differ based upon specific disease characteristics and prior therapies to inform dose selection and future clinical trial design. Specifically:

Cohort 2: CLL harboring mutations in BTK other than C481S with prior BTKi and BCL-2i

Mutations at residues other than C481S (eg, T474 and L528) are enriched in patients with disease progression following treatment with ncBTKi and confer resistance to this emerging treatment modality (Blombery 2022; Jackson 2023). This is in contrast to pre-clinical studies of NX-5948, in which robust BTK degradation and cell killing activity was maintained despite the presence of these mutants (Noviski 2023). Despite their importance, these mutations remain uncommon compared to the canonical BTKi resistance mutation C481S in the R/R CLL population (Bennett 2024). While double-exposed patients with non-C481S BTK mutations can enroll in Cohort 1, their current numbers in the NX-5948 data set are below 30% (Shah 2024).

PK/PD modeling of BTK degradation predicts coverage for wild type BTK at all dose levels. Coverage for individual BTK mutations suggests a minimum dose level of 200 mg for target degradation coverage. A dose level of 600 mg is anticipated to provide the greatest likelihood of target degradation coverage for known BTK mutations including C481S.

Given the anticipated growing incidence of BTKi resistance mutations to both covalent and ncBTKi drugs, there is a compelling need to enrich for such patients in the randomized dose assessment. Adequate clinical data from this population will mitigate the risk of selecting a sub-therapeutic dose.

Cohort 3: CLL with prior ncBTKi and BCL-2i naïve

Post ncBTKi patients currently represent a small percentage of patients with relapsed/refractory CLL and are almost exclusively patients with multiply relapsed disease, their numbers will continue to increase as more patients receive ncBTKi, including in earlier lines. While ncBTKi represent an effective approach to treat R/R CLL, these compounds are not curative with disease progression frequently driven by the development of non-C481S BTK mutations.

One option available to this population following relapse is the use of BCL-2is. This class of compounds while active in CLL have treatment associated toxicities that impact their use. Furthermore, BCL-2i resistance is driven by changes such as alteration in the expression of BCL-2 and other BCL-2 family members, BCL-2 mutations, and tumor micro-environmental factors. Because these changes occur outside of the BCR signaling pathway, they contribute to a treatment resistant CLL phenotype that is less responsive to further BTK targeted therapies.

Thes findings are the basis of a commonly used treatment strategy for CLL which leverages sequential BTKi, covalent followed by covalent or ncBTKi without intercurrent exposure to a BCL-2i. By saving BCL-2i for use when the BTKi directed treatments are no longer active, this approach limits the development of resistance via alternate pathways (Huntington 2024).

Similarly, limiting the evaluation of NX-5948 in patients previously treated with ncBTKi to post-BCL-2i populations runs a significant risk of confounding interpretation of its activity.

Conversely, because NX-5948 targets the same signaling pathway as BTKi, BCL-2i naïve patients who progress following NX-5948 would be expected to retain BCL-2i as a viable treatment option in a similar fashion to patients treated with serial BTK inhibitors.

Enrolling a BCL-2i naïve population in Cohort 3 (as well as Cohorts 4 and 5 discussed below) has the potential to inform future treatment sequencing and provide key data for design of future clinical trials.

Cohort 4: 2L CLL/SLL patients with prior BTKi, and documented TP53 mutation and/or del 17p, and BCL-2i naïve

Cohort 5: 2L+ CLL/SLL patients with prior BTKi and BCL-2i naïve

With the availability of BCL-2-targeted therapies, patients whose disease progresses following upfront BTKi therapy for CLL have more highly effective options than in the past. For some patient populations, however, the utility of BCL-2i is limited either due to efficacy (Cohort 3) or tolerability (Cohort 4).

One such group is patients whose disease harbors TP53 and/or 17p deletions (Cohort 4). Despite limited impact on response to BTK-targeted therapies aberrations in p53 signaling are associated with inferior outcomes in CLL patients treated with conventional chemoimmunotherapy or BCL-2 inhibitor-targeted therapies (Kwok 2021). For this reason, CLL patients with TP53 mutations and/or del 17p cytogenetics following upfront BTKi treatment have limited options for 2L therapy and would benefit from novel treatment approaches in earlier lines. Preliminary data from NX-5948-301 demonstrates that NX-5948 has clinical activity in 3L+ CLL patients with TP53 mutations (Linton 2024). This suggests a potential role for NX-5948 in this patient population following relapse post-BTKi therapy. By evaluating a discrete homogenous population Cohort 4 will provide data to inform future clinical studies.

While highly effective in CLL, BCL-2i therapy is associated with increased on target toxicities in post-BTKi patients including prolonged cytopenias, life-threatening infections, tumor lysis syndrome, and increased risk of development of Richter's Transformation (Jones 2018). These toxicities and the required ramp up approach to dosing impact the tolerability and feasibility of dosing for patients with CLL (Venetoclax PI). Because of these risks, not all patients will be deemed eligible to receive BCL-2i treatment by their treating physician. These same patients are also unlikely to tolerate other treatment approaches with significant associated toxicities such as CAR-T therapy further limiting their treatment options.

For these reasons, in Cohorts 4 and 5, requiring prior BCL-2i treatment for patients with unmet medical need would further reduce their access to potentially effective therapy.

Furthermore, because BCL-2is target alternate signaling pathways, they would be expected to remain a viable salvage therapy for patients in this cohort if they failed to respond to or progressed following treatment with NX-5948.

By generating data in CLL patients with fewer prior lines of therapy, Cohorts 4 and 5 should provide support for more effective design of future pivotal trials. The preliminary data for NX-5948 treatment in 3L+ CLL supports this investigation (Linton, 2024; Shah 2024).

Cohort 15: BTKi-naïve CLL/SLL

While there is currently an array of effective therapies available for the treatment of CLL, treatment for the vast majority to patients is unlikely to be curative (NCCN Guidelines for CLL/SLL 2025). A key question that remains is how best to sequence available therapies to maximize duration of responses while minimizing the development of treatment resistance that may impact future therapeutic options.

This is particularly relevant for BTK-targeted therapies, which now include covalent and ncBTKi, as well as investigational BTK degraders such as NX-5948. Studies of patients who progressed following continuous treatment with BTKi have shown that progression is frequently driven by the development of BTK-resistance mutations that interfere with the binding of either covalent or ncBTKi (Chirino 2023). Because of the reliance on BTK signaling in these patients, one treatment approach for CLL utilizes BTK-targeted agents sequentially, conserving other approaches for patients whose disease is no longer dependent on BCR signaling (Huntington 2024).

BTK degraders such as NX-5948 have demonstrated activity in the presence of BTK-resistance mutations (Montoya 2024), making them an attractive option in patients who have developed resistance following treatment with covalent and ncBTKi.

A lack of dependence of tight binding to BTK raises the possibility that degraders may be able to avoid driving the development of resistance mutations in the first place. This in turn may provide a significantly longer response to therapy in patients who have not received a BTKi. In order to test this hypothesis, Cohort 15 will enroll patients without prior exposure to a BTKi, including a limited number of treatment-naïve patients.

Cohort 16: CLL with secondary wAIHA

On its own, autoimmune hemolytic anemia (AIHA) is a life-threatening disease and may be considered primary (50%) or secondary (50%) to an underlying condition (Autore 2021). Several

conditions have been associated with secondary immune-related cytopenia, including viral infections, autoimmune and connective tissues diseases, and lymphoproliferative disorders. The pathogenesis of AIHA is based on the development of autoantibodies to red blood cells with or without activation of complement. Depending on the type of immunoglobulin detected with the condition, it is designed warm if IgG-related and cold if associated with IgM. AIHA is particularly common among patients with CLL, with an incidence of approximately 10% to 15% of patients at diagnosis and up to a third over the duration of the disease (Autore 2021). Patients with CLL who also present with secondary underlying AIHA have been severely underserved in clinical trials and there are currently no approved treatment regimens for these patients.

Treatment for CLL with secondary wAIHA is the same as for other primary or secondary forms of the condition. First-line (1L) therapy of wAIHA typically consists of short-term corticosteroids, with or without rituximab, or splenectomy based on the patient's age and comorbidities. Although steroids can produce responses in approximately three-quarters of patients, recurrence occurs in 80% to 85% of patients once steroids are discontinued (Zanella 2014). Subsequent lines of immunosuppressive treatment result in responses in 40% to 60% of patients, but again relapse occurs in 70% to 80% of patients. Therefore, there is an unmet need for additional treatment options for CLL with secondary wAIHA (Montillo 2017; Rogers 2016a). Historically, many trials with BTKi have excluded patients with CLL with AIHA.

There is evidence that BTK-targeting therapies may provide benefit to these patients as observed by data from a retrospective analysis of CLL trials with ibrutinib reporting 22 patients with CLL with a history of AIHA receiving treatment for their ongoing cytopenia; 86% were able to discontinue this therapy (Rogers 2016b). In addition, the BTKi rilzabrutinib demonstrated a 64% overall hemoglobin response rate in wAIHA, with sustained responses in 41% of patients, providing further evidence of a role of BTK in this immune-related cytopenia (Cooper 2024)

These data support the evaluation of BTKi as a therapy for CLL with secondary wAIHA and are the basis for the addition of this cohort to understand the potential activity of BTK degradation in these patients.

Cohort 17: CLL/SLL with CNS involvement

CNS involvement is a rare manifestation of CLL. While leukemia cells are found in the CNS in postmortem examinations of patients with CLL at prevalence ranging from 7% to 71%, the clinical significance of these findings remains unclear. One retrospective single center study examined 172 patients, out of a total 4174 (4%) patients with CLL at that center, who underwent evaluation for neurologic symptoms. Of the 172 patients, 18 (9% of evaluated, 0.4% of total) had CNS CLL and a further 15 patients (9% evaluated, 0.3% overall) had CNS involvement of Richters Syndrome (Strati 2016)

Due to its rarity, there is no standard approach to treatment of CNS CLL. Furthermore, patients with CLL with CNS involvement are typically excluded from clinical trials, with most evidence for potential approaches limited to case studies or small case series (Kalac 2007; Pohar 1993). Therefore, both novel approaches to therapy, as well as opportunities to consistently evaluate their response to therapy, remain a significant unmet need for these patients.

NX-5948 crosses the blood-brain barrier where it is both detectable in the cerebrospinal fluid (CSF) and demonstrates clinical activity in B-cell malignancies (Hansen 2024). Furthermore, as of 10 October 2024, 5 patients with CNS involvement of their CLL have been treated with NX-5948. At

the time of data cut-off, 4 of the 5 remained on study, with the final patient having discontinued treatment due to systemic disease progression (Shah 2024). The addition of a discrete cohort of patients with CLL/SLL with CNS disease will both offer this patient population access to a novel therapy as well as allow for a rigorous assessment of NX-5948 activity in the CNS.

2.6.1.2. Rationale for WM Cohorts 9 and 14: 2L WM and treatment-naïve WM unfit for chemoimmunotherapy

Despite a frequently indolent course, WM remains an incurable disease. Frontline therapy for WM has consisted of anti-CD20 based chemoimmunotherapy (ie, bendamustine plus rituximab; dexamethasone plus rituximab and cyclophosphamide are the most commonly used), proteosome inhibitors (bortezomib, ixazomib, and carfilzomib) and BTKi (ibrutinib and zanubrutinib) (NCCN Guidelines for WM 2025).

While high ORRs of > 80% were observed with bendamustine plus rituximab and dexamethasone plus rituximab and cyclophosphamide, significant toxicities (including cytopenia, infections, and secondary malignancies) were observed with these agents (Kastritis 2015; Martin 2017; Rummel 2013). Proteosome inhibitor-based regimens have also demonstrated high ORR in WM, however these also present with their own toxicities. Bortezomib-based regimen is associated with peripheral neuropathy (Gavriatopoulou 2017) while carfilzomib, a neuropathy-sparing proteosome inhibitor, is associated with cardiac toxicity (cardiomyopathy) (Meid 2017). Ibrutinib and zanubrutinib are both approved for the treatment of WM, and also used in the frontline setting with an ORR of 94% (Tam 2020). While the overall safety profile favors zanubrutinib, the following BTK-associated toxicities were still observed: atrial fibrillation/flutter (2% vs 15.3%), hemorrhage (48.5% vs 59.2%), major hemorrhage (5.9% vs 9.2%), and hypertension (10.9% vs 17.3%). A higher rate of neutropenia was seen with zanubrutinib (29.7% vs 13.3%). Importantly, WM most frequently occurs in older patients, with a median age at the time of diagnosis of 70 years. In addition to their advanced age, this patient population frequently has comorbidities that make them particularly vulnerable to treatment-related toxicities, limiting their therapeutic options. Understanding the sequencing of treatment to maximize the durability of clinical responses is key to improving their outcomes. In addition to the treatment-limiting toxicities, the development of resistance to BTKi, which commonly occur as C481 mutations and PLCG2 missense mutations (Chen 2018), has relevant clinical implications. Although BTK is a proven and effective target in WM, novel mechanisms of action, such as BTK degradation, are needed in order to overcome these resistance mutations, potentially maximizing the benefits of BTK as a validated therapeutic target.

In the R/R setting, no standard approach to therapy currently exists for WM and current guidelines are based on prior treatments (NCCN Guidelines for WM 2025). Subsequent treatment includes a different regimen and drug class that had not been previously utilized. For instance, patients who previously received BTKi are usually treated with chemoimmunotherapy or proteosome inhibitor-based regimens. To date, there are few treatment options and limited prospective studies evaluating next-line treatment in patients who failed BTKi.

As of 10 October 2024, 13 patients with WM have been treated with NX-5948. The median number of prior lines of therapy was 3 (range, 2 to 5). All had prior chemoimmunotherapy and all had prior BTKi. Nine patients were evaluated for efficacy and the ORR was 77.8%; this was achieved early at Week 8 (first response assessment) and was associated with a significant reduction in IgM level compared to baseline (O'Connor 2024).

The high ORR in this population of heavily pretreated patients who have failed standard treatments, combined with the overall safety profile of NX-5948 (see above), provides a rationale for the evaluation of patients with R/R WM in the 2L setting following treatment with a BTKi, as well as in treatment-naïve patients who are unfit for chemoimmunotherapy. Studies evaluating novel approaches to the treatment of WM are needed to develop effective therapies for this incurable disease.

2.6.1.3. Rationale for Cohort 13: 2L, BTKi-naïve PCNSL Cohort

While PCNSL is a potentially curable disease, for patients whose disease is refractory to upfront methotrexate-based therapy or who relapse following initial response, the outcomes remain poor. No standard of care (SoC) currently exists for R/R PCNSL. BTKi have demonstrated high response rates in studies in this population demonstrating a role for targeting BCR signaling in this disease, but limited durability and significant toxicity remain major hurdles (Ferreri 2023). Significant improvements in outcomes are likely to require combination therapies incorporating agents that are both active and well tolerated.

BTKi are among the classes of small molecules that have been investigated as a potential therapy in PCSNL. Covalent BTKi have been demonstrated to have single-agent activity in the relapsed/refractory setting leading to the inclusion of ibrutinib in the NCCN guidelines for the treatment of PCNSL and to the approval of tirabrutinib in Japan (Narita 2021).

Despite promising activity, responses to single agent BTKi are often short lived and their inclusion in combination therapy can be associated with significant toxicities (Grommes 2019). How best to utilize BTK targeted therapies remains an ongoing question and area of unmet medical need.

Preliminary data from Phase 1a has shown that NX-5948 presents a tolerable safety profile with no CNS patient-specific safety signals (see Section 2.5). Furthermore, NX-5948 is detectable in the CSF of patients with CNSL and has anti-tumor activity in patients who have been exposed to multiple salvage therapies, including BTKi (Hansen 2024). Understanding the nature and degree of NX-5948 activity in these patients has been confounded due to their prior exposure to BTKi which in turn increases their likelihood of having developed resistance agents targeting the BCR signaling pathway.

Understanding the safety and activity of NX-5948 in patients with CNSL immediately following upfront therapy, and in the absence of prior BTKi exposure provide key data to inform future studies of the potential for BTK degraders in this group of patients for whom novel and effective therapies remain a significant unmet medical need.

2.7. Summary of Overall Risks and Benefits

Nonclinical data available for NX-5948 support possible toxicities, including reversible subcutaneous hemorrhage, mucosal effects, and decreased red cell mass (anemia/cytopenia). The toxicity profile of NX-5948 in non-human primates, at doses up to 100 mg/kg for 28 consecutive days, is generally consistent with BTK inhibition.

Toxicities associated with BTKi include an increased risk of bleeding, particularly for patients receiving concurrent anticoagulation. Additional adverse events (AEs) associated with use of BTKi include: serious and opportunistic infections, cytopenias, cardiovascular complications such as cardiac arrhythmias and hypertension, and tumor lysis syndrome (TLS). Refer to Section 6.6 for further detail on possible risks for NX-5948 and those associated with BTKi.

The NX-5948-301 protocol contains entry criteria, dose-limiting toxicity (DLT) definitions, and Safety Review Committee (SRC) oversight to minimize risk to study participants. Patients will be monitored for signs and symptoms of toxicities throughout the study.

Phase 1a evaluation of potential DLTs and decisions on dose escalation will be determined by the SRC. The SRC will be comprised of representatives of the Sponsor, the Sponsor's Medical Monitor, the Contract Research Organization's (CRO's) Medical Monitor (if applicable), and at a minimum, a quorum of Investigators with patients under review. One or more relevant external medical experts may be added at the committee's discretion. During Phase 1b, an Independent Data Monitoring Committee (IDMC) will provide oversight.

Given that the main expected adverse effects of NX-5948 are anticipated to be similar to BTKi, NX-5948 has an acceptable risk/benefit profile for use in patients with R/R malignant disease.

3. STUDY OBJECTIVES AND ENDPOINTS

Objectives and endpoints for Phase 1a, Phase 1b Part 1, and Phase 1b Part 2 are summarized in Table 1, Table 2, and Table 3 respectively.

Table 1 Phase 1a (Dose Escalation) – Objectives and Endpoints

Objectives	Endpoints
Primary • To evaluate the safety and tolerability of BTK degrader NX-5948, when taken orally, in adult patients with R/R B-cell malignancies. • To establish the maximum tolerated dose (MTD) and/or recommended Phase 1b dose(s) of NX-5948 in adult patients with R/R B-cell malignancies.	• Incidence of treatment-emergent adverse events (TEAEs); Grade 3, 4, 5 TEAEs; serious adverse events (SAEs); TEAEs leading to study drug discontinuation; deaths due to TEAEs, and all deaths. • Changes from baseline in safety parameters. • Incidence of DLTs.

Objectives	Endpoints
Secondary • To characterize the PK profile of NX-5948. • To characterize the PD profile of NX-5948. • To assess preliminary anti-tumor activity of NX-5948.	• NX-5948 PK parameters in plasma (C_{max}, time at which C_{max} is observed [T_{max}], half-life, $AUC_{0\text{-last}}$, area under the concentration time curve to the end of a dosing period [$AUC_{0\text{-tau}}$], minimum concentration [C_{min}], accumulation ratio). • Changes from baseline of BTK levels in B cells. • ORR. • Complete response (CR) rate. o For patients with CLL, CR/complete response with incomplete recovery (CRi). • Time to first response. • Duration of response (DOR). • Progression-free survival (PFS). • Time to next therapy.

Exploratory
• To assess OS.
• To explore mechanisms of response/resistance.
• To characterize any PK/PD relationships with anti-tumor activity, biomarkers, and/or safety.
• To evaluate NX-5948 metabolism.
• To assess the impact of NX-5948 on cognitive function.
	• OS.
	• Biomarkers of the BTK signaling and resistance pathways and target engagement, eg, including and not limited to:
	  o Evaluation of immune cell phenotypes and BCR signaling in peripheral blood mononuclear cells (PBMCs).
	  o Circulating cytokine and chemokine levels.
	  o Next generation sequencing (NGS): retrospective evaluation of gene expression, mutation status, and allele frequency.
	• Identification of potential NX-5948 metabolites.
	• Mini Mental State Exam, 2nd Edition: Brief Version (MMSE-2: BV).

Table 2 Phase 1b Part 1 (Safety Expansion) – Objectives and Endpoints

Objectives	Endpoints
Primary • To evaluate the anti-tumor activity of NX-5948 in least 2 dose levels in patients with CLL/SLL selected for Phase 1b safety expansion. • To evaluate the anti-tumor activity of NX-5948 in patients with NHLs (MCL, MZL, DLBCL/HGBL, FL, PCNSL/SCNSL) or WM based on the dose levels selected for Phase 1b safety expansion. • To further characterize and compare the safety and tolerability of at least 2 dose levels of NX-5948 in patients with CLL/SLL based on the dose(s) selected in Phase 1a for Phase 1b safety expansion. • To further characterize the safety and tolerability of NX-5948 in patients with NHLs (MCL, MZL, DLBCL/HGBL, FL, PCNSL/SCNSL) or WM based on the dose levels selected in Phase 1a for Phase 1b safety expansion. • To identify the optimized dose for patients with CLL/SLL based on safety, tolerability, anti-tumor activity, and PK/PD profile for further evaluation of dose expansion cohorts.	• ORR as assessed by Investigator. • Incidence of TEAEs; Grade 3, 4, 5 TEAEs; SAEs; TEAEs leading to discontinuation; deaths due to TEAEs; all deaths; and changes in safety parameters.
Secondary • To further characterize the PK profile of at least 2 dose levels of NX-5948. • To further characterize the PD profile of at least 2 dose levels of NX-5948. • To further assess anti-tumor activity of NX-5948 in at least 2 dose levels selected for Phase 1b safety expansion.	• NX-5948 PK parameters (C_{max}, T_{max}, half-life, $AUC_{0\text{-last}}$, $AUC_{0\text{-tau}}$, C_{min}, accumulation ratio, CSF:plasma concentration ratio) in plasma and CNS (for patients with CNS disease only). • Changes from baseline of BTK levels in B cells. • CR rate. o For patients with CLL, CR/CRi. • Time to first response.

Objectives	Endpoints
	• DOR. • PFS. • Time to next therapy.

Exploratory
 	
• To assess OS.
• To evaluate NX-5948 metabolism.
• To explore mechanisms of response/resistance.
• To characterize any PK/PD relationships with anti-tumor activity, biomarkers, and/or safety.
• To assess the impact of NX-5948 on cognitive function.
• To further assess efficacy of NX-5948 in CLL/SLL subsets defined by mutations associated with clinical risk and/ or treatment resistance
• To assess the efficacy of NX-5948 in CLL with secondary wAIHA in Cohort 16
	• OS.
	• Biomarkers of the BTK signaling and resistance pathways and target engagement, eg, including and not limited to:
	  o Evaluation of immune cell phenotypes and BCR signaling in PBMCs.
	  o Circulating cytokine and chemokine levels.
	  o NGS: retrospective evaluation of gene expression, mutation status, and allele frequency.
	  o Minimal residual disease (MRD): evaluation of disease burden to understand depth and durability of response by NX-5948.
	  o Explore patient-reported outcome (PRO) measure for overall impact of treatment toxicity (using Functional Assessment of Chronic Illness Therapy [FACIT-GP5] single item) for each cohort in Phase 1b Part 1.
	• Identification of potential NX-5948 metabolites.
	• MMSE-2: BV.
	• ORR and DOR for CLL/SLL mutation subsets including but not limited to BTK, PLCG2, TP53
	• In patients with CLL with secondary wAIHA:
	  o Percentage of patients with AIHA complete response (CR^{AIHA}), AIHA partial response (PR^{AIHA}), and overall AIHA response as assessed by Investigator

Objectives	Endpoints
	○ Probability of maintained response ○ DOR ○ Time to response ○ Relapse-free survival: Time from start of treatment until death or relapse of AIHA ○ Frequency of packed red blood cell (PRBC) transfusion while receiving NX-5948

Table 3 Phase 1b Part 2 (Cohort Expansion) – Objectives and Endpoints

Objectives	Endpoints
Primary • To further evaluate the efficacy of NX-5948 in patients with CLL/SLL at the optimized dose identified in Phase 1b Part 1.	• ORR as assessed by Investigator.
Secondary • To further assess efficacy of NX-5948 at the optimized dose identified in Phase 1b Part 1. • To further characterize the safety and tolerability of NX-5948 in patients with CLL. • To further characterize the PK profile of NX-5948 at the optimized dose identified in Phase 1b Part 1. • To further characterize the PD profile of NX-5948 at the optimized dose identified in Phase 1b Part 1.	• Rate of CR/CRi. • Time to first response. • DOR. • PFS. • Time to next therapy. • Incidence of TEAEs; Grade 3, 4, 5 TEAEs; SAEs; TEAEs leading to discontinuation; deaths due to TEAEs; all deaths; and changes in safety parameters. • NX-5948 PK parameters (C_{max}, T_{max}, half-life, $AUC_{0\text{-last}}$, $AUC_{0\text{-tau}}$, C_{min}, accumulation ratio, CSF:plasma concentration ratio) in plasma and CNS (for patients with CNS disease only). • Changes from baseline of BTK levels in B cells.

Objectives	Endpoints

Exploratory
• To assess OS.
• To evaluate NX-5948 metabolism.
• To explore mechanisms of response/resistance.
• To characterize any PK/PD relationships with anti-tumor activity, biomarkers, and/or safety.
• To assess the impact of NX-5948 on cognitive function.
• To further assess efficacy of NX-5948 in CLL/SLL subsets defined by mutations associated with clinical risk and/ or treatment resistance.
	• OS.
	• Biomarkers of the BTK signaling and resistance pathways and target engagement, eg, including and not limited to:
	  o Evaluation of immune cell phenotypes and BCR signaling in PBMCs.
	  o Circulating cytokine and chemokine levels.
	  o NGS: retrospective evaluation of gene expression, mutation status, and allele frequency.
	  o MRD: evaluation of disease burden to understand depth and durability of response by NX-5948.
	  o Explore PRO measure for overall impact of treatment toxicity (using FACIT-GP5 single item) for each cohort in Phase 1b Part 2.
	• Identification of potential NX-5948 metabolites
	• MMSE-2: BV.
	• ORR and DOR for CLL/SLL mutation subsets including but not limited to BTK, PLCG2, TP53.

4. INVESTIGATIONAL PLAN

4.1. Overall Study Design

This is an FIH, open-label, multicenter, Phase 1a/1b study evaluating the safety, tolerability, and anti-tumor activity of NX-5948 in adult patients with R/R B-cell malignancies. As shown in Figure 2, the study will be divided into: Phase 1a (Dose Escalation) followed by Phase 1b Part 1 (Safety Expansion) and Phase 1b Part 2 (Cohort Expansion).

Phase 1a (Dose Escalation) will evaluate the safety and tolerability of NX-5948 in patients with these indications:

- R/R CLL, SLL

- DLBCL with the following subgroups
  - DLBCL, not otherwise specified (NOS), GCB, ABC (includes transformed indolent lymphoma [eg, Grade 3b/transformed FL] and Richter-transformed DLBCL)
  - HGBL with *MYC* and *BCL-2* and/or *BCL-6* rearrangements, HGBL, NOS

(note: other subgroups of DLBCL or large B-cell lymphoma (LBCL) based on the 2016 WHO classification of lymphoid malignancies [Swerdlow 2016] are excluded)

- FL (Grade 1–3a; eligibility for systemic treatment as determined by the Groupe d'Etude des Lymphomes Folliculaires [GELF] criteria; Section 15.8)
- MCL
- MZL (eligible subtypes include EMZL, MALT, NMZL, and SMZL)
- WM
- PCNSL

All patients in Phase 1a must have received at least 2 prior lines of therapy (except for patients with PCSNL, who are required to have received at least 1 prior line of therapy) and secondary CNS involvement in any disease indications listed is permitted. Dose escalation will occur independently between CLL/SLL and the NHL (DLBCL, FL, MCL, MZL, PCNSL) or WM cohorts because of differences in disease biology between these B-cell malignancies (eg, circulating cells and bone marrow involvement may put patients with CLL at higher risk of TLS/intravascular cell death, differences in prevalence of disease-related cytopenias in CLL relative to NHL patients with aggressive or indolent disease, etc.). As such, the safety profile of drugs developed to treat these diseases may differ as well as the regimen (dose and schedule).

Phase 1b Part 1 (Safety Expansion) will further investigate the safety, tolerability, and anti-tumor activity of NX-5948 in patients with B-cell malignancies at the recommended dose(s) selected in Phase 1a.

In total, up to 360 patients may be enrolled in Phase 1b Part 1. Patients will be enrolled into up to 17 safety expansion cohorts based on disease type, prior treatments, and/or molecular features, and dosing schedules. The specific requirements for each cohort are as described in Section 5.1.

The Phase 1b Part 1 dose for each indication, as determined by the SRC, will be no greater than the MTD established in Phase 1a, or the Phase 1a maximum administered dose (MAD) if the MTD is not reached.

In Cohort 1, at each dose level, a Simon 2-stage approach will be used (Simon 1989). Once lack of futility is established in Phase 1b Part 1 and available safety data has been reviewed by sponsor and the IDMC, additional patients may be enrolled into Phase 1b Part 2. The dose(s) that will be considered for Phase 1b Part 2 initiation will be based on the totality of the data and the Sponsor's discretion. The design characteristics of Simon 2-stage are documented in the Section 4.2.

Dose optimization of at least 2 dose levels selected in Phase 1a (Dose Escalation) will be evaluated in patients with CLL/SLL, with 1 dose that does not exceed the MTD (if reached) or MAD, and 1 or more lower doses at which preliminary clinical activity was observed. Based on evolving data, additional intermediate dose levels may be explored in CLL. If intermediate dose levels open after those currently planned, the randomization schedule will be adjusted accordingly. The selection of dose levels for dose optimization will be determined by the SRC, and will be based on the totality of the available data, which may include PK, PD, safety, tolerability, anti-tumor activity, and exposure-response analysis in keeping with FDA guidance *"Optimizing the Dosage of Human Prescription Drugs and Biological Products for the Treatment of Oncologic Diseases"* (FDA 2023). Data from other CLL cohorts (Cohorts 2 to 5, and 15 to 17) may be included in the identification of the optimized dose.

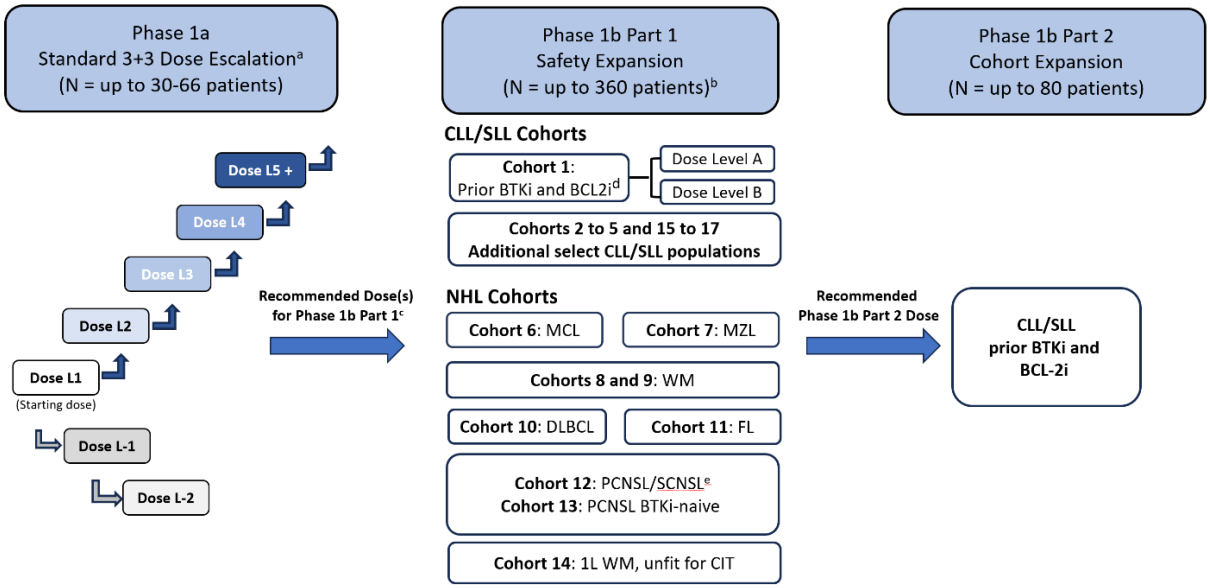
The totality of clinical data observed in the safety expansion cohorts for NHLs will be considered for any future dose optimization and/or further dose expansion in a particular indication(s). Dose level(s) used in each disease type will be evaluated in combination with safety, tolerability, PK/PD, and anti-tumor activity in order to select an optimized dose (or evaluate an alternative dose) for further development.

Phase 1b Part 2 (Cohort Expansion) will further investigate the anti-tumor activity of NX-5948 in patients with R/R CLL/SLL at the optimized dose(s) identified in Phase 1b Part 1. An additional 80 patients with CLL/SLL (post-BTKi and post-BCL-2 inhibitor) may be enrolled. ORR will be utilized to further assess the anti-tumor activity of NX-5948 in this population, in addition to its safety, tolerability, and PK/PD properties.

Patients treated during Phase 1b Part 1 who also meet the criteria for 1 of the Phase 1b Part 2 cohorts may be considered as part of the evaluable patients for that cohort. The Sponsor will determine the correct cohort assignment for the patient based on the diagnosis and prior treatment history information provided during eligibility assessment.

For more details on Phase 1b, see Section 4.1.4.

Figure 2 Study Schematic



a. In each of the parallel dose escalations for CLL/SLL and NHL/WM indications in Phase 1a.
b. Up to 20 patients per disease indication except for CLL/SLL Cohort 1, which will be up to 20 patients per dose level. For CNSL (Cohort 12), up to maximum of 10 patients with SCNSL.
c. Recommended doses may be different for CLL/SLL and NHLs. Recommended doses could be MTD as established in Phase 1a. If MTD is not reached in Phase 1a, MAD, as determined by the SRC, may be used. For CLL/SLL, 2 or more dose levels, none of which are to exceed MTD/MAD, will be evaluated.
d. Patients enrolled in Cohort 1 will be randomized to 1 of the dose levels.
e. SCNSL meeting criteria for any indication described (with the exception of CLL/SLL) with secondary CNS involvement of their disease.

Abbreviations: 1L, first-line; BCL-2i, B-cell lymphoma-2 inhibitor; BTKi, Bruton's tyrosine kinase inhibitor; CIT, chemoimmunotherapy; CLL, chronic lymphocytic leukemia; CNSL, central nervous system lymphoma; DLBCL, diffuse large B-cell lymphoma; FL, follicular lymphoma; L, level; MAD, maximum administered dose; MCL, mantle cell lymphoma; MTD, maximum tolerated dose; MZL, marginal zone lymphoma; NHL, non-Hodgkin lymphoma; PCNSL, primary central nervous system lymphoma; SLL, small lymphocytic lymphoma; SCNSL, secondary central nervous system lymphoma; SRC, Safety Review Committee; WM, Waldenström macroglobulinemia.

A summary of events and assessments are provided by visit in the Schedule of Assessments (Section 15.3; Table 16). Only assessments required for the study are listed in the tables; any additional assessments or repeat assessments should be performed as indicated and according to clinical practice.

All AEs will be monitored and recorded until 30 days after the last dose of study treatment or until initiation of another anti-cancer therapy, whichever occurs first. In addition, patients with unresolved AEs or abnormal laboratory values may be contacted by site personnel by telephone for follow-up of these events. Any SAE assessed as related to study drug that is ongoing after 30 days from the last dose of study drug will be monitored and followed until stability, resolution, death, or study withdrawal, whichever comes first. AEs will be graded according to National Cancer Institute's (NCI) Common Terminology Criteria for Adverse

Events (CTCAE) v5.0. Additional details on AE reporting are provided in Section 8, including details on the AE reporting period in Section 8.3.

4.1.1. Discussion of Study Design

Dose escalation in patients with R/R B-cell malignancies is a standard method in early phase clinical development of novel anti-cancer treatments. Dose optimization and safety expansion will allow a more thorough investigation of the safety, PK/PD, and biological effects of study drug NX-5948.

4.1.2. Selection of Study Population

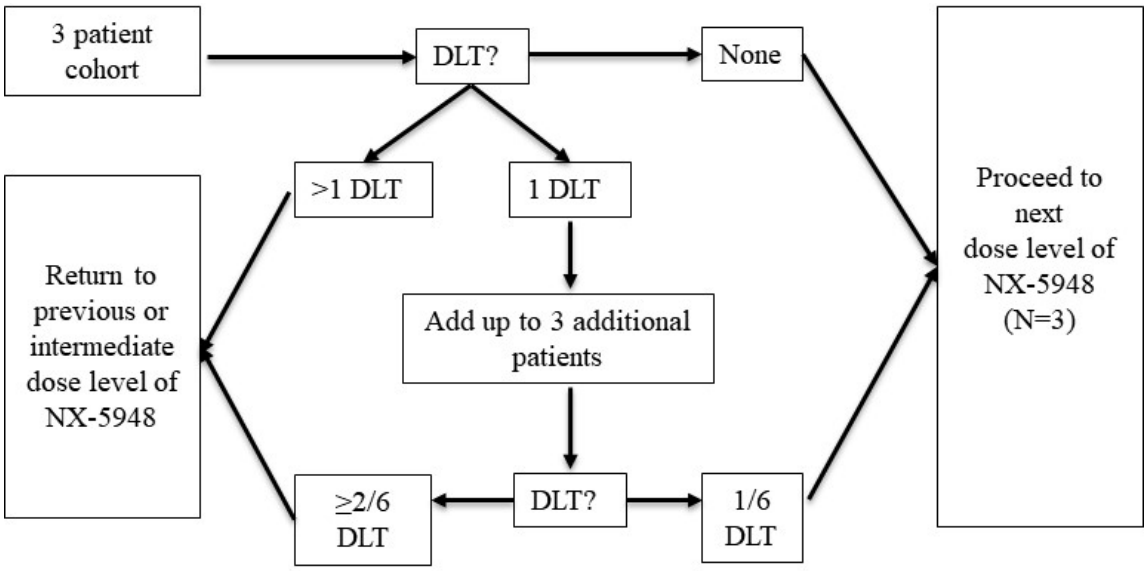
Adult patients with R/R B-cell malignancies who have received appropriate prior systemic therapies will be selected to participate in this study (exceptions: Cohort 14, which will enroll patients with treatment-naïve WM; and Cohort 15, which will enroll a limited number of patients with treatment-naïve CLL/SLL).

Additional criteria for participation in the study are provided in Section 5.1.1, and Section 5.1.2. For rationale for inclusion of specific populations see Section 2.6.1.

4.1.3. Phase 1a

Phase 1a of the study will utilize a standard 3+3 dose escalation design of 3 patients per dose level (as shown in Figure 2 and Figure 3) in the absence of DLTs as shown in Table 5. The study will commence with accrual to Dose Level (DL) 1 (50 mg/day). Patients at DL 1 will be treated for a minimum of 28 days (1 cycle of treatment) before enrolling any additional patients and/or subsequent dose levels. AEs will be assessed per the NCI CTCAE v5.0. The scheduled assessments at each clinic visit are shown in Section 15.3.

Figure 3 Decision Schema for Phase 1 Dose Escalation (3+3 Design)



Abbreviations: DLT, dose-limiting toxicity.

Patients in Phase 1a will attend the following scheduled clinic visits:

- Screening visit (within 28 days prior to starting study drug).
- Treatment with NX-5948 will be taken orally in 28-day treatment cycles. On-treatment visits will be conducted in Cycle 1 on:
  - Days 1, 2, 8, 15, and 22.
  - DLT assessment period is the first 28 days of treatment with NX-5948 (Cycle 1 Day 1 up to and including Cycle 1 Day 28).
- Cycle 2 on-treatment visits will be conducted on:
  - Days 1, 2, 8, 15, and 22.
- Cycle 3 and 4 on-treatment visits will be conducted on:
  - Days 1 and 15.
- Subsequent cycles (Cycle 5 and beyond) have on-treatment visits conducted on:
  - Day 1.
- In the event of a treatment interruption, planned visits should still occur with assessments conducted in alignment with the Schedule of Assessments (Section 15.3). There will not be a pause in the study calendar in the event of a treatment interruption.
- An End of Treatment (EOT) visit up to 3 days following the last dose of study drug. If EOT cannot occur within 3 days of last dose of study drug due to AE or treatment interruption, alternative scheduling for EOT and all subsequent follow-up visits may be discussed with the Sponsor.
- A 15-day Post-Treatment Safety Follow-Up visit will be performed 15 (± 5) days after last dose of study drug or prior to initiation of another anti-cancer therapy, whichever occurs first. If another anti-cancer therapy has already been started, the 15-day visit is not required.
- A 30-day Post-Treatment Safety Follow-Up visit will be performed 30 (± 7) days after last dose of study drug or prior to initiation of another anti-cancer therapy, whichever occurs first. If another anti-cancer therapy has already been started, the 30-day visit is not required.
- Long-term Follow-up contact/visit: patients will be followed every 12 weeks after the 30-day Post-Treatment Safety Follow-Up visit until death, withdrawal of consent, or loss to follow-up. Long-term Follow-up can be via clinic visit, phone call to the patient or referring physician, or

other method deemed appropriate by the site, and should assess survival, disease progression, and subsequent therapy.

4.1.3.1. NX-5948 Dose Escalation Decision Schema

The starting dose for DL 1 is 50 mg/day. Once the patients at DL 1 have completed 28 days (Cycle 1), in the absence of DLTs and upon agreement of the SRC, escalation may proceed to DL 2. Subsequent dose level cohorts following completion of DL 1 are planned as shown in Table 4.

Table 4 NX-5948 Planned Dose Levels in Phase 1a Escalation

Dose Level	NX-5948 Dose (mg) Taken Daily in a 28-day Cycle
1 (Starting Dose)	50
2	100
3	200
4	300
5	450
6	600
7	800
8	1000
Dose Level	**NX-5948 Dose (mg)**
-1	25 (Daily in a 28-day cycle)
-2	25 (Every-other-day in a 28-day cycle)

Assuming no (0) DLTs at any given dose level (see Section 4.1.3.2), accrual may advance to the next dose level. A 3-patient cohort with 1 DLT during Cycle 1 will be expanded to 6 patients.

If ≥ 33% of patients (ie, ≥ 2) have a DLT in any dose level cohort during Cycle 1, then accrual at that dose will stop and a lower dose cohort will be expanded up to 6 patients. This lower dose level cohort may be at the previous lower dose (if 6 patients have not already been accrued at that lower dose) as shown in Table 5 or at an intermediate dose level subject to SRC approval. If ≥ 33% of patients have a DLT in DL 1, a lower dose level and/or alternative dosing design may be explored following SRC approval.

Table 5 Decision Schema for NX-5948 Phase 1a Dose Escalation

If DLT was observed in:	Then:
0 in 3 patients	Proceed with enrollment in next dose level cohort.
1 of 3 patients	Enroll up to 3 additional patients in current dose level cohort.
1 of 6 patients	Proceed with enrollment in next dose level cohort subject to SRC approval.

≥ 2 of 3–6 patients	Do not escalate; enroll up to 3 additional patients in a lower (previous or intermediate) dose level cohort, subject to SRC approval.

Abbreviations: DLT, dose-limiting toxicity; SRC, Safety Review Committee.

To accommodate logistics of enrollment at multiple centers, over-enrollment of a 3-patient dose level cohort by an additional patient may be allowed provided no DLT has been observed in that dose level cohort. If a DLT occurs, a total of 6 evaluable patients will be enrolled. All patients will be considered for DLT assessments.

During dose escalation, up to a maximum of 12 backfill patients per previously cleared dose levels may be enrolled in order to further explore safety and tolerability, with approval of the Sponsor. Any AEs that may occur in these patients during Cycle 1 will be considered as DLT evaluable, and the SRC will consider these AEs when determining the recommended Phase 1b dose(s).

The following criteria apply in order to be evaluable for dose level cohort DLT/MTD decisions:

- If a patient in Phase 1a misses more than 25% of doses for Cycle 1 (eg, > 7 doses of NX-5948 in 28 days) during the DLT period for reasons other than an AE, additional patient(s) may be enrolled to provide adequate data for dose escalation decision making.
- Patients withdrawn from dosing during the DLT period in any dose level cohort for reasons unrelated to NX-5948 safety may be replaced in order to have 3 or 6 patients evaluable for dose escalation (dependent on dose level cohort size) as described in Table 5.

4.1.3.2. DLT

DLTs are defined as TEAEs or laboratory abnormalities occurring during the first cycle (28 days) of study drug administration during Phase 1a Dose Escalation that meet 1 of the NCI CTCAE v5.0 criteria per Table 6 unless clearly and incontrovertibly due to the underlying disease or an extraneous cause and assessed as not related to study drug by the Investigator. Toxicities occurring in treatment Cycles 2 or later (Phase 1a) or in Phase 1b will be reviewed and their impact on dose escalation and dosing frequency assessed.

Table 6 DLT Definition

Type of AE	DLT definition (NCI CTCAE V5.0)
Any	• Grade 5.
Hematological	• Grade 4 neutropenia lasting > 5 days (if grade worsened from baseline). • Grade ≥ 3 febrile neutropenia. • Grade ≥ 3 thrombocytopenia with Grade ≥ 2 hemorrhage.

Type of AE	DLT definition (NCI CTCAE V5.0)
	• Grade ≥ 4 thrombocytopenia lasting ≥ 7 days or any requirement for platelet transfusion.
Non-hematological	Any Grade 3 or 4 non-hematological toxicity that occurs despite adequate medical intervention, **except for the following**: • Grade 3 vomiting, and/or diarrhea lasting < 48 hours. • Grade 3 fatigue that improves to Grade ≤ 2 within 5 days. • Grade 3 electrolyte imbalances lasting < 72 hours. • Grade 3 elevations in AST/ALT with duration < 72 hours. • Any isolated transient laboratory value without clinical sequalae.
Other	• Omission of > 25% of planned doses of NX-5948 due to an AE. • Any other DLT as decided by SRC.

Abbreviations: AE, adverse event; ALT, alanine aminotransferase; AST, aspartate aminotransferase; CTCAE, Common Terminology Criteria for Adverse Events; DLT, dose-limiting toxicity; NCI, National Cancer Institute; SRC, Safety Review Committee.

4.1.3.3. Intrapatient Dose Escalation

After completing 2 cycles of study drug treatment, intrapatient dose escalation may occur at the discretion of the Principal Investigator (PI) and in consultation with the Medical Monitor, once a higher dose level has been cleared and reviewed by the SRC.

Any related AEs from prior NX-5948 administration must have improved to Grade ≤ 1 or baseline before intrapatient dose escalating. Intrapatient dose escalation may only occur up to a maximum of 2 times, always after completing 2 cycles of study drug treatment, up to the highest dose that has been cleared at the discretion of the PI in consultation with the Medical Monitor. The scheduled assessments at each clinic visit are shown in Section 15.3.

4.1.3.4. MTD

MTD is defined as the highest dose level with < 33% DLTs during Phase 1a of the study. This will be determined by the SRC after review of all available data. The recommended Phase 1b dose(s) will be determined using all available PK, PD, safety, anti-tumor activity, and tolerability data. The recommended Phase 1b dose(s) will not be higher than the MTD.

Indication -specific MTDs/recommended Phase 1b dose(s) may be observed during dose escalation. Once an MTD/recommended Phase 1b dose(s) has been identified for 1 indication, the Sponsor may continue to concurrently enroll patients into the Phase 1a portion of the study for other indications.

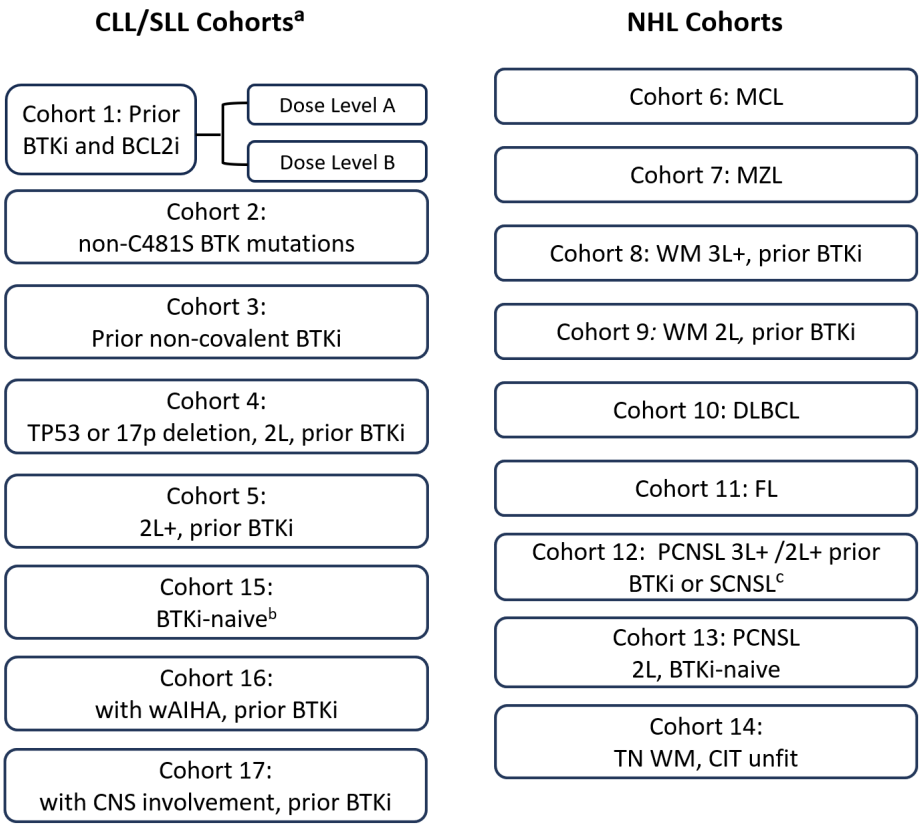
4.1.4. Phase 1b

4.1.4.1. Phase 1b Part 1 (Safety Expansion)

Once a recommended dose(s) for further study has been established in Phase 1a, up to 17 safety expansion cohorts based on disease type, prior treatments, and/or molecular features,

and dosing schedules (described in Section 5.1.1) will be recruited to further assess safety, PK, PD, and preliminary anti-tumor activity of study drug (Figure 4).

Figure 4 CLL/SLL and NHL Cohorts in Phase 1b Part 1 (Safety Expansion)



a. Patients in CLL/SLL Cohort 1 and Cohort 2 should have prior exposure to both a BTKi and BCL-2i (or be deemed ineligible for a BCL-2i). Cohorts 3-5 should be BCL-2i naïve.
b. A limited number of patients with treatment-naïve CLL/SLL may enroll in Cohort 15.
c. SCNSL meeting criteria for any indication described (with the exception of CLL) with secondary CNS involvement of their disease.

Abbreviations: BCL-2i, B-cell lymphoma-2 inhibitor; BTK, Bruton's tyrosine kinase; BTKi, Bruton's tyrosine kinase inhibitor; CIT, chemoimmunotherapy; CLL, chronic lymphocytic leukemia; CNS, central nervous system; DLBCL, diffuse large B-cell lymphoma; FL, follicular lymphoma; MCL, mantle cell lymphoma; MZL, marginal zone lymphoma; NHL, non-Hodgkin lymphoma; PCNSL, primary central nervous system lymphoma; SCNSL, secondary central nervous system lymphoma; SLL, small lymphocytic lymphoma; TN, treatment-naïve; TP53, tumor protein p53; wAIHA, warm autoimmune hemolytic anemia; WM, Waldenström macroglobulinemia.

Cohorts 1 to 5 and 15 to 17: CLL/SLL – 8 cohorts of CLL/SLL patients are planned (Figure 4) consisting of the following populations (see Section 2.6.1 for population selection rationale and Section 5.1.1 for additional eligibility information):

Cohort 1: Prior exposure to both BTKi and BCL-2i (unless deemed ineligible for BCL-2i by Investigator at time of study enrollment)

Cohort 2: Prior exposure to both BTKi and BCL-2i (unless deemed ineligible for BCL-2i by Investigator at time of study enrollment) and documented BTK mutation other than C481S (eg, T474 or L528) within 6 months prior to study entry

Cohort 3: Prior exposure to ncBTKi and BCL-2i naïve

Cohort 4: Patients with documented TP53 mutation or 17p deletion and 1 prior line of therapy that included a BTKi and BCL-2i naïve

Cohort 5: Patients with at least 1 prior line of therapy that included a BTKi and BCL-2i naïve

Cohort 15: Patients with no prior exposure to a BTKi. A limited number of patients with treatment-naïve CLL/SLL may enroll in this cohort.

Cohort 16: Patients with at least 1 prior of line of therapy that included a BTKi **and** active, uncontrolled secondary wAIHA following 1L+ therapy for AIHA

Cohort 17: Patients with CLL/SLL with CNS involvement, prior exposure to BTKi

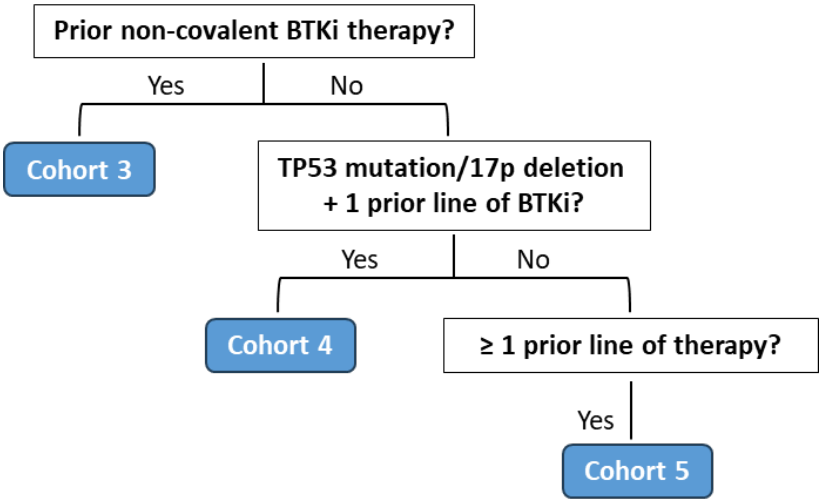
At least 2 dose levels of NX-5948 will be tested in Cohort 1 to optimize dosing for future study, with up to 20 patients enrolled per dose level. Additional intermediate dose levels may be explored based on evolving data and recommendation from the IDMC. If intermediate dose levels open after those currently planned, the randomization schedule will be adjusted accordingly.

Enrollment of Cohorts 2 to 5 and 15 to 17 may start after Cohort 1 has been filled, with each enrolling up to 20 patients per cohort. These additional safety expansion cohorts are designed to assess the safety, tolerability, and activity of NX-5948 in clinically meaningful patient subsets. Cumulative data from Phase 1a and available data from Phase 1b Part 1 Cohort 1 will be reviewed prior to initiating subsequent safety expansion cohorts. Cohorts 2 to 5 and 15 to 17 may be opened or filled based on emerging data.

Patients in Cohorts 2 to 5 and 15 to 17 may be allocated to dose levels tested in Cohort 1 using a non-blinded randomization (see Section 11.2.5), provided that the dose level(s) does not have any safety or activity signal that precludes further enrollment. Randomization will be implemented as warranted following review of the available results from Phase 1a and Phase 1b Part 1 Cohort 1 at the time the cohorts are activated.

For patients who meet multiple cohort criteria, refer to Figure 5 for decision-making schema for cohort assignment.

Figure 5 Phase 1b Part 1 CLL/SLL Cohorts 3 to 5 (Prior BTKi, BCL-2i-naïve) Assignment Decision Schema[a]



a. All patients are required to be BCL-2i naïve. Patient allocation may also be based on cohort availability.

Abbreviations: BCL-2i, B-cell lymphoma-2 inhibitor, BTKi, Bruton's tyrosine kinase inhibitor; TP53, tumor protein p53.

Cohorts 6 to 14: NHL/WM

Patients with NHL/WM will be enrolled into Phase 1b cohorts based on underlying diagnosis and treatment history (Figure 4) (see Section 5.1.1 for additional details regarding eligibility criteria).

Cohorts representing non-overlapping populations: MCL, MZL, WM (1L/chemoimmunotherapy unfit, 2L, and 3L+), DLBCL, and FL may be opened concurrently or sequentially based on emerging data.

Cohort 12 will enroll both PCSNL patients (3L+ or 2L with prior BTKi) and SCNSL patients. SCNSL patients are defined as any indication described (with the exception of CLL/SLL) meeting criteria for the appropriate cohort with secondary CNS involvement of their disease.

Cohort 13 will enroll PCNSL following upfront therapy (2L) and with no prior exposure to a BTKi.

4.1.4.2. Phase 1b Part 2 (Cohort Expansion)

Following the review of maturing data from Phase 1a and available data from Phase1b Cohort 1 after the completion of Cohort 1 enrollment, and the identification of a recommended Phase 1b Part 2 dose, up to an additional 80 patients with CLL/SLL (post-BTKi, post-BCL-2 inhibitor unless deemed ineligible for a BCL-2 inhibitor by Investigator at time of study enrollment) may be enrolled. ORR as assessed by Investigator will be utilized to further assess the anti-tumor activity of NX-5948 in this population in addition to its safety, tolerability, and PK/PD properties.

Patients in Phase 1b will attend the following scheduled clinic visits:

- Screening visit (within 28 days prior to starting study drug).
- Oral NX-5948 will be administered in 28-day treatment cycles. On-treatment visits will be conducted in Cycle 1 on:
  - Days 1, 2, 8, 15, and 22.
- Cycle 2 on-treatment visits will be conducted on:
  - Days 1, 2, 8 15, and 22.
- Cycle 3 and 4 on-treatment visits will be conducted on:
  - Days 1 and 15.
- Subsequent cycles (Cycle 5 and beyond) have on-treatment visits conducted on:
  - Day 1.
- In the event of a treatment interruption, planned visits should still occur with assessments conducted in alignment with the Schedule of Assessments (Section 15.3). There will not be a pause in the study calendar in the event of a treatment interruption.
- An EOT visit up to 3 days following the last dose of study drug. If EOT cannot occur within 3 days of last dose of study drug due to AE or treatment interruption, alternative scheduling for EOT and all subsequent follow-up visits may be discussed with the Sponsor.
- A 15-day Post-Treatment Safety Follow-Up visit will be performed 15 (± 5) days after last dose of study drug or prior to initiation of another anti-cancer therapy, whichever occurs first. If another anti-cancer therapy has already been started, the 15-day visit is not required.
- A 30-day Post-Treatment Safety Follow-Up visit will be performed 30 (± 7) days following the last dose of study drug or prior to initiation of another anti-cancer therapy, whichever occurs first. If another anti-cancer therapy has already been started, the 30-day visit is not required.
- Long Term Follow-up contact/visit: patients will be followed every 12 weeks after the 30-day Post Treatment Safety Follow-Up visit until death, withdrawal of consent, or loss to follow-up. Long Term Follow-up can be via clinic visit, phone call to the patient or referring physician, or other method deemed appropriate by the site, and should assess survival status, disease progression, and subsequent therapy.

4.2. Stopping Rules

Phase 1a: Enrollment will be paused for the review of safety events deemed by the SRC (Phase 1a) to constitute a significant safety risk at any point during the conduct of the study. These may include, but are not limited to, toxic death or incidence of ≥ 33% DLT-level AEs including patients who are beyond the DLT observation window.

Phase 1b: The IDMC will assess additional criteria to pause enrollment if >33% of treated patients develop any of the following that are at least possibly related to NX-5948:

- Any Grade ≥ 3 cardiac event
- Any Grade ≥ 4 infections
- Any Grade ≥ 3 hemorrhagic event
- Any Grade ≥ 4 febrile neutropenia
- Any Grade ≥ 4 thrombocytopenia

The relevant study governance and oversight body (refer to Section 12 of the protocol) will complete regular reviews of the data from Phase 1a and Phase 1b data through study completion. These reviews will assess safety data (interval and complete) as well as the activity data to recommend cohort and study continuation or termination. If recruitment is suspended for safety reasons noted above or others, then approval by the IDMC and the Competent Regulatory Authority will be required prior to restarting the study and applicable communications will be made to inform the Health Authority.

The stopping rules related to futility using Simon 2-stage for the two dose levels in Cohort 1 are tested independently with a 1-sided type I error rate of 5% and power of 80%. The null hypothesis for this cohort tests the true response rate is 35% against the alternative hypothesis is 55%.

- In Stage 1, if 8 (r1) or fewer responses are observed among 20 (n1) patients, the dose from cohort 1 will be stopped.
- In Stage 2, only one of the two doses proceeds to phase 2 such that 80 additional subjects are enrolled. If 46 (r) or more responses among the cumulative of 100 (n) patients are observed, reject the null hypothesis and proof of concept is claimed.

For additional details, see Sections 11.3.2 and 11.3.3.

4.3. End of Treatment

Patients may discontinue study treatment at any time upon their request. Discontinuation of study treatment does not automatically require withdrawal from the study. In addition, patients should cease further study treatment if any of the following are observed:

- Documented and unequivocal disease progression based on indication-specific disease response criteria (unless meeting exceptions below).

- Patients who are deriving overall clinical benefit with equivocal progression on imaging may continue to receive study treatment based on Investigator judgment until the next scheduled tumor assessment to confirm or refute progressive disease.
- Patients with radiographic disease progression meeting the applicable criteria below may remain on study treatment until they demonstrate clinical symptoms or radiographic evidence of disease progression above that which is present at time of initial disease progression (next scheduled radiographic assessment following a dose change) or until they meet other study discontinuation criteria. Specifically:
  1. Patients with NHL or WM must have enrolled in Phase 1a at a level below Phase 1b dose (600 mg) and patients with CLL/SLL must have randomized to the lower of the 2 dose levels (200 mg) during dose optimization or enrolled in Phase 1a at doses below the upper Phase 1b dose.
  2. Patients must meet dose escalation intrapatient dose escalation criteria as described in Phase 1a (Section 4.1.3.3); the lower dose must be due to assignment at study entry not dose reduction due to toxicity.
  3. Patient must have ongoing clinical benefit as assessed by the PI/treating physician and be assessed to have potential to benefit from escalation to maximum administered dose (600 mg).
  4. For Cohort 16, failure of wAIHA to respond or relapse of wAIHA does not require study discontinuation in the absence of progression of underlying malignancy.

- Patient experiences a serious or intolerable AE or clinically significant laboratory abnormality that meets criteria for treatment discontinuation (see Section 6.7) and/or jeopardizes patient safety if study treatment is continued.
- Use of protocol-defined prohibited anti-cancer therapy.
- Patient requires medication or medical procedure prohibited by the protocol.
- Symptomatic deterioration attributed to disease progression.
- Patient becomes pregnant (refer to Section 8.6 for details on the reporting procedures to follow in the event of pregnancy).
- Discontinuation of study treatment is in the best interest of the patient based on Investigator and/or Sponsor assessment.

Sponsor must be notified of all discontinuations of study treatment as soon as possible.

The Investigator must make every effort to perform the tests and evaluations listed for the EOT visit, 15-day Post-Treatment Follow-up visit, and 30-day Post-Treatment Follow-up visit. These data will be recorded as they comprise an essential evaluation that needs to be done prior to discharging any patient from study.

4.4. Withdrawal Criteria or End of Study

Patients may withdraw their consent to participate in the study at any time without prejudice. The Investigator must withdraw any patient who requests to be withdrawn. A patient's participation in the study may be discontinued at any time at the discretion of the Investigator in accordance with clinical judgment.

Sponsor must be notified of all patient withdrawals from the study as soon as possible.

Besides death, reasons for which the Investigator or Sponsor may withdraw participation from the study include, but are not limited to, the following:

- Withdrawal of consent.
- Patient does not adhere to study requirements specified in the protocol.
- Patient is lost to follow-up.
- Patients may be removed from study if considered by the Investigator or Sponsor to be in the best interest of the patient.
- Study termination by the Sponsor.

For patients who withdraw, procedures and assessments for the EOT, 15-day and 30-day Post-Treatment Safety Follow-up visits should be performed (unless the patient withdraws consent to do so).

If a patient fails to return for scheduled visits, a documented effort must be made to determine the reason. If the patient cannot be reached by telephone, a certified letter should be sent to the patient (or the patient's legally authorized representative, if appropriate) requesting contact with the Investigator. This information should be recorded in the study records.

The Investigator or designee must explain to each patient, before enrollment into the study, that for evaluation of study results, the patient's protected health information obtained during the study may be shared with the study Sponsor, regulatory agencies, and ethics committees (ECs)/Institutional Review Boards (IRB). It is the Investigator's (or designee's) responsibility to obtain written permission to use protected health information per country-specific regulations, such as the Health Insurance Portability and Accountability Act (HIPAA) in the US, from each patient, or if appropriate, the patient's legally authorized representative. In the EU and UK, the patient's informed consent form (ICF) will specifically include statements relating to the General Data Protection Regulation (GDPR). If permission to use protected health information is withdrawn, it is the Investigator's responsibility to obtain a written request, to ensure that no further data will be collected from the patient and the patient will be removed from the study.

4.5. Patient Identification and Replacement of Patients

Each patient will be assigned a unique patient identification (ID) number. This unique patient ID will not contain any information that can be directly linked to the patient. It will be on all applicable electronic case report form (CRF) pages and will be used throughout the patient's study participation and used on all study documentation related to the patient.

Individual Screening lab results outside of eligibility range may be repeated within the 28-day Screening window and will not constitute a rescreening or require a new unique patient ID number. If any procedure(s)/assessment(s) are repeated within 28 days of the signing of the original informed consent, it is only necessary to perform the procedure(s)/assessment(s) that did not originally meet the eligibility criteria; all other initial Screening procedures/assessments do not need to be repeated. If Screening/rescreening is delayed more than 28 days from the signing of the original informed consent, patients must be re-consented and repeat all Screening procedures/assessments that fall outside of a 28-day window prior to Cycle 1 Day 1 (unless otherwise specified; see Table 16 for additional details on safety assessments prior to Cycle 1 Day 1). Patients can be rescreened more than once with Medical Monitor approval.

Patients who provide informed consent and subsequently do not meet eligibility criteria or withdraw consent prior to treatment are considered screen failures and are not enrolled. Study site personnel should ensure that the source record includes documentation for the screen failure reason (eg, medical history, eligibility criteria, procedures performed).

Investigative sites will maintain a log of all screened patients who were reviewed and evaluated for study participation. Information collected on the Screening log should include limited information, such as the date of Screening, date the patient was enrolled, or the reason why the patient was screen failed.

Patient ID numbers assigned for patients who become screen failures are not to be reused.

If a patient in Phase 1a misses more than 25% of the doses for Cycle 1 (eg, > 7 doses of NX-5948 in 28 days) during the DLT period for reasons other than an AE, additional patient(s) may be enrolled to provide adequate data for dose escalation decision making. Patients withdrawn from dosing during the DLT period in any dose level cohort for reasons unrelated to NX-5948 safety may be replaced in order to have 3 or 6 patients evaluable for dose escalation (dependent on dose level cohort size) as described in Table 5.

4.6. Duration of Patient Participation

Patients will receive treatment in repeated 28-day cycles. Patients should remain on treatment until they experience documented disease progression, unacceptable toxicity, until they meet any of the withdrawal criteria (Section 4.4), or until the study is terminated by the Sponsor.

4.7. End of Study

The study can be suspended or terminated at any time by the Sponsor, at the recommendation of the SRC, or by applicable regulatory authorities. If the study is suspended or terminated for safety reasons, Nurix will promptly inform the Investigator and will also inform the regulatory authorities of the suspension or termination of the study and the reasons for the action. The Investigator may be informed of additional procedures to be followed in order to ensure adequate protection of patients. The Investigator will be responsible for promptly informing the EC/IRB, other applicable regulatory committees, and study patients of the suspension or early termination of the study, including the reasons for suspension or termination.

The end of study is defined as when the last patient has either been in Long-term Follow-up for at least 1 year or has progressed, died, withdrawn from study, or been lost to follow-up, whichever occurs first. Sponsor will notify the applicable regulatory agencies in accordance with local requirements. Anticipated duration of the study is approximately 6 years.

5. PATIENT POPULATION

A total of up to approximately 572 patients are planned to be enrolled. In Phase 1a (Dose Escalation) approximately 30 to 66 evaluable patients (ie, that meet DLT evaluable criteria and up to a maximum of 12 patients per previously cleared dose level) in each of the parallel dose escalations for CLL/SLL and NHL/WM with histologically confirmed R/R B-cell malignancies who have received at least 2 prior lines of therapy (or at least 1 prior line of therapy for patients with PCNSL), and for whom no other therapies are known to provide clinical benefit, are planned to be enrolled over a period of approximately 18 to 24 months. The total number of patients is dependent on the number of dose levels investigated and will include at least 6 patients at the dose(s) selected for Phase 1b for additional safety, PD/PK parameters, and anti-tumor activity.

Once recommended Phase 1b dose(s) for NX-5948 monotherapy have been established in CLL and NHL, preliminary anti-tumor activity of NX-5948 will be investigated in up to 360 patients total in up to 17 B-cell malignancy safety expansion cohorts (up to 20 response-evaluable patients per dose level or per cohort) (described in Section 4.1). Following dose optimization, up to 80 additional CLL patients with prior exposure to inhibitors of BTK and BCL-2 may be enrolled at the recommended dose to further assess anti-tumor activity.

5.1. Entry Criteria

Individuals eligible to participate in this study must meet all of the inclusion criteria and none of the exclusion criteria at Screening and on Cycle 1 Day 1.

5.1.1. Inclusion Criteria

Deviations from inclusion and exclusion criteria are not allowed because they can potentially jeopardize the scientific integrity of the study, regulatory acceptability, or patient safety. Therefore, adherence to the criteria as specified in the protocol is essential. Any questions

regarding patient eligibility and entry criteria should be discussed with the Medical Monitor prior to enrolling the patient.

1. Each patient must sign an ICF indicating that he or she understands the purpose of procedures required for the study and is willing to participate in the study. Consent is to be obtained prior to the initiation of any study-related tests or procedures that are not part of standard-of-care for the patient's disease.

2. Patients must be ≥ 18 years of age.

3. Patients in **Phase 1a (Dose Escalation)** must have 1 of the following histologically confirmed R/R B-cell malignancies:

   - R/R CLL, SLL
   - DLBCL of the following subgroups:
     - DLBCL NOS, GBC, ABC (includes transformed indolent lymphoma [eg, Grade 3b/transformed FL] and Richter-transformed DLBCL)
     - HGBL with *MYC* and *BCL-2* and/or *BCL-6* rearrangements, HGBL NOS

     (note: other subgroups of DLBCL or LBCL from the 2016 WHO classification of lymphoid malignancies [Swerdlow 2016] are excluded).
   - FL (Grade 1–3a; eligibility for systemic treatment as determined by GELF criteria; Section 15.8)
   - MCL
   - MZL (eligible subtypes include EMZL, MALT, NMZL, and SMZL)
   - WM
   - PCNSL

4. Patients in **Phase 1a (Dose Escalation)** must meet the following:

   a. For non-PCNSL indications, received at least 2 prior lines of therapy and have no other available therapies known to provide clinical benefit.

   b. For PCNSL, received at least 1 prior line of therapy.

5. Patients in **Phase 1b (Safety Expansion and Cohort Expansion)** must have 1 of the following histologically documented B-cell malignancies, must meet criteria for systemic treatment, and have prior treatments and/or molecular features described for their assigned cohort:

   <u>CLL Cohorts:</u>

   Cohort 1 and Phase 1b Part 2:

- CLL or SLL with prior exposure to both a BTKi and a BCL-2i, unless deemed ineligible for a BCL-2i by Investigator at time of study enrollment.

Cohort 2:

- CLL or SLL patients who meet eligibility criteria for Cohort 1 and Phase 1b Part 2 (described above) **and** have documentation of a mutation in BTK other than C481S (eg, T474 or L528) within 6 months prior to study enrollment.

Cohort 3:

- CLL or SLL following treatment with an ncBTKi and BCL-2i naïve.

Cohort 4:

- CLL or SLL with documented TP53 mutation and/or 17p deletion following upfront therapy that included a BTKi and BCL-2i naïve.

Cohort 5:

- CLL or SLL following at least 1 line of therapy that included a BTKi and BCL-2i naïve.

  (Note: For Cohorts 3 to 5, if patients meet multiple cohort criteria, refer to Section 4.1.4 for decision-making schema).

Cohort 15:

- 1L+ CLL or SLL requiring treatment with no prior exposure to a BTKi.
  - A minimum of 6 (33%) and no more than 10 (50%) treatment-naïve patients may enroll in this cohort.
  - All other patients in this cohort: 2L+ with prior exposure to non-BTKi therapy, including BCL-2i.
- Are intolerant to SoC treatment, or
- Refused SoC treatment, or
- Have been offered SoC and investigational medicinal product (IMP) but refused SoC in favor of IMP (as per consenting requirements), or
- The patient's treating physician considers that the lack of SoC treatment is not detrimental for the patient.

Cohort 16:

- BTKi-exposed R/R CLL or SLL requiring treatment, with secondary wAIHA.
- Patients must have:

- Received at least 1 line of therapy for CLL/SLL and at least 1 line of therapy for wAIHA.

  (Note: In the event a therapeutic modality was intended to treat **both** CLL/SLL requiring treatment per International Workshop on CLL (iwCLL) **and** wAIHA requiring treatment, it may be utilized to meet both treatment requirements).

- Anemia with hemoglobin ≤ 10 g/dL.
- Active wAIHA requiring treatment demonstrated by laboratory evidence of immune mediated hemolysis defined as:
  - Positive direct antiglobulin test (DAT).
  - Haptoglobin below the lower limit of normal (LLN).
  - Minimum of one of the following:
    - Lactate dehydrogenase above the upper limit of normal (ULN).
    - Increased indirect bilirubin.
    - Reticulocytosis appropriate for degree of anemia.
- Uncontrolled AIHA defined as one of the following:
  - Relapsed after previous treatment with corticosteroids (with or without rituximab), BTKi, or splenectomy.
  - Steroid resistant as evidenced by failure to obtain a hematologic response within 3 weeks on at least 1 mg/kg prednisone or prednisolone.
  - Steroid dependent (on a dose of ≤ 20 mg/day of prednisone or prednisolone) and on stable doses for ≥ 2 weeks.
- Note: Patients with multiple co-occurring autoimmune cytopenias (eg, Evans syndrome) are not eligible to enroll in Cohort 16.

Cohort 17:

- BTKi-exposed R/R CLL or SLL requiring treatment due to one of the following:
  - Synchronous progression of systemic CLL/SLL and progression or new onset of CNS involvement **OR**
  - Progression of systemic disease in the presence of measurable ongoing CNS involvement **OR**
  - New or worsening CNS disease in the presence of systemic disease that is stable or demonstrating evidence of clinical response to therapy.

NHL/WM Cohorts:

Cohort 6 (MCL):

- Non-blastoid MCL with prior exposure to a BTKi and an anti-CD20 monoclonal antibody (mAb)-based chemoimmunotherapy regimen.

Cohort 7 (MZL):

- MZL (EMZL, MALT, NMZL, SMZL) with prior exposure to an anti-CD20 mAb-based chemoimmunotherapy regimen and an additional line of therapy.

Cohort 8 (3L+ WM):

- WM with prior exposure to a BTKi and an additional line of therapy.

Cohort 9 (2L WM):

- WM following upfront therapy with a BTKi.

Cohort 10 (DLBCL):

- DLBCL of the following subgroups with prior exposure to an anthracycline (unless previously deemed ineligible to receive), an anti-CD20 mAb-based chemoimmunotherapy regimen, and an additional line of therapy:
  - Transformed indolent lymphoma (eg, Grade 3b/transformed FL).
  - Richter-transformed DLBCL.

  (Note: other subgroups of DLBCL or LBCL from the WHO classification of lymphoid malignancies [Swerdlow 2016] are excluded).

Cohort 11 (FL):

- FL (Grade 1–3a; eligibility for systemic treatment as determined by the GELF criteria; Section 15.8) with prior exposure to an anti-CD20 mAb-based chemoimmunotherapy regimen and an additional line of therapy.

Cohort 12 (PCNSL/SCNSL):

- PCNSL following at least 1 prior line of therapy that included a BTKi (2L+) or following 2 or more prior lines of therapy (3L+).
- SCNSL meeting criteria for any indication described (with the exception of CLL/SLL) with secondary CNS involvement of their disease.

  (Note: CLL/SLL patients with CNS involvement will enroll in Cohort 17).

Cohort 13 (PCSNL):

- PCNSL following upfront therapy and with no prior exposure to a BTKi (2L).

Cohort 14 (1L WM):

- Patients with treatment-naïve WM deemed unfit for chemoimmunotherapy by Investigator at time of study entry.
- Are intolerant to SoC treatment, or
- Refused SoC treatment, or
- Have been offered SoC and IMP but refused SoC in favor of IMP (as per consenting requirements), or
- The patient's treating physician considers that the lack of SoC treatment is not detrimental for the patient.

6. Patients must have measurable disease per response criteria specific to the malignancy (ie, iwCLL, Lugano Classification of Lymphoma response criteria, WM response criteria, or International PCNSL Collaboration Group criteria; Section 15.4), which may include imaging by computed tomography (CT), CT/positron emission tomography (PET) scan, or magnetic resonance imaging (MRI) as per respective response criteria.
   a. For patients with NHL: must have either at least 1 metabolically active lesion defined as fluorodeoxyglucose (FDG)-avid (5PS score of 4 or 5) as assessed by PET-CT or at least 1 target lesion that is > 1.5 cm for lymph nodes or > 1.0 cm for extranodal lesion(s) in the longest diameter as assessed by CT.
   b. For patients with WM, serum monoclonal IgM paraprotein > 0.5 g/dL.
   c. Patients with SCNSL (Cohort 12) must have measurable disease per International PCNSL Collaboration Group criteria, as evidenced by at least one of the following: CNS disease on MRI, detectable in CSF, or on ophthalmologic exam. The presence or absence of systemic disease should be assessed and documented during screening, but systemic disease is not required for enrollment.
7. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 (0 to 2 for patients with PCNSL and SCNSL).
8. Anticipated life expectancy of at least 12 weeks from the time of Screening.
9. Adequate organ and bone marrow function as follows:

Laboratory Parameter	Value
Hematology	
Hemoglobin	> 90 g/L (9 g/dL; 5.59 mmol/L) or > 70 g/L (7 g/dL; 4.34 mmol/L) if anemia is due to bone marrow involvement or AIHA per Investigator assessment. A minimum 5-day transfusion-free interval is required.

Laboratory Parameter	Value
Absolute neutrophil count (ANC)	> 1.0 ×10^9/L unless neutropenia is due to bone marrow involvement per Investigator assessment, in which case there is no minimum ANC requirement. Granulocyte colony stimulating factor (G-CSF) is allowed for those with bone marrow involvement of disease.
Platelets	Platelets > 50 × 10^9/L (or > 30 ×10^9/L if thrombocytopenia is due to bone marrow or splenic involvement). For patients with PCNSL or SCNSL: platelets > 50 × 10^9/L regardless of etiology. For all patients, a minimum 14-day transfusion-free interval is required prior to first dose. For patients undergoing lumbar puncture, see Section 5.1.3.
Coagulation (international normalized ratio [INR] and partial thromboplastin time [PTT] or activated partial thromboplastin time [aPTT])	≤ 1.5 × ULN (Patients undergoing lumbar puncture must have either PTT or aPTT < ULN). See Section 5.1.3 for rationale.
Pancreatic	
Serum amylase	≤ 1.5 × ULN
Serum lipase	≤ 1.5 × ULN
Hepatic	
Aspartate aminotransferase (AST) and alanine aminotransferase (ALT)	≤ 2.5 × ULN
Total bilirubin	≤ 1.5 × ULN (or ≤ 5 × ULN for patients with Gilbert's syndrome) except for patients with indirect hyperbilirubinemia and evidence of hemolysis (haptoglobin below LLN) who may enroll in the absence of laboratory or clinical evidence of liver dysfunction.
Renal	
Calculated glomerular filtration rate (GFR)	> 45 mL/min/1.73 m^2 Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) creatinine (see Section 15.5)
Cardiac	
QT interval	Triplicate electrocardiogram (ECG) average QT interval corrected for heart rate (QTcF) < 450 msec, measured by Fridericia's formula [QTcF = QT/(RR^0.33)]

Laboratory Parameter	Value
	Note: For patients with pacemaker or bundle branch block, the QTcF should be < 475 msec.
Pulmonary	
Blood oxygen saturation (SpO_2)	> 92% on room air

10. Women of childbearing potential (WOCBP) must agree to use highly effective contraception (failure rate of < 1% per year) during the study from Screening through 6 months after the last dose of study drug (see Section 6.6.4). WOCBP must agree to not donate eggs (ova, oocytes) during the same timeframe. WOCBP and contraception methods are defined as per the 2020 Clinical Trial Facilitation Group (CTFG) recommendations and provided in Section 15.9 (CTFG 2020).

11. Male patients with WOCBP partners must agree to use highly effective contraception and barrier contraception (condom; defined as per CTFG 2020 [Section 15.9]) during the study from Screening through 3 months after the last dose of study drug. Men must agree to not donate sperm during the same timeframe.

12. Patients must be willing and able to adhere to the prohibitions and restrictions specified in this protocol.

5.1.2. Exclusion Criteria

Individuals who meet any of the following exclusion criteria at Screening will not be eligible to participate in the study:

1. Known or suspected active prolymphocytic leukemia or Richter's transformation to Hodgkin's lymphoma prior to enrollment.

2. Prior treatment for the indication under study for anti-cancer intent that includes:

    a. Radiotherapy within 2 weeks of planned start of study drug (excluding limited palliative radiation).

    b. Prior systemic chemotherapy within 2 weeks of planned start of study drug.

    c. Prior mAb therapy within 4 weeks of planned start of study drug, except for patients enrolling in Cohort 16 (CLL with secondary wAIHA) for whom a 16-week washout period is required.

    d. Prior small molecule therapy within 5 half-lives or 2 weeks (whichever is shorter) of planned start of study drug.

    e. Autologous or allogeneic stem cell transplant within 100 days prior to planned start of study drug.

f. Chimeric antigen receptor (CAR) T-cell therapy within 100 days prior to start of study drug (within 60 days prior to start of study drug for Phase 1b).

g. Use of systemic corticosteroids outside of dosing limits described below and within the 7 days prior to initiation of study treatment excepting those used as prophylaxis for radio diagnostic contrast. Patients with CNSL (including both primary and secondary CNSL): no greater than 40 mg/day prednisone, or equivalent; CNSL patients using > 20 mg/day prednisone, or equivalent, must be clinically stable at that dose for 7 days. All other diagnoses: no greater than 20 mg/day prednisone or equivalent.

h. Use of systemic immunosuppressive drugs other than systemic corticosteroids for any medical condition within 60 days prior to first dose of study drug.

i. Previously treated with a BTK degrader.

3. Active, uncontrolled autoimmune hemolytic anemia (except for patients enrolling in Cohort 16) or active, uncontrolled autoimmune thrombocytopenia.

4. Unable to swallow capsules or malabsorption syndrome, disease or procedure significantly affecting gastrointestinal function, or resection of the stomach or small bowel or ulcerative colitis, inflammatory bowel disease, or partial or complete bowel obstruction likely to interfere with the delivery, absorption, or metabolism of study drug.

5. Patients with active graft-versus-host disease (GVHD) or who have required anti-GVHD treatment or prophylaxis within 60 days of planned start of study drug.

6. History of known or suspected uncontrolled seizure disorder not related to CNS involvement of underlying malignancy.

7. Patient has any of the following:

   a. Myocardial infarction, unstable angina, unstable symptomatic ischemic heart disease, or placement of a coronary arterial stent within 6 months of planned start of study drug.

   b. Uncontrolled atrial fibrillation or other clinically significant arrhythmias, conduction abnormalities, or New York Heart Association (NYHA) class III or IV heart failure within 6 months of planned start of study drug.

   c. Thromboembolic events (eg, deep vein thrombosis, pulmonary embolism, or symptomatic cerebrovascular events), stroke, or intracranial hemorrhage within 6 months of planned start of study drug.

   d. Any other significant cardiac condition (eg, pericardial effusion, restrictive cardiomyopathy, severe untreated valvular stenosis, severe congenital heart disease, or persistent uncontrolled hypertension defined as systolic blood

pressure > 160 mmHg or diastolic blood pressure > 100 mmHg despite optimal medical management) within 6 months of planned start of study drug.

8. Bleeding diathesis, or other known risk for acute blood loss.
9. History of Grade ≥ 2 hemorrhage within 28 days of planned start of study drug.
10. Patients requiring ongoing treatment with warfarin or an equivalent vitamin K antagonist within 168 hours prior to first dose of study drug.
11. Toxicities from previous anti-cancer therapies must have resolved to baseline levels or to Grade 1 (except for alopecia, hypothyroidism with adequate replacement therapy, hypopituitarism with adequate replacement therapy, peripheral neuropathy, or hematologic parameters meeting inclusion criteria).
12. Active known concurrent malignancy or malignancy other than the one under study within the past 3 years. (Exceptions include, but are not limited to, patients with more recent history of basal or squamous cell skin cancer, superficial bladder cancer, or carcinoma in situ of the cervix or breast may enroll if they have undergone curative therapy and have no evidence of disease).
13. Patient who has known allergies, hypersensitivity, or intolerance to components of study drug.
14. Patient who requires TLS prophylaxis and has known allergy to both xanthine oxidase inhibitors and rasburicase.
15. Pregnant, breastfeeding, or planning to become pregnant while enrolled in this study or within 6 months after the last dose of study drug; for WOCBP, negative pregnancy status must be confirmed by pregnancy test at Screening and within 24 hours of first dose of study drug.
16. Patient is a man who plans to father a child while enrolled in this study or within 3 months after the last dose of study drug.
17. Patient has had major surgery (eg, requiring general anesthesia) within 4 weeks before the planned first dose of study drug, or will not have fully recovered from surgery, or has surgery planned during the time the patient is expected to participate in the study or within 4 weeks after the last dose of study drug. Note: patients with minor planned surgical procedures to be conducted under local anesthesia may participate.
18. Active, significant bacterial, fungal, parasitic, or viral infection including:
    a. Current active liver disease from any infectious cause, including hepatitis B (hepatitis B virus surface antigen [HbsAg] positive), and hepatitis C (hepatitis C virus [HCV] antibody positive, confirmed by detectable HCV ribonucleic acid). Exceptions: Patients with hepatitis B virus (HBV) who are negative for HBV deoxyribonucleic acid (DNA) by quantitative polymerase chain reaction (PCR) are eligible irrespective of serology findings; as are, patients with a history of hepatitis C with undetectable viral levels following treatment.

b. Infection with HIV-1 or HIV-2. Exception: patients with well-controlled HIV (eg, CD4 > 350/mm^3 and undetectable viral load) are eligible.

c. Active viral reactivation (eg, cytomegalovirus [CMV] or Epstein-Barr virus [EBV]).

19. Active, uncontrolled diabetes (eg, hemoglobin A1c > 9%, symptomatic, with or without end organ complications, etc.) as assessed by the Investigator (note: poor glycemic control in the context of disease modifying corticosteroid treatment is not exclusionary).

20. Any other medical or psychiatric condition or social situation that, in the opinion of the Investigator, would compromise patient safety, or interfere with the objectives, the protocol, or completion of treatment per protocol.

21. Vaccinated with a live vaccine within 28 days prior to the first dose of study drug or completion of any vaccinations within 14 days prior to first dose of study drug. Live vaccines are prohibited throughout study treatment and vaccinations of any kind are prohibited during the DLT period (Phase 1a) or Cycle 1 (Phase 1b).

22. Administration of any strong CYP3A inhibitors or inducers within 14 days or 5 half-lives, whichever is longer, and any moderate CYP3A inhibitors or inducers within 168 hours, or 5 half-lives, whichever is longer, prior to first dose of study drug. (See Section 6.8.2.2).

23. Administration of any inhibitors of P-glycoprotein or breast cancer resistance protein (BCRP) transporters within 48 hours prior to first dose of study drug (see Section 6.8.2.3).

24. Administration of a proton pump inhibitor (prescription or over-the-counter) within 168 hours prior to first dose of study drug. See Section 6.8.2.4 for allowed use of alternative agents.

25. Use of biotin (ie, vitamin B7) or supplements containing biotin higher than the daily adequate intake of 30 µg within 72 hours prior to Screening and during study treatment (NIH 2022; Note: patients who switch from a high dose to a dose of 30 µg/day or less are eligible for study entry).

26. Dual treatment with an anticoagulant and antiplatelet agent is prohibited within 2 weeks of treatment initiation and while on study.

5.1.3. Other Eligibility Criteria Considerations

Rationale for including patients with PCNSL and any indication with CNS involvement is as follows:

- While patients with CNS involvement of disease have a higher risk of catastrophic outcome if they experience a severe bleed in the CNS, pre-clinical data in NX-5948-treated monkeys demonstrated no effects on coagulation parameters and non-dose limiting findings of hemostasis with reversibility in the skin/subcutis. The risk of hemorrhage is known for BTKi with and without the use of antithrombotic agents (see Section 6.6.2.1 for more detailed information). Although the risk of hemorrhage

is unknown with NX-5948, with safety precautions and close monitoring in place for signs of bleeding, the unmet need for this disease is significant (see Section 2.1.8). With the lack of effective treatment options for patients with R/R CNSL, the potential benefits patients might receive from NX-5948 outweigh the risks.

Rationale for removing the inclusion thresholds for neutrophils and for lowering those for hemoglobin from 9 to 7 g/dL in patients with bone marrow involvement and/or AIHA is as follows:

- iwCLL guidelines state the following, "Evaluation of hematological toxicity in patients with CLL must take into consideration that many patients have low blood cell counts at the initiation of therapy due to underlying bone marrow infiltration or AIHA. Therefore, the standard criteria used for solid tumors cannot be applied because many CLL patients then would be considered to have Grade 2–4 hematological toxicity at the initiation of treatment. Furthermore, absolute blood neutrophil counts are rarely used at the initiation of therapy to modify the treatment dose because these values typically are unreliable in CLL patients with lymphocytosis." The same rationale also applies for patients with lymphoma who have bone marrow involvement. Therefore, the inclusion thresholds for hematologic parameters will be lowered for patients who have bone marrow involvement to allow for the inclusion of patients with typical CLL and other lymphomas who have lower blood cells counts prior to initiating therapy.

Rationale for allowing patients with coagulation parameters ≤ 1.5 × ULN onto the study is as follows:

- Performing a lumbar puncture carries risks of spinal hematoma and intracranial hemorrhage, especially in patients with pre-existing coagulopathy. Although the incidence is low, morbidity is high. Suspending anti-thrombotic therapy also has risks for ischemia and thromboembolism. Therefore, guidelines from the British Society of Haematology suggest the following parameters are met prior to performing a lumbar puncture: platelet count > 40 x 10^9 L, INR ≤ 1.5 × ULN, and aPTT ≤ ULN (Dodd 2018). To mitigate the risk of serious protocol-related bleeding and consistent with site-specific national guidelines, thrombocytopenia should be corrected with platelet transfusion prior to lumbar punctures. Transfusion of platelets solely to increase platelet counts prior to a lumbar puncture will not be considered a DLT.

Rationale for allowing patients who have received small molecule therapy within 5 half-lives or 2 weeks or (whichever is shorter) onto the study.

- Utilizing a 2-week washout for prior systemic therapy is standard. However, patients who have been treated with BTKi may experience extremely rapid disease progression most commonly due to mutations of the C481 residue, which abrogates the binding of covalent inhibitors (Quinquenel 2019; Sedlarikova 2020; Woyach 2017). The half-life of ibrutinib, the most commonly prescribed BTKi for CLL, is 4–6 hours (5 × = ~30 hours). Once a covalent inhibitor such as ibrutinib is stopped, patients can progress extremely rapidly in as soon as 3 days. To mitigate the risk of rapid disease progression to the patient, the washout period for small molecule therapy has been reduced to 5 half-lives or 2 weeks (whichever is shorter).

Patient eligibility may require additional or repeat assessments such as safety labs, vital signs, or ECG during the Screening period.

To assess any potential impact on patient eligibility with regard to safety, refer to Section 6.6 for detailed information regarding risks, precautions, contraindications, AEs, and other significant data pertaining to the study drug used in this study.

6. TREATMENTS

6.1. Dispensation

Sponsor and/or its designee will provide the study site with a supply of NX-5948 sufficient for the completion of the study.

6.2. Identity of Investigational Product

6.2.1. Product Characteristics and Labeling

NX-5948 is an anti-cancer drug and, as with other potentially toxic compounds, caution should be exercised when handling NX-5948. Specific instructions on dispensation will be provided in the Study Pharmacy Manual. NX-5948 study drug will be supplied as 25 mg, 50 mg, 100 mg, and 150 mg capsules for oral administration QD under fasting conditions.

6.2.2. Packaging and Labelling

NX-5948 capsules are manufactured, packaged, and labeled according to Good Manufacturing Practice (GMP) and Good Clinical Practice (GCP). Refer to the Study Pharmacy Manual for further details.

6.2.3. Storage and Handling

NX-5948 capsules should be stored at room temperature 20–25ºC (68–77ºF); excursions are permitted between 15ºC and 30ºC and transient spikes up to 40ºC are permitted as long as the spikes do not exceed 24 hours. Refer to the Study Pharmacy Manual for further details.

6.3. Method of Assigning Patients to Treatment Groups

For those patients who meet all of the eligibility criteria, an Enrollment Form will be completed and submitted with the results of the Screening evaluations to the Sponsor or designee for review. Eligible patients will then be assigned to a treatment cohort and may be enrolled. See the Investigator Site File for additional details on the enrollment process.

6.4. Selection of Doses Used in the Study

To determine DLTs, MTD, and the recommended Phase 1b dose(s), this trial will employ a traditional 3+3 design, which has been the most widely used dose escalation design in oncology for the past 2 decades and is considered the standard approach to the Phase 1 evaluation of new anti-cancer agents (Eisenhauer 2015).

The starting dose of NX-5948 in Phase 1a will be 50 mg/day (see Section 2.4 for the starting dose rationale). Planned dose escalation levels for NX-5948 are shown in Table 4.

To this end, the Sponsor will conduct a dose escalation followed by safety and cohort expansion. The totality of all available safety, PK, PD, and preliminary anti-tumor activity data will be evaluated to guide further development. The Sponsor may open DL -1 and DL -2 cohorts to further explore safety, PK, PD, and preliminary anti-tumor activity outside the need to open for safety concerns. For additional dosing information during the dose escalation and the dose expansion portions of the study, see Sections 4.1.3 and 4.1.4. Additional patients may be allocated in parallel to these respective dose levels. If enrolled, the SRC will review all DLT evaluable safety data.

6.5. NX-5948 Administration

NX-5948 capsules should be taken orally with water as defined for each cohort. As the effect of food on absorption is currently unknown, all patients should fast for at least 2 hours before and 2 hours after taking NX-5948. In addition, on intensive PK collection days (Cycle 1 Day 1 and Cycle 2 Day 1), patients should fast for a minimum of 8 hours before and for 2 hours after taking NX-5948.

NX-5948 should be taken at approximately the same time each day. NX-5948 capsules should be swallowed whole and not crushed, chewed, or dissolved in water. A dosing period of up to 30 minutes is permissible if required by the number of capsules to be taken or as convenient for the patient. Missed doses should not be taken as a double dose at the next administration. Patients who vomit their dose should be instructed NOT to make up that dose. Doses that are vomited should not be replaced.

To evaluate the impact of food on NX-5948 PK, PD, and tolerability, a pilot food effect will be investigated in Healthy Volunteers. Based on emerging clinical data (eg, safety, tolerability, activity, and/or PK and PD data) related to this pilot food effect investigation and in consultation with the SRC and Medical Monitor, timing and/or food content restrictions (eg, fasting) for all patients may be modified in a future protocol amendment.

Patients will be treated with NX-5948 in 28-day cycles.

6.6. Risk Information for NX-5948

6.6.1. Contraindications

NX-5948 is contraindicated in patients with previously demonstrated hypersensitivity to any of the components of the product.

6.6.2. Possible Clinical Risks For NX-5948

At this time, there are no compound-specific identified risks for NX-5948. See Section 2.5 for preliminary safety data.

The possible clinical risks in this section are informed from the nonclinical toxicity studies as well as selected known risks associated with BTKi that may be related to the proposed mechanism of action. Please see Section 2.3 and the Investigator's Brochure for additional details.

6.6.2.1. Hemorrhage

In nonclinical toxicity studies, target organs with histopathological findings relevant for hemorrhage in mice included lymph nodes (mandibular), and in monkeys included lymph nodes (mandibular), pancreas, skin/subcutis, and stomach.

Relevant findings in mandibular lymph nodes of NX-5948-treated mice included scattered macrophages characterized by cytoplasmic pigment that most likely represented hemosiderin, and in mandibular lymph nodes of NX-5948-treated monkeys included minimal or mild erythrocytosis suggestive of draining hemorrhage. Findings in the pancreas, skin/subcutis, and stomach were limited to NX-5948-treated monkeys, and included minimal or mild angiectasis/hemorrhage in the pancreatic islets of some males (all NX-5948 dose levels) and some females (≥ 10 mg/kg/day), and microscopic hemorrhage in the skin/subcutis that was sporadic and minimal to mild in males and more frequent or severe (minimal to moderate) in females at > 10 mg/kg/day; microscopic skin/subcutis findings correlated with clinical and macroscopic observations in the skin. Findings in the stomach included macroscopic observations of discolored focal areas and microscopic submucosal hemorrhage. At the end of the treatment-free recovery periods, there were no noteworthy microscopic findings in the target organs described above. The effects of NX-5948 in monkeys described above are considered potentially relevant to human risk. However, reversibility upon cessation of dosing suggests the possible risk in NX-5948-treated patients will be clinically manageable.

Although no NX-5948-related effects on coagulation parameters were noted in nonclinical toxicity studies at any dose level evaluated, test article-related disruption of hemostasis was considered a possible cause of hemorrhage-related findings in the skin/subcutis and mandibular lymph nodes. The effect of BTK inhibition on platelet function and bleeding risk is not clear. In NX-5948-treated monkeys, findings in the skin/subcutis suggestive of an effect on hemostasis were not dose-limiting and were reversible upon cessation of dosing.

In addition to findings from nonclinical toxicity studies for NX-5948, hemorrhage is a known risk for BTKi. Fatal and serious hemorrhagic events have occurred in patients with hematologic malignancies treated with BTKi. Major hemorrhage (serious or Grade ≥ 3 bleeding or any CNS bleeding) occurred in 3.0–4.0% of patients, with fatal hemorrhage occurring in 0.1–0.4% of 3867 patients exposed to BTKi in clinical trials. Bleeding events of any grade, excluding bruising and petechiae, occurred in 22–39% of patients. Use of antithrombotic agents concomitantly with BTKi may further increase the risk of hemorrhage. In clinical trials, major hemorrhage occurred in 2.7–3.1% of patients taking BTKi without antithrombotic agents and 3.6–6.1% of patients taking BTKi with antithrombotic agents. The addition of antiplatelet therapy with or without anticoagulants had a risk of hemorrhage of 4.4%.

Overall, while the risk of hemorrhage is unknown with NX-5948, the nonclinical toxicity observations for NX-5948 along with AEs associated with the therapeutic class of BTKi suggests that the risk of hemorrhage is possible. The risks and benefits of antithrombotic or antiplatelet agents when co-administered with NX-5948 must be taken into account and discussed with the Medical Monitor. If antithrombotic or antiplatelet agents are initiated while the patient is receiving NX-5948, the patient should be closely monitored for signs of bleeding. If a patient requires a major surgical procedure while taking NX-5948, discuss with the Medical Monitor to consider if a temporary interruption of NX-5948 is appropriate.

Please see Section 6.8.2.1 for information on permitted/prohibited anticoagulant and antiplatelet concomitant medications.

6.6.2.2. Mucosal Effects

In nonclinical toxicity studies, noteworthy microscopic findings in the stomach were limited to NX-5948-treated monkeys. In monkeys administered NX-5948 days, mucosal damage (erosion/ulcer) or evidence of prior mucosal damage (atrophy or epithelial or glandular hyperplasia) was occasionally noted in the stomach of males and/or females at all dose levels evaluated. In addition, 2 male animals (1 at 10 mg/kg/day and 1 at 100 mg/kg/day) had minimal to mild vascular necrosis in the gastric submucosa, 1 male (10 mg/kg/day) had multifocal thrombosis in the submucosa, and 1 female (30 mg/kg/day) had moderate necrosis of the fundic submucosa. These findings were considered likely NX-5948-related. Sporadic incidences of hemorrhage or erosion/ulceration were noted in some NX-5948-treated animals in the small and/or large intestine, but these findings were considered not clearly related to NX-5948 administration. At the end of the treatment-free periods, there were no noteworthy microscopic findings in the stomach (or intestine). The effects of NX-5948 on the stomach in monkeys are considered potentially relevant to human risk. However, reversibility upon cessation of dosing suggests the possible risk in NX-5948-treated patients will be clinically manageable.

6.6.2.3. Serious and Opportunistic Infections

The possible risk of serious and opportunistic infections is informed solely from the known risks of infections for BTKi. Fatal and serious infections, including opportunistic

infections, have been observed in patients taking BTKi, even in the absence of profound neutropenia. Serious or Grade ≥ 3 infections (bacterial, viral, or fungal) were observed in 19–21% of patients exposed to BTKi in clinical trials, most often due to respiratory tract infections (11% of all patients; 6% pneumonia). These infections occurred mostly in the absence of severe neutropenia, with neutropenia-related infections observed in approximately 2% of patients. Opportunistic infections have included, but were not limited to, HBV reactivation, fungal pneumonia, *Pneumocystis jirovecii* pneumonia, EBV reactivation, CMV, and progressive multifocal leukoencephalopathy.

6.6.2.3.1. Hepatitis B Reactivation

Patients should be tested for both HbsAg and hepatitis B core antibody (HbcAb). If either test is positive while on study, quantitative PCR tests should be performed. Such patients should be treated per SoC for the entire duration of the study. Such patients should also be monitored with frequent liver function tests and HBV DNA PCR at least every 2 months. If the HBV DNA level becomes detectable during this monitoring, management should be reviewed with an appropriate specialist with experience managing hepatitis and consideration given to halting study drug treatment.

6.6.2.4. Cytopenias

In nonclinical toxicity studies, noteworthy findings included effects on hematology parameters in both mice and monkeys of minimal decreases in red cell mass (red blood cell count, hemoglobin, hematocrit) with no dose dependency. At the end of the treatment-free recovery periods, hematology parameters in mice demonstrated partial recovery, while in monkeys there were no noteworthy hematology findings with the exception of 1 female (30 mg/kg/day) in which red cell mass remained decreased with a robust reticulocyte response.

In addition to findings from nonclinical toxicity studies for NX-5948, cytopenias are known risks for BTKi. Patients with B-cell malignancies who take BTKi experienced Grade 3 or 4 cytopenias, including neutropenia (23%), thrombocytopenia (8%), anemia (3–8%), and lymphopenia (7%).

Please see Section 15.3 for information on hematology monitoring. If hematology parameters demonstrate Grade ≥ 3 cytopenias based on severity grading by CTCAE, increase frequency of monitoring as clinically appropriate.

6.6.2.5. Cardiac Arrythmias

The possible risk of cardia arrythmia is informed solely from the known risks of cardiac arrhythmias for BTKi. Fatal and serious cardiac arrhythmias have occurred with BTKi. Grade ≥ 3 ventricular tachyarrhythmias occurred in 0.2% of patients and Grade ≥ 3 atrial fibrillation and atrial flutter occurred in 1.1–4% of patients who received BTKi in clinical trials. Patients with cardiac risk factors, hypertension, acute infections, and a previous history of cardiac arrhythmias may be at increased risk.

6.6.2.6. Hypertension

The possible risk of hypertension is informed solely from the known risk of hypertension for some BTKi. Nearly 20% of patients taking BTKi developed hypertension with 8% developing Grade ≥ 3 hypertension. The median time to development of hypertension was 5.9 months in 1 study.

6.6.2.7. Tumor Lysis Syndrome

The possible risk of TLS is informed solely from the known risk of TLS for some BTKi. TLS has occurred in patients with B-cell malignancies taking BTKi.

Please see Section 6.8.3 for information on required TLS prophylaxis for patients with DLBCL, CLL/SLL, and MCL.

6.6.3. Drug Interactions

PK of NX-5948 may be altered by strong or moderate inducers or inhibitors of CYP3A, and inhibitors of P-glycoprotein and BCRP; refer to Sections 6.8.2.2 and 6.8.2.3 for information on prohibited concomitant medications relating to these drug interactions.

In addition, NX-5948 exhibits pH dependent solubility; refer to Section 6.8.2.4 for information on permitted and prohibited acid reducing concomitant medications.

6.6.4. Reproductive and Developmental Toxicity

No reproductive or developmental toxicity studies have been conducted to date, and the potential for NX-5948 to cause fetal harm or to be excreted in breast milk is unknown.

WOCBP must use highly effective contraception (failure rate of < 1% per year) during the study from Screening through 6 months after the last dose of study drug. WOCBP must agree to not donate eggs (ova, oocytes) during the same timeframe.

All men with WOCBP partners must agree to use highly effective contraception <u>and</u> barrier contraception (condom) during the study from Screening through 3 months after the last dose of study drug. Even with medical confirmation of successful vasectomy, barrier method (condom) use is indicated to prevent possible direct exposure of a developing fetus to active drug ingredient in seminal fluid. Men must agree to not donate sperm during the same timeframe.

Refer to Section 15.9 for the definitions of WOCBP and acceptable contraception methods.

For additional detailed guidance and definitions, please refer to the 2020 CTFG Recommendations related to contraception and pregnancy testing in clinical trials (CTFG 2020).

6.6.5. Overdose

There are currently no clinical or nonclinical data available relevant to overdose.

In the event of an overdose, patients should be observed for symptomatic side effects and maximum monitoring and supportive care should be initiated, as indicated. If the overdose is recent and substantial, gastric lavage or induced emesis may be considered.

Please refer to Section 8.4 for guidance on safety reporting for AEs associated with overdose.

6.7. Dose Modification of NX-5948

Patients will receive daily doses of NX-5948 in 28-day treatment cycles, until they experience documented disease progression, unacceptable toxicity, until they meet any of the withdrawal criteria, or until the study is terminated by the Sponsor.

Cycles are 28 days in duration regardless of dose interruption.

The Sponsor should be notified in a timely manner of dose holds and in advance of dose modifications for systems update. In addition, in the event of a significant clinical event leading to treatment interruption or need for dose modification PK samples for trough levels of NX-5948 should be collected when feasible (Section 7.5.3).

Phase 1a Cycle 1:

Dose delays/reductions during the DLT period are not advised, but are permitted.

Dose delays/modifications will be considered a DLT if the dose was delayed or modified for a toxicity that is considered related to study drug (ie, not related to the patient's underlying disease, other medical condition or concomitant medications).

A patient who experiences a DLT may have NX-5948 dosing held for as per Table 7 and Table 8 to allow for recovery (to Grade 1 or less or baseline if baseline is Grade 2 or above).

The patient may restart therapy if it is considered in his/her best interest to continue therapy and with documented Sponsor approval.

Upon restarting, the patient **may** have the dose reduced per Table 7 and Table 8.

Phase 1a Cycle 2 and beyond and Phase 1b all cycles

In Cycle 2 and beyond for patients enrolled in Phase 1a and throughout the study for patients enrolled in Phase 1b, dose delays and modifications for safety purposes may be implemented.

Dose hold and modification guidelines for Grade 3 and 4 NX-5948 related AEs are provided in Table 7 (hematologic AEs) and Table 8 (non-hematologic AEs).

For hematologic events, dose modification decision/action may be impacted by the underlying disease and bone marrow involvement (e.g. cytopenias caused by disease). Sponsor's Medical Monitor should be notified when considering dose modifications (treatment interruptions and/or reductions) for hematological AEs in patients with bone marrow involvement of their disease (Table 7).

All Grade 3/4 non-hematologic events will be reviewed by the Medical Monitor in consultation with the Investigator and action will be taken with study drug as outlined in Table 8. In addition, please refer to Section 8.4 for criteria for permanent discontinuation of study drug in association with liver test abnormalities.

In the case of a patient who experiences a clinically significant AE not related to NX-5948 that requires dose interruption (e.g., due to need for time-limited use of prohibited medications or recovery from a surgical procedure), NX-5948 dosing may be held for up to 28 days.

If the patient requires a dose delay of >28 days, the patient will have treatment permanently discontinued unless there is a compelling clinical rationale articulated by the Investigator and approved by the Sponsor.

Table 7 Suggested Dose Modification Scheme for NX-5948-related Hematological AEs in Escalation Cohorts and Expansion Cohorts – Phase 1a Cycle 2+ and all Phase 1b Cycles

NX-5948-related Hematological AEs	Action to be Taken with NX-5948[a]
Grade 2 – *any occurrence*	No action with study drug.
Grade 3 Febrile Neutropenia Grade 4 thrombocytopenia lasting more than 10 days (if non-disease related and ≥ 1 grade below baseline) Grade 4 neutropenia lasting more than 10 days (if non-disease related and ≥ 1 grade)	**First occurrence:** • Interrupt treatment. • If the AE resolves to Grade ≤ 1 or baseline, (if baseline cytopenia is Grade > 1) within 7 days, resume study drug at the same dose. • If the AE does not resolve to Grade ≤ 1 within 7 days, continue to hold study drug. Patient may resume study drug at next lower dose if AE resolves to ≤ 2 within 21 days • If AE does not resolve to ≤ 2 within 21 days discontinue study drug and notify the Medical Monitor. **Second occurrence:** • Interrupt treatment. • Patient may resume study drug at next lower dose if the AE resolves to Grade ≤ 1 or baseline within 10 study days. • If the AE does not resolve to Grade ≤ 1 or baseline within 10 study days, discontinue study drug and notify the Medical Monitor. **Third Occurrence:** • Discontinue study drug and notify the Medical Monitor.
Grade 4 Febrile Neutropenia	**First occurrence:** • Interrupt treatment.

NX-5948-related Hematological AEs	Action to be Taken with NX-5948[a]
	• If the AE resolves to Grade ≤ 1 within 10 days, patient may resume study drug at next lower dose • If the AE dose not resolve to Grade ≤ 1 or baseline within 10 study days, discontinue study drug and notify the Medical Monitor. **Second Occurrence:** • Discontinue study drug and notify the Medical Monitor.
Grade ≥ 3 thrombocytopenia with Grade > 2 hemorrhage	**First Occurrence:** • Discontinue study drug and notify the Medical Monitor.
[a] Dose modification decision/action may be impacted by the underlying disease and bone marrow involvement. Please consult the Medical Monitor when considering dose modifications (treatment interruptions and/or reductions) for hematological AEs in patients with bone marrow involvement of their disease. Abbreviations: AE, adverse event.	

Table 8 Suggested Dose Modification Scheme for NX-5948-related non-hematological AEs in Escalation Cohorts and Expansion Cohorts

NX-5948-related non-hematologic AEs	Action to be Taken with NX-5948
Grade 2 – *any occurrence*	No action with study drug.

NX-5948-related non-hematologic AEs	Action to be Taken with NX-5948

Grade 3 – *first occurrence of AE* [a]	Action with study drug:
	• Interrupt treatment:
	• For transient, isolated laboratory abnormalities with no clinical sequalae:
	  o If the AE resolves to Grade ≤ 2 or baseline grade, whichever is higher, within 10 days, resume study drug at a lower dose.
	  o If AE does not resolve to Grade ≤ 2 or baseline grade, whichever is higher, within 10 days, permanently discontinue study drug.
	• For all other non-hematologic AEs
	  o If the AE resolves to Grade ≤ 1 or baseline grade, whichever is higher, within 10 days, resume study drug at a lower dose.
	  o If AE does not resolve to Grade ≤ 1 or baseline grade, whichever is higher, within 10 days, permanently discontinue study drug.

Grade 3 – *second occurrence of AE*	Action with study drug: • Permanently discontinue study drug
Grade 4 – *any occurrence of AE*	Action with study drug: • Permanently discontinue study drug
Abbreviations: AE, adverse event; QTc, QT interval corrected. [a] In the case of patients experiencing a Grade 3 cardiac AE, sites may consider evaluation by a cardiologist prior to rechallenge with NX-5948. For EU only: cardiology evaluation is required prior to NX-5948 rechallenge in patients with Grade 3 prolonged QTc.	

6.8. Prior and Concomitant Medications

All prescription and over-the-counter medications, herbal/homeopathic remedies, and nutritional supplements taken by a patient during the 28 days prior to Cycle 1 Day 1 will be recorded on the designated CRF. At a minimum, the drug generic name, dose amount, route of administration, start date, and stop date will be recorded in the source documents and the CRF. The Investigator (or other treating physicians) may prescribe additional medications during the study, as long as the prescribed medication is not prohibited by the protocol. In the event of an emergency, any required medications may be prescribed without prior approval, but the Medical Monitor or designee must be notified of the use of any contraindicated medications immediately thereafter. Any concomitant medications added or discontinued during the study should be recorded on the applicable CRF, including indication, and start and end dates of treatment, as applicable.

6.8.1. Permitted Concomitant Medication

Supportive medications may be provided prophylactically or therapeutically at the Investigator's discretion. Growth factors may be given during the trial at the discretion of the Investigator. In addition, growth factors are to be prescribed per institutional policy, and in alignment with American Society of Clinical Oncology 2006 Recommendations for the Use of White Blood Cell Growth Factors (Smith 2006).

Allowed medications may include oral anti-emetics, such as prednisone (up to 20 mg daily for non-CNSL indications and up to 40 mg for CNSL), metoclopramide or granisetron, and anti-diarrheal medications, such as loperamide hydrochloride. Bisphosphonates and the mAb denosumab are also permitted for treatment, or prophylaxis, of bone metastases as per local standards of care.

Medications known to be substrates of the organic cation antiporter: multidrug and toxin extrusion 1 (MATE1) should be used with caution in patients receiving NX-5948. NX-5948 may lead to altered PK of these medications (increased levels) via inhibition of MATE1 activity. MATE1 substrates include, but are not limited to, the following: metformin, cimetidine, acyclovir, ganciclovir, quinidine, lamivudine, oxaliplatin, procainamide, guanidine, and topotecan. The Medical Monitor should be notified prior to initiation of or upon changes in the dose of MATE1 substrates. Please contact the Medical Monitor if any questions or concerns arise regarding the use of this class of medications.

External beam radiotherapy will be allowed, following consultation with the Medical Monitor, for the treatment of isolated non-target bone or soft tissue lesions for no more than 5 fractions (unless more are approved in advance by the Medical Monitor) to reduce fracture risk or to relieve symptoms.

Intrathecal chemotherapy is allowed as determined by Investigator for indications with CNS involvement and given per institutional guidelines.

6.8.2. Prohibited Concomitant Medication

Patients may not receive any other systemic anti-cancer therapy that is directed at the tumor under study while on study, inclusive of investigational therapies and radiotherapy (focal palliative radiation is permitted following discussion with the Medical Monitor). Bone marrow transplant is prohibited.

The following medications are prohibited prior to the first dose of study medication and during the study.

- Radiotherapy within 2 weeks of planned start of study drug (excluding limited palliative radiation).
- Prior systemic chemotherapy within 2 weeks of planned start of study drug.

- Prior anti-cancer mAb therapy within 4 weeks of planned start of study drug or within 16 weeks for patients enrolling in Cohort 16.
- Prior small molecule therapy within 5 half-lives or 2 weeks (whichever is shorter) of planned start of study drug.
- Autologous or allogeneic stem cell transplant within 100 days prior to planned start of study drug.
- Anti-GVHD treatment or prophylaxis within 60 days prior to planned start of study drug.
- CAR T-cell therapy within 100 days prior to start of study drug (within 60 days prior to start of study drug for Phase 1b).
- Ongoing treatment with warfarin or an equivalent vitamin K antagonist within 168 hours prior to start of study drug.
- Vaccination with a live vaccine within 28 days prior to the first dose of study drug and at any time on study.
- Any vaccination within 14 days prior to first dose of study drug and during the DLT period (Phase 1a) or Cycle 1 (Phase 1b).
- Use of systemic corticosteroids within 7 days prior to first dose of study drug, > 20 mg/day prednisone or equivalent for patients without CNSL, or > 40 mg/day prednisone or equivalent for patients with CNSL, except for prophylaxis for radiodiagnostic contrast reactions. Patients with CNSL using > 20 mg/day prednisone or equivalent must be clinically stable at that dose for 7 days.
- Use of immunosuppressive drugs other than systemic corticosteroids within 60 days prior to first dose of study drug.
- Use of strong CYP3A inducers or inhibitors within 14 days or 5 half-lives, whichever is longer, prior to first dose of study drug.
- Use of moderate CYP3A inducers or inhibitors within 168 hours or 5 half-lives, whichever is longer, prior to first dose of study drug. (In the event that a patient must receive time-limited therapy with a moderate or strong CYP3A inhibitor or inducer for treatment of an emergent condition please notify Sponsor and manage as per protocol Section 6.8.2.2.)
- Use of inhibitors of P-glycoprotein or BCRP transporters within 48 hours prior to first dose of study drug. If a patient must receive time-limited therapy with inhibitors of P-glycoprotein or BCRP transporters while on study treatment, the time between the last dose of that substance and

study drug administration must be at least 48 hours. See Section 6.8.2.3 for examples of prohibited substances in this category.

- Use of a proton pump inhibitor (prescription or over-the-counter) within 168 hours prior to first dose of study drug. See Section 6.8.2.4 for allowed use of alternative agents.
- Use of biotin (ie, vitamin B7) or supplements containing biotin higher than the daily adequate intake of 30 µg within 72 hours prior to Screening and during study treatment (NIH 2022). A total daily dose of ≤ 30 µg is allowed.
- Dual treatment with an anticoagulant and antiplatelet agent within 2 weeks of treatment initiation and while on study.

6.8.2.1. Anticoagulant/Antiplatelet Medication

Treatment with warfarin or an equivalent vitamin K antagonist is prohibited within 168 hours of starting study drug and while on study. Dual treatment with an anticoagulant and antiplatelet agent is prohibited within 2 weeks of treatment initiation and while on study. If initiation of anticoagulation is clinically indicated while the patient is on study treatment, please notify the Sponsor and discuss with the Medical Monitor.

6.8.2.2. CYP3A Strong or Moderate Inducers and Inhibitors

Patients may not be treated with substances that are known strong or moderate inducers or inhibitors of CYP3A (see Table 9 below), including drugs, herbs, nutritional supplements, or grapefruit- or Seville orange-containing products as they may alter the PK of NX-5948. If the patient is taking any of these restricted substances at the time of Screening, then the time between the last dose or consumption of that substance and study drug administration must be at least 14 days or 5 half-lives, whichever is longer, for strong inhibitors or inducers; and at least 168 hours or 5 half-lives, whichever is longer, for moderate inhibitors or inducers.

In the event that a patient must receive <u>time-limited</u> therapy with a moderate or strong CYP3A inhibitor or inducer for treatment of an emergent condition, please notify Sponsor and manage as follows:

<u>CYP3A inhibitors</u>:

All Cycles:

- Hold NX-5948 during treatment and for
  - <u>Strong inhibitors</u>: an additional 14 days or 5 half-lives whichever is **longer after last dose of inhibitor**.
  - <u>Moderate inhibitors</u>: an additional 168 hours or 5 half-lives whichever is **longer after last dose of inhibitor**.

CYP3A inducers:

Cycles 1 and 2:

- Hold NX-5948 during treatment and for
  - Strong Inducers: an additional 14 days or 5 half-lives whichever is **longer after last dose of inducer**.
  - Moderate inducers: an additional 168 hours or 5 half-lives whichever is **longer after last dose of inducer**.

Cycle 3 and later:

- Continue NX-5948 during treatment.
- No washout required.

Table 9 Examples of Prohibited CYP450 Enzyme Interaction Medications[a]

CYP450 Enzyme	Medication Name
Strong CYP3A Inducers	carbamazepine, ethotoin, mephenytoin, phenobarbital, phenytoin, primidone, rifabutin, rifampin, and St. John's Wort
Moderate CYP3A Inducers	bosentan, efavirenz, etravirine, phenobarbital, and primidone
Strong CYP3A Inhibitors	amprenavir, ataznavir, boceprevir, clarithromycin, conivaptan, fosamprenavir, indinavir, itraconazole, ketoconazole, lopinavir, nefazodone, posaconazole, ritonavir, saquinavir, telaprevir, telithromycin, and voriconazole
Moderate CYP3A Inhibitors	aprepitant, ciprofloxacin, conivaptan, crizotinib, cyclosporine, diltiazem, dronedarone, erythromycin, fluconazole, fluvoxamine, imatinib, tofisopam, and verapamil

a. This is not intended to be an exhaustive list; the Investigator should consider each concomitant medication and potential for drug-drug interactions. For reference, please consult the US Food and Drug Administration (FDA) Drug Development and Drug Interactions Table of Substrates, Inhibitors and Inducers at: https://www.fda.gov/drugs/drug-interactions-labeling/drug-development-and-drug-interactions-table-substrates-inhibitors-and-inducers. For additional information, please consult the Flockhart Table of drug interactions available at the following Indiana University Department of Medicine Clinical Pharmacology website at: https://drug-interactions.medicine.iu.edu/MainTable.aspx or the University of Washington Drug Interaction Database at: https://www.druginteractionsolutions.org/solutions/drug-interaction-database/.

6.8.2.3. Inhibitors of P-glycoprotein and BCRP

Patients may not be treated with any substances at doses at which they are clinically relevant inhibitors of P-glycoprotein or BCRP (see Table 10 below), including drugs, herbs, or nutritional supplements as they may alter the PK of NX-5948. If the patient is taking any of

these restricted substances at the time of Screening or administered any of these restricted substances while on study treatment, the time between the last dose of that substance and study drug administration must be at least 48 hours.

Table 10 Examples of Prohibited Transporter Inhibitor Medications[a]

Medication Class	Medication Name
P-glycoprotein Inhibitors	
Anti-anginal	ranolazine
Anti-arrhythmic	amiodarone, dronedarone
Anti-infectives	azithromycin, clarithromycin, erythromycin, gentamicin, tetracycline
Anti-fungal	ketoconazole, itraconazole
Anti-hypertensive	carvedilol, diltiazem, verapamil
Hepatitis C	glecaprevir/pibrentasvir, sofosbuvir/velpatasvir/voxilaprevir
HIV	cobicistat, indinavir, lopinavir/ritonavir
Immunosuppressant	cyclosporine
Kinase inhibitor	lapatinib, vemurafenib
BCRP Inhibitors	
Hematopoietic agents	eltrombopag
Herbal supplement	curcumin
Immunosuppressant	cyclosporine

a. This is not intended to be an exhaustive list; the Investigator should consider each concomitant medication and potential for drug-drug interactions. For reference, please consult the US FDA Drug Development and Drug Interactions Table of Substrates, Inhibitors and Inducers at: https://www.fda.gov/drugs/drug-interactions-labeling/drug-development-and-drug-interactions-table-substrates-inhibitors-and-inducers. For additional information, please consult the Flockhart Table of drug interactions at: https://drug-interactions.medicine.iu.edu/MainTable.aspx or the University of Washington Drug Interaction Database at: https://www.druginteractionsolutions.org/solutions/drug-interaction-database/. For further information on drug transporters, please consult the SOLVO Biotechnology Drug Interaction Table of Human Transporters at: https://www.solvobiotech.com/knowledge-center/transporters-a-z.

6.8.2.4. Proton Pump Inhibitors and Other Acid-reducing Agents

NX-5948 exhibits pH dependent solubility and therefore attention should be paid to concomitant dosing of medications that elevate the pH of the stomach to treat gastroesophageal reflux disease. If a patient requires treatment for gastroesophageal reflux disease, then a holistic approach should be applied if possible (eg, diet, etc.). If treatment is needed, the following regimens will minimize the effects of absorption.

Patients may not be treated with any proton pump inhibitors (prescription or over-the-counter) as NX-5948 exhibits pH dependent solubility. If the patient is taking any of these restricted substances at the time of Screening, the time between the last dose of that substance and study drug administration must be at least 168 hours.

Alternative acid-reducing agents may be used:

- H2 antagonists (eg, famotidine) are permitted at least 12 hours prior to study drug administration and at least 4 hours after study drug administration;
- Antacids (eg, calcium carbonate) are permitted at least 4 hours prior to or 4 hours following study drug administration; and
- Alginates are prohibited at least 4 hours prior to or 4 hours following study drug administration.

6.8.2.5. High Dose Biotin Supplements

Patients should avoid taking high doses (> 30 µg/day) of biotin-containing supplements (eg, vitamin B7, CoEnzyme R, vitamin H).

In November 2019, the US Food and Drug Administration (FDA) issued a safety communication on biotin interference intended for healthcare professionals (FDA 2019). Briefly, the communication made the following recommendations:

- Talk to your patients about any biotin supplements they may be taking, including supplements marketed for hair, skin, and nail growth.
- Be aware that many laboratory tests, including but not limited to cardiovascular diagnostic tests and hormone tests that use biotin technology are potentially affected, and incorrect test results may be generated if there is biotin in the patient's specimen.
- Communicate with the laboratory conducting the testing if your patient is taking biotin.
- If a laboratory test result does not match the clinical presentation of your patient, consider biotin interference as a possible source of error.
- Know that biotin is found in multivitamins, including prenatal multivitamins, biotin supplements, and dietary supplements for hair, skin, and nail growth in levels that may interfere with laboratory tests.
- Report to the laboratory test manufacturer and the FDA if you become aware of a patient experiencing an AE following potentially incorrect laboratory test results due to biotin interference.

A threshold "high level" biotin dose has not yet been defined by the FDA.

Study Investigators should discuss with study patients about any biotin supplements they may be taking. For patients taking biotin supplements at doses > 30 µg/day, the Investigator should assess if it is possible to discontinue use of these supplements. In cases where the

Investigator believes a patient is taking high doses of biotin and discontinuation of this supplementation is not in the best interest of the patient, the Investigator should contact the Medical Monitor to assess the potential impact on study laboratory test results.

6.8.3. Tumor Lysis Syndrome Prophylaxis

For patients with DLBCL, CLL/SLL, and MCL, TLS prophylaxis is required as per institutional standards.

6.8.4. Dietary or Other Protocol Restrictions

NX-5948 capsules should be dosed orally with water as defined for each cohort. As the effect of food on absorption is currently unknown, patients should fast for at least 2 hours before and 2 hours after taking NX-5948, except for on intensive PK collection days (Cycle 1 Day 1 and Cycle 2 Day 1) when patients should fast for a minimum 8 hours before and 2 hours after taking the dose of NX-5948. There are no other dietary restrictions or specific requirements in this study. However, a fasting glucose test is required at Screening or prior to dosing on Cycle 1 Day 1. In the event of a clinically significant elevated glucose level on-study under non-fasted conditions, a fasted sample should be obtained.

6.9. Other Protocol-specified Materials

Central laboratory kits will be provided for sample collection, shipment, and storage for PK and PD analyses.

7. STUDY VISIT SCHEDULE AND PROCEDURES

The schedule of visits and timing of procedures are detailed in Section 15.3. The acceptable visit windows are noted in the tables.

7.1. Patient Entry Procedures

The study will be discussed with the patient, and a patient wishing to participate must give informed consent before the patient may be evaluated for eligibility. After informed consent has been obtained, the Investigator determines whether the patient is eligible for enrollment, including meeting the inclusion criteria for the study (see Section 5.1.1) and establishing that none of the exclusion criteria apply (see Section 5.1.2). All Screening procedure results and relevant medical history must be available and reviewed by the Investigator before eligibility can be determined. No eligibility waivers will be granted by the Sponsor.

Prior to enrollment, the study site personnel will provide the results of the Screening evaluations to the Sponsor/Sponsor's representative by submitting an Eligibility Review Packet (ERP) via secure transmission, for review by the medical monitor or designee during the screening period. The duration and frequency of this process will be defined in the medical monitoring plan and communicated to the sites.

The ERP will include information to support that the patient meets the inclusion/exclusion criteria. This information will include, but is not limited to, patient medical records, significant medical history, concomitant medications, underlying cancer disease history, pathology diagnosis including any relevant molecular and/or cytogenetic studies, information from relevant imaging reports (eg, CT, PET-CT, MRI, as applicable), and prior anti-cancer therapies for the underlying malignancy.

Per the patient privacy/confidentiality provisions in this protocol, all materials in the ERP should be coded with the patient's study ID number, and all reports should be redacted to protect the patient's personal data in accordance with local laws and regulations and Nurix requirements (eg, direct identifiers such as names, initials, and full date of birth should never be shared, and personal data not specifically required by the protocol should never be included or otherwise provided).

The medical monitor or designee will review the ERP and confirm the assigned treatment cohort and/or dose level randomization if it is determined that the Investigator has accurately assessed that all inclusion criteria have been met and that none of the exclusion criteria apply; the sponsor representative will communicate their assessment to the site via email. If the medical monitor raises concerns about the eligibility of a potential patient or the contents of the ERP, the Investigator is responsible for reviewing and responding to the medical monitor's concerns and/or correcting the contents of the ERP, as appropriate. A patient may not start on study drug until the site receives the medical monitor's or designee's confirmation of patient eligibility. Each site is responsible for maintaining the record of the review correspondence according to the site's policy. The sponsor's medical monitor will support the Investigator and/or site staff by answering any queries or questions relating to protocol eligibility criteria.

Sponsor verification of patient eligibility will also be managed by way of source data verification in accordance with ICH E6.

7.2. Screening, Treatment, and Follow-up Periods

All patients must provide written informed consent. During the consent process, the person obtaining consent must inform the patient of all elements of the study. No protocol-specific procedures, including Screening procedures, are to be performed until the patient has signed and dated an EC/IRB-approved ICF. The Screening period begins with the signing and dating of the ICF.

Screening procedures are to be performed within 28 days prior to Cycle 1 Day 1 unless otherwise specified in the Schedule of Assessments (Section 15.3). All scheduled study visits (with ± windows) are described in Section 15.3.

The EOT visit will take place within 3 days after the last dose of NX-5948 whenever possible.

The 15-day and 30-day Post-Treatment Safety Follow-Up visits will take place approximately 15 (± 5) days and 30 (± 7) days, respectively, after last dose of study drug or prior to starting any new anti-cancer therapy, whichever occurs first (see Section 4.4 for reasons for discontinuation or withdrawal of a patient from the study). If a new therapy is

started immediately after the last dose of NX-5948, the 15-day and 30-day Post-Treatment Safety Follow-up visits may not be completed.

Patients will be followed every 12 weeks after the 30-day Post-Treatment Safety Follow-up visit until death, withdrawal of consent, or loss to follow-up. Survival follow-up contact can be via clinic visit, phone call to the patient or referring physician, or other method deemed appropriate by the site, and should assess survival, disease progression, and subsequent therapy.

Patients who discontinue study treatment without documented disease progression should have their disease monitored by B-cell malignancy-specific response criteria per SoC until disease progression or starting any new anti-cancer therapy, whichever occurs first. Refer to the Schedule of Assessments in Section 15.3.

7.3. Unscheduled Visits

Patients may be brought into the clinic for unscheduled visits as deemed appropriate by the Investigator.

7.4. COVID-19 Public Health Emergency Impact on Protocol Assessments

During study conduct, challenges may arise, for example, from quarantines, site closures, travel limitations, interruptions to the supply chain for the investigational product, or other considerations if site personnel or trial patients become infected with the SARS-Cov-2 virus and develop coronavirus disease 2019 (COVID-19). These challenges may lead to difficulties in meeting protocol-specified procedures, including administration of the study drug or adhering to protocol-mandated visits and laboratory/diagnostic testing. During the course of study conduct, protocol modifications may be required, and there may be unavoidable protocol deviations due to COVID-19 illness and/or COVID-19 public health control measures. In such cases, these incidences should be documented.

Since trial participants potentially may not be able to come to the investigational site for protocol-specified visits, alternative methods for safety assessments (eg, phone contact, virtual visit, telemedicine, alternative location for assessment, including local labs or imaging centers) may be implemented when necessary and feasible, and would be sufficient to assure the safety of trial participants.

7.5. Description of Procedures

7.5.1. Safety Procedures

The initial safety analysis will be a comprehensive evaluation of AEs and/or toxicity per patient, presented by dose, tumor type cohort, and overall, based on:

- DLT(s) as judged by the SRC.

- TEAEs by NCI CTCAE v5.0, including treatment-emergent SAEs, Grade 3, 4, 5 TEAEs, TEAEs leading to study drug discontinuation, deaths due to TEAEs, and all deaths.
- Physical examination and vital sign findings.
- Occurrence of cumulative TEAEs, treatment-emergent SAEs, and Grade 3, 4, 5 TEAEs.
- Clinical chemistry, hematology/coagulation, and urine analysis tests.
- ECG results.
- Concomitant medication use.

7.5.1.1. Adverse Event Assessments

The determination, evaluation, and reporting of AEs will be performed as outlined in Section 8.

7.5.1.2. Medical History

Medical history includes clinically significant diseases, surgeries, cancer history, and all medications (eg, prescription drugs, over-the-counter drugs, herbal/homeopathic remedies, nutritional supplements) used by the patient within 28 days prior to the first dose of study drug on Cycle 1 Day 1 and history of treatment for the primary diagnosis, including prior systemic, radiation treatment, and surgical treatment. Date of last prior cancer treatment must be documented. Radiographic studies performed prior to study entry may be collected for review by the Investigator. Molecular cytogenetics data from testing of peripheral blood lymphocytes will be collected as part of cancer history.

7.5.1.3. ECOG Performance Status

An assessment of performance status will be performed using the ECOG performance status scale to quantify the patient's general well-being and ability to perform activities of daily life (Section 15.1). For patients with PCNSL, an assessment based on Karnofsky performance score (Section 15.6) will be performed at Screening.

7.5.1.4. Weight, Height, and Demographic Information

Height and weight will be measured during Screening (within 28 days prior to the first dose of study drug). Weight will be obtained at additional time points as specified in the Schedule of Assessments (Section 15.3). Patients should have weight and height measured while wearing indoor clothing and with shoes removed.

Demographic data, including sex, year of birth and/or age, and race/ethnicity, will be collected. Demographic factors such as age, sex, race, and ethnicity could influence the effects (safety and efficacy) of medicines and the risk/benefit assessment in different populations. Race and ethnicity

data are collected in accordance with ICH E5 and ICH E17. This information is used to understand whether race and ethnicity may have a role in influencing the PK, safety, and/or efficacy of the study drug. Results from race and ethnicity evaluations or other demographic based analyses will be included in approved labeling for the marketed product to provide guidance, as applicable, on the safety and efficacy variations (if any) linked to certain patient subpopulations as well as any potential dose adjustments for those populations, as supported by the data. Collection of race and ethnicity from the study participants is critical to support the effect of race and ethnicity on PK, safety and/or efficacy.

7.5.1.5. Vital Signs

Vital sign measurements include seated measurement of diastolic and systolic blood pressure, heart rate, respiratory rate, pulse oximetry, and temperature. For each patient, temperature should be recorded using the same modality throughout the entire study. Patients should be sitting at rest for 5 minutes before blood pressure is measured. When post-dose vital sign measurements are scheduled at the same time as an ECG and blood draw, the vital signs will be obtained prior to the ECG and blood draw, and the blood draw will be collected at the scheduled time. Pulse oximetry will be taken at all vital sign assessments during Cycle 1 Day 1, and pre-dose on Day 1 of each subsequent cycle. The schedule for vital sign assessment and the required observation period following NX-5948 administration are described in Section 15.3.

7.5.1.6. Physical Exams

A complete physical examination will occur at the Screening visit and will include head, ears, eyes, nose, mouth, thyroid, skin, heart, and lung examinations, lymph nodes, gastrointestinal, skeletal, and neurological systems. Other weekly physical examinations as specified in the Schedule of Assessments (Section 15.3) may be targeted, at the discretion of the Investigator, to focus on signs and symptoms reported by the patient to assess for clinically significant changes from baseline. Safety assessments, including the physical exam, may be performed up to 96 hours prior to Day 1 dosing of each cycle beginning with Cycle 2.

For patients with CNSL, a detailed ophthalmologic exam should be performed at Screening per International PCNSL Collaboration Group recommendations. This should include a dilated fundoscopic examination to evaluate vitreous, retinal, or optic nerve involvement. Ophthalmologic exams performed within 6 weeks of Cycle 1 Day 1 that were completed as SoC may be used for baseline. Follow-up ophthalmologic examinations are only required in patients with ocular disease at Screening or upon the development of new ocular symptoms.

7.5.1.7. Fitzpatrick Skin Color Assessment

The Fitzpatrick skin color assessment (Figure 6) will be administered during Screening as noted in the Schedule of Assessments (Section 15.3).

7.5.1.8. Mini Mental State Examination, 2nd Edition: Brief Version (MMSE-2: BV)

The MMSE-2: BV (Folstein 2010) should be administered and scored during Screening and at subsequent disease assessments as specified in the Schedule of Assessments (Section 15.3). Sites will be provided with paper MMSE-2: BV questionnaires (both a red form and a blue form) and with an MMSE-2 User's Manual. The person responsible for administering the exam should alternate between the red form and the blue form each time it is administered. Site staff should be trained by reviewing the MMSE-2 User's Manual and documenting their training prior to administering the test to patients.

7.5.1.9. Clinical Laboratory Assessments

Specific visits for obtaining clinical laboratory assessment samples are outlined in the Schedule of Assessments (Section 15.3). Laboratory samples should be collected at the clinical study site, if possible. Reference ranges for all safety parameters will be provided by the site to the Sponsor. The scheduled clinical laboratory tests are listed in Table 11 and will be performed by local laboratory. Patients should be assessed for eligibility prior to dosing on Cycle 1 Day 1. To accommodate logistics, safety labs may be obtained within 96 hours of first dose on Cycle 1 Day 1 and assessed for eligibility and do not need to be repeated on Cycle 1 Day 1. However, Cycle 1 Day 1 hematology should be completed for baseline establishment but does not need to be evaluated for eligibility purposes if previously collected within 96 hours of first dose. Serologies may be obtained at any time during the Screening period and assessed for eligibility purposes (EBV, CMV, HIV, HBV/HCV). Serum amylase and lipase may be obtained at any time during the Screening period and assessed for eligibility purposes. Samples for Cycles 2 and beyond may be obtained up to 96 hours prior to dosing.

Table 11 Clinical Laboratory Tests

Hematology	**Clinical Chemistry**	**Other**
• Red blood cell count (RBC) • White blood cell count (WBC) with differential (including absolute neutrophils, basophils, eosinophils, lymphocytes, monocytes) • CD4 counts if a patient with HIV • Hemoglobin • Hematocrit • Platelet count • Mean corpuscular volume (MCV) • Reticulocyte count • Haptoglobin • Direct antiglobulin test (DAT; Cohort 16 CLL with secondary wAIHA; at Screening only for all other CLL/SLL and WM cohorts)	• Bicarbonate • Blood urea nitrogen (BUN) or urea • Calcium • Chloride • Creatinine • Estimated glomerular filtration rate (eGFR; CKD-EPI creatinine calculation; Section 15.5) • Glucose (fasting required at Screening or Cycle 1 Day 1 [pre-dose] only; fasting status is optional for all other visits) • Magnesium • Phosphorous or phosphate • Potassium • Sodium	• Pregnancy test • Serum IgM and serum IgM monoclonal protein by immunofixation (WM patients only) • Serum immunoglobulins (IgG, IgA, IgM) • Uric acid • Lactate dehydrogenase (LDH) • C-Reactive Protein • Lipase • Amylase • Thyroid function: thyroid stimulating hormone (TSH), if abnormal, obtain triiodothyronine (T3) and thyroxine (T4) (free or total) • β-2-microglobulin (CLL patients at Screening only) • CSF cytology and protein (CNSL patients undergoing CSF collection only)
Coagulation • INR • PTT or aPTT **Liver Function Tests (LFTs)** • ALT • AST • Albumin • Alkaline phosphatase • Total bilirubin • Indirect bilirubin • Total protein	**Viral Serology** • HIV (Type 1 and Type 2) • Hepatitis panel, according to Centers for Disease Control and Prevention criteria *(provision for PCR, as appropriate)* • HbcAb • HbsAg • Hepatitis B surface antibody (HbsAb) • HCV antibody • CMV (PCR or serologies) • EBV (PCR or serologies)	**Urinalysis** • Appearance • Color • pH • Specific gravity • Ketones • Leukocyte esterase • Protein • Glucose • Bilirubin or urobilinogen • Blood • Microscopic examination of specimen will be performed only if the results of the urinalysis dipstick evaluation are positive

In the event of a significant safety event (eg, DLT or study drug-related Grade ≥ 3 AE/SAE) throughout the study, an additional PK and PD flow cytometry sample should be taken as close to the event as possible.

Any abnormal test results determined to be clinically significant by the Investigator should be repeated (at the Investigator's discretion) until the cause of the abnormality is determined, the value returns to baseline or to within normal limits, or the Investigator determines that the abnormal value is no longer clinically significant. Patients who have an increase not only in transaminases but also elevated bilirubin will be required to return to the clinic for repeat testing within 3 days.

In the event of clinically significant elevated glucose under non-fasted conditions, a fasted sample should be obtained per institutional guidelines and repeated as clinically indicated.

In the event of clinical suspicion for *Aspergillus* infection, a galactomannan or β-D-Glucan test may be obtained and repeated as clinically indicated.

All abnormal clinical laboratory results should be reviewed by the Investigator and documentation of this review should be present in the site files. Each clinically significant laboratory result should be recorded as an AE; whenever possible, the diagnosis, if known, associated with abnormalities in clinical laboratory tests that are considered clinically significant by the Investigator will be recorded on the AE CRF.

7.5.1.10. 12-Lead Electrocardiogram

Single or triplicate 12-lead ECGs will be performed locally at the time points specified in the Schedule of Assessments (Section 15.3). Patients should rest in the supine or semi recumbent position for at least 5 minutes before each 12-lead ECG recording is started. ECGs should be recorded digitally and in triplicate (approximately 10 seconds per ECG over a 5-minute period) at Screening, Cycle 1 Day 1, Cycle 1 Day 8, and Cycle 2 Day 1. On subsequent cycles, a pre-dose (within 30 minutes prior to dosing), single tracing ECG will be performed. When a post-dose ECG is scheduled at the same time as a blood draw, the ECG will be obtained prior to the blood draw; the blood draw will be collected at the scheduled time. ECGs will be performed using the site's own machine and read locally by a qualified physician. Results will be collected and included in the electronic CRF.

If patients have a prolonged QT interval corrected (QTc; > 500 msec) measured by Fridericia's formula [QTcF = QT/(RR^0.33)] on study treatment, the ECG should be repeated and confirmed. Patients who are taking concomitant medications that are known to prolong the QTc interval should have a repeat ECG after stopping the medication. Patients at Screening should have QTcF < 450 msec. For patients with pacemaker or bundle branch block, the QTcF should be < 475 msec. All abnormal ECG findings must be documented by the Investigator as clinically significant or not.

Repeat ECGs should be conducted to evaluate other abnormal findings as clinically indicated.

7.5.1.11. Echocardiogram or Multiple Gated Acquisition Scan

An echocardiogram or multiple gated acquisition scan (MUGA) will be performed at Screening, and on study if clinically indicated. Echocardiogram or MUGA done within 6 weeks of Cycle 1 Day 1 can be used for Screening.

7.5.1.12. Pregnancy Testing

For WOCBP, a serum pregnancy test is required at Screening. Female patients with a positive pregnancy test at Screening do not meet eligibility criteria for enrollment. A urine pregnancy test will be performed and resulted (by local laboratory) within 24 hours of first dose on Cycle 1 Day 1 prior to initial dosing of study medication. If positive, a confirmatory serum pregnancy test will be performed. Additional pregnancy tests will be performed every cycle (4 weeks) on study, and if clinically indicated at any visit in which pregnancy status is in question. If treatment with study drug is stopped for 14 days or more, WOCBP should have a negative pregnancy test prior to restarting drug. For WOCBP, pregnancy tests are also required at the EOT and 30-day Post-Treatment Safety Follow-up visits.

Refer to Section 8.6 for details on the reporting procedures to follow in the event of pregnancy.

7.5.1.13. Patient Reported Outcomes

A PRO (FACIT-GP5) will be used to evaluate patient tolerability to study treatment for all patient cohorts enrolled in Phase 1b Part 1. The tool will be administered in paper format by study site personnel (eg, study coordinator or research nurse) in accordance with the timepoints specified in the Schedule of Assessments (Table 16). If feasible, patients should complete the FACIT-GP5 question before study drug administration and before performing any other study procedures on that visit day.

FACIT-GP5 is a summary measure of the overall impact of treatment toxicity, based upon its association with the number and degree of AEs in clinical trials. The single item has demonstrated a significant relationship to overall quality of life and has been shown to be associated with clinician-reported AEs (Pearman 2018). All patients will respond to the question “I am bothered by the side effects of treatment” over the past 7 days utilizing a 5-point scale (0=not at all, 1 =a little bit, 2=somewhat, 3=quite a bit, 4=very much).

7.5.2. Efficacy Procedures

For all patients, disease assessments will be performed at Weeks 8, 16, 24, and thereafter every 12 weeks (–7 day window), and at time of clinical suspicion of disease progression. PCNSL/SCNSL patients will undergo an additional disease assessment between Cycle 1 Day 22 and Cycle 1 Day 28. Disease assessments should be obtained prior to Day 1 of the next cycle. During Screening for patients with WM, a bone marrow biopsy and/or aspirate will be required.

During study treatment, for patients with CLL/SLL and WM, a bone marrow biopsy and/or aspirate will be performed to confirm CR, as required per iwCLL and WM disease-specific response criteria, respectively.

For patients with DLBCL with disease determined to be FDG-avid by PET-CT at Screening, a bone marrow biopsy and/or aspirate will not be required to confirm a complete metabolic response/CR. However, because the sensitivity of PET-CT for bone marrow involvement is less precise for particular lymphomas, patients with MCL, MZL, and FL should obtain a bone marrow biopsy and/or aspirate to confirm CMR/CR as recommended by the Lugano response criteria for malignant lymphomas (Barrington 2014). Marrow biopsies may be repeated in patients with persisting cytopenia(s) to uncover disease vs treatment-related causes. Please refer to Section 15.4 for detailed response criteria for the diseases included in this study.

Disease assessment should occur at the timepoints indicated (also refer to Schedule of Assessments in Table 16) until patient has documented progressive disease, initiates another anti-cancer therapy, death, or withdraws from the study, whichever occurs first.

ORR, CR rate (CR/CRi for patients with CLL), time to first response, DOR, PFS, and time to next therapy will be assessed based on the following B-cell malignancy indication-specific criteria and require the following radiographic imaging modality(s) in addition to other laboratory tests and/or diagnostic procedures that may be required per respective disease-specific response criteria for assessment of overall disease burden at baseline and during the study:

- iwCLL criteria (Hallek 2018) for CLL/SLL.
  - CT of diagnostic quality (eg, intravenous contrast recommended) of chest, abdomen, and pelvis should be performed at Screening and during study treatment at the timepoints specified.
  - In the event that a patient is unable to undergo diagnostic quality CT scan due to tolerability concerns (e.g. contrast allergy), MRI may be utilized as an alternative with a consistent modality continued while enrolled on study
  - PET-CT is **not** indicated per iwCLL and should not be utilized for disease assessments
- Lugano Classification of Lymphoma response criteria (Cheson 2014) for DLBCL, FL, MCL, and MZL.
  - PET-CT from skull base to mid-thigh (CT portion of diagnostic quality with use of intravenous contrast recommended) should be performed at Screening, and if result is determined to be FDG-avid, during study treatment at the timepoints specified.
  - CT of diagnostic quality of neck, chest, abdomen, and pelvis (eg, intravenous contrast recommended) should be performed in lieu of PET-CT if disease is determined not to be FDG-avid at Screening and during study treatment at the timepoints specified.

- WM response criteria (Owen 2013) for WM.
  - CT with or without contrast should be performed at Screening to assess for lymphadenopathy, splenomegaly, and/or extranodal disease.
  - If measurable extramedullary disease is present at baseline, imaging studies should be performed during study treatment to confirm response category when CR, very good PR or PR are otherwise met at any particular disease assessment timepoint.
- International PCNSL Collaboration Group criteria (Abrey 2005) for PCNSL.
  - Brain MRI with gadolinium enhancement should be performed at Screening and during study treatment at the timepoints specified.
- If feasible, OS will be calculated using a Kaplan-Meier method.

For PCNSL and SCNSL patients:

- Patients with PCSNL/SCNSL will have their first disease assessment between Cycle 1 Day 22 and Cycle 1 Day 28; and will follow the same disease assessment schedule as other indications thereafter.
- Disease assessment should include a detailed ophthalmologic examination (recommend to be performed by neuro-ophthalmologist) at Screening and during study treatment only in those patients with documented ocular lymphoma at baseline or development of new ocular symptoms. Screening lumbar punctures with CSF cytology are required for baseline disease assessment. In the case of patients with positive CSF cytology at study entry, lumbar puncture (or Ommaya reservoir, if available) with CSF cytology should be performed as part of each disease assessment until there is no evidence of disease in the CSF.
- SCNSL patients should complete the appropriate imaging modality for response assessment of their systemic disease during Screening and if systemic disease is present at the relevant time points thereafter. For SCNSL patients with isolated CNS involvement (ie, no evidence of systemic disease at baseline), radiographic assessment of systemic recurrence or progression may be based on the clinical judgment of the Investigator.

For Cohort 16 (CLL with secondary wAIHA):

- AIHA response criteria (Barcellini 2012).
  - CR^{AIHA} is defined as hemoglobin >12 g/dL and normalization of all hemolytic markers

  - PR^{AIHA} is defined as hemoglobin = 10-12 g/dL or at least 2 g/dL increase in hemoglobin, and no transfusion requirement
  - Overall AIHA response is defined as CR^{AIHA} + PR^{AIHA}
- Maintained response is defined as maintenance of response for more than 6 months after the beginning of treatment with the discontinuation of concomitant immunosuppressive therapy (Peñalver 2010)
- Duration of response (DOR) is defined as time from initial response (CR^{AIHA} or PR^{AIHA}) until relapse or death (Jaime-Perez 2019)
- Time to response as analyzed by Kaplan-Meier method (Jaime-Perez 2019)
- Relapse-free survival is defined as time from start of treatment until death or relapse of AIHA as analyzed per Kaplan-Meier method (Barcellini 2012)
- Frequency of packed red blood cell (PRBC) transfusion while receiving NX-5948 will be measured

Imaging studies may be centrally reviewed. If central review is initiated, details will be provided in an Imaging Manual.

7.5.3. Pharmacokinetic Assessments

Blood samples will be collected for determination of the concentrations of NX-5948 in plasma as described in Section 15.3. The Cycle 1 and Cycle 2 sampling will be used for a full PK profile; subsequent sampling will be used to determine trough steady-state concentrations of NX-5948 and to characterize single-dose/multiple-dose PK, dose proportionality, and time-dependence of PK. In addition, PK samples for trough levels of NX-5948 should be collected prior to any dose modifications. Furthermore, exploratory analysis of metabolites of NX-5948 may be conducted in intensive PK plasma samples. Refer to the Study Laboratory Manual for more information on sample volumes, processing, and shipping.

CSF samples will be collected from patients with CNSL at steady state for determination of the concentration of NX-5948 in the CNS, and to determine CSF:plasma concentration ratios. In the event that CSF is being collected for clinical purposes, additional unscheduled CSF samples may be collected, when feasible.

Note: In the event of a significant safety event (eg, DLT or study drug-related Grade ≥ 3 AE/SAE) throughout the study, an additional PK and PD flow cytometry sample should be taken as soon as possible after the onset of the event, if possible.

Additional samples of blood (up to 3 per week) may be drawn for analysis of NX-5948 levels if considered by the Investigator and Sponsor to be informative to understand safety or efficacy of NX-5948. In the event of a treatment interruption, PK samples should still be collected.

PK parameters will be derived using non-compartmental analysis, and will include C_{max}, T_{max}, half-life, $AUC_{0\text{-last}}$, $AUC_{0\text{-tau}}$, C_{min}, and accumulation ratio. Plasma to CSF concentration ratios

will also be calculated. Additional PK parameters may be derived as appropriate. In addition, the concentrations of NX-5948 in these samples may be used for population PK analysis, evaluation of the dose-exposure relationship, and evaluation of exposure-response relationships. These additional analyses may be reported separately.

7.5.4. Pharmacodynamic Assessments

PD biomarkers will be explored in blood of all patients at baseline and on study treatment at the timepoints described in Section 15.3. Lymph node biopsies will be taken to explore proof of NX-5948 mechanism, any PK/PD relationships, and any correlation between PD markers at sites of disease and peripheral blood. Lymph node biopsies are optional during Phase 1a dose escalation in DLT-evaluable cohorts. In Phase 1b, lymph node biopsies will be mandatory, if feasible, for up to 10 patients per safety expansion cohort (excluding WM cohorts), where 10 non-biopsy patients have been enrolled. For patients with WM, a bone marrow aspirate and biopsy should be utilized for these assays due to limited nodal involvement. Therefore, samples consisting of both a bone marrow aspirate and biopsy will be mandatory if feasible for up to 10 patients enrolled in WM cohorts in Phase 1b safety expansion cohorts.

CSF samples will be taken from all patients with CNS involvement (primary and secondary) at Screening, steady state, and EOT to evaluate mechanisms of response or resistance.

In the event of a treatment interruption, PD samples should still be collected.

Biomarkers of the BTK signaling and target engagement for example include, and are not limited to:

- FLOW: Flow cytometry to assess targeted protein degradation in circulating blood.
- PBMC: Evaluation of immune cell phenotypes and BCR signaling from circulating blood.
- Cytokines: Circulating cytokine and chemokine levels.
- NGS: Retrospective evaluation of gene expression, mutation status, and allele frequency.
- Biomarkers of the BTK signaling and resistance pathways, and target engagement.
- Exploratory PD circulating tumor (ct)DNA CSF (for patients with PCNSL or secondary CNS involvement).
    - A buccal swab will be collected at Screening for a baseline germline analysis required as a reference sample for the exploratory PD ctDNA analysis.

- Exploratory MRD (for patients with CLL/SLL only): A validated Laboratory Developed Test (LDT) flow cytometry assay will be used to evaluate disease burden of patients. Evaluation of NX-5948 in driving durability of response (ie, timing to undetectable disease and time to MRD relapse). MRD assessment is for exploratory purposes only.

8. REPORTING ADVERSE EVENTS

8.1. Adverse Events

For this protocol, a reportable AE is any untoward medical occurrence (eg, sign, symptom, illness, disease, or injury) in a patient who has taken study drug or other protocol-imposed intervention, regardless of attribution. This includes the following:

- AEs not previously observed in the patient that emerge during the course of the study.
- Pre-existing medical conditions judged by the Investigator to have worsened in severity or frequency or changed in character during the study.
- Complications that occur as a result of non-drug protocol-imposed interventions (eg, AEs related to protocol-specified Screening procedures).

A study drug-related AE is any AE for which there is a reasonable possibility that study drug caused the AE. “Reasonable possibility” means there is evidence to suggest a causal relationship between study drug and the AE.

Whenever possible, the AE term should be recorded as the diagnosis rather than the symptoms, signs, or a procedure.

Events that **do not** meet the definition of an AE include:

- Medical or surgical procedures (eg, surgery, endoscopy, tooth extraction, transfusion); however, new or worsening conditions that lead to procedures should be reported as AEs;
- Standard of care screening procedures not associated with the study (eg, routine screening mammography or colonoscopy);
- The cancer indication being studied, or expected progression of that cancer indication, even within 30 days of last dose and/or if serious criteria is met. Disease progression is to be recorded on specific tumor assessment pages rather than the AE page of the CRF; however, if there is uncertainty about whether the study drug contributed to or accelerated progression of the disease, then disease progression should be reported as

an AE and recorded on both the specific tumor assessment pages and the AE page of the CRF;

- Signs and symptoms solely due to expected progression of the underlying disease should not be reported as AEs; however, if there is uncertainty about whether the study drug contributed to signs and symptoms of disease progression, they may be reported as AEs;
- Abnormal laboratory results that the Investigator does not consider to be clinically significant are not AEs; however, if an abnormal laboratory result requires treatment (eg, transfusion) or results in study drug dose reduction, interruption, or discontinuation, it should be considered clinically significant and be reported as an AE;
- Pre-existing diseases or conditions present or detected before the start of study drug administration that do not worsen; and
- Situations where an untoward medical event has not occurred (eg, planned hospitalization for an elective procedure, with elective defined as known or planned at the time of signing of the informed consent).

The AE/SAE reporting period is described in Section 8.3.

If a non-serious AE remains unresolved at the conclusion of the study, the Investigator will assess whether continued follow-up of the AE is warranted, and the results of this assessment must be documented. Resolution is defined as the return to baseline status or stabilization of the condition with the expectation that it will remain chronic. The Investigator should follow all unresolved AEs until the events are resolved or stabilized, the patient is lost to follow-up, or it has been determined that the study drug or participation is not the cause of the AE. Outcome of AEs (with dates) should be documented on the appropriate CRF page(s) and in the patient's medical record.

The criteria for determining, and the reporting of, SAEs is provided in Section 8.2.

The Investigator responsible for the care of the patient or qualified designee will assess AEs for severity, for relationship to study drug, and for seriousness (refer to Section 8.2 for SAE definition).

Severity (as in mild, moderate, or severe) is not equivalent to seriousness, which is based on patient/event outcome or action criteria usually associated with events that pose a threat to a patient's life or functioning. Seriousness (not severity) serves as a guide for defining regulatory reporting obligations.

The Investigator will determine the severity of each event by using NCI CTCAE v.5.0. AEs that do not have a corresponding CTCAE term will be assessed according to the general guidelines for grading used in the NCI CTCAE v5.0 as described in Table 12.

Table 12 Adverse Event Severity Grading

Grade	Description
1	Mild: asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated.
2	Moderate: minimal, local, or non-invasive intervention indicated; limiting age-appropriate instrumental activities of daily living (ADL).[a]
3	Severe or medically significant but not immediately life-threatening: hospitalization or prolongation of hospitalization indicated; disabling; limiting self-care ADL.[b]
4	Life threatening or debilitating: consequences; urgent intervention indicated.
5	Death related to AE.

[a] Instrumental ADL refer to the following examples: preparing meals, shopping for groceries or clothes, using the telephone, managing money.

[b] Self-care ADL refer to the following examples: bathing, dressing and undressing, feeding oneself, using the toilet, taking medications, not bedridden.

The study Investigator responsible for the patient's care (or qualified designee) will assess causality of reportable AEs/SAEs. The causal attribution guidance in Table 13 will be applied.

Table 13 Causal Attribution Guidance

Is there a reasonable possibility that the AE/SAE was caused by the study drug based on facts (evidence) or arguments to suggest a causal relationship?	
YES (Possible, probable, or definite)	If there is a plausible temporal relationship between the onset of the AE/SAE and study drug administration and the AE/SAE cannot be readily explained by the patient's clinical state, intercurrent illness, or concomitant therapies; and/or the AE/SAE follows a known pattern of response to study drug; and/or the AE/SAE abates or resolves upon discontinuation of study drug or dose reduction and, if applicable, reappears upon re-challenge.
NO (Not related)	If data are available to identify a clear alternative cause for the AE/SAE other than study drug, such as the patient's clinical state, concomitant therapy, and/or other interventions. OR The AE/SAE has no plausible temporal relationship to administration of study drug (eg, cancer diagnosed 2 days after first dose of study drug).

Abbreviations: AE, adverse event; SAE, serious adverse event.

The Investigator's assessment of causality for individual AE reports is part of the study documentation process. Regardless of the "Yes" or "No" causality assessment for individual AE/SAE reports, Sponsor will promptly evaluate all reported safety information against

cumulative product experience to identify and expeditiously communicate possible new safety findings to Investigators and applicable regulatory authorities.

8.2. Serious Adverse Events

An SAE is any untoward medical occurrence that meets 1 or more of the following criteria:

- Is fatal.
- Is life threatening.
  - Note: Life-threatening refers to an event that places the patient at immediate risk of death. This definition does not include a reaction that, had it occurred in a more severe form, might have caused death.
- Requires or prolongs inpatient hospitalization.
- Results in persistent or significant disability or incapacity.
- Is a congenital anomaly or birth defect in the child or fetus of a patient exposed to study drug prior to conception or during pregnancy.
- Is an important medical event or reaction.
  - Note: Important medical event or reaction should only be selected as serious criteria when no other serious criteria apply and based upon appropriate medical judgement, not taking action may jeopardize the patient or may require intervention to prevent 1 of the other outcomes listed in the definition above.

The following are NOT considered SAEs in this clinical study:

- A visit to the emergency room or other hospital department of less than 24 hours duration that does not result in admission (unless considered an “important medical event” or “life-threatening”).
- Elective surgery, planned prior to signing study consent.
- Medical/surgical hospital admission for purpose other than remedying an ill health state and which was planned prior to entry into the study (eg, to support frequent laboratory studies, TLS prophylaxis medications as part of institutional standard protocols). Appropriate documentation is required in these cases.
- Routine health assessment requiring hospital admission for baseline/trending of health status (eg, routine colonoscopy).

- Hospital admission encountered for another life circumstance that carries no bearing on health status and requires no medical/surgical intervention (eg, lack of housing, economic inadequacy, care-giver respite, family circumstances, administrative).
- Hospital admission or other medical occurrences (such as prolonged hospitalization or death) for signs and symptoms solely due to the malignant disease under study (including associated signs and symptoms of disease progression). Changes in disease are reported separately.
- Hospital admission solely for study biopsies and without an associated AE.
- Hospital admission solely due to progression of underlying malignancy without any other associated AE.

The AE/SAE reporting period is described in Section 8.3.

Any SAE, whether or not considered related to study drug, must be reported immediately (within 24 hours of knowledge of the event) to Sponsor Pharmacovigilance by email using the study-specific SAE Form. Each SAE must also be reported on the appropriate CRF page(s). Investigators should not wait to collect information that fully documents the event before notifying Sponsor Pharmacovigilance of an SAE. The Sponsor may be required to report certain SAEs to regulatory authorities within 7 calendar days of being notified about the event; therefore, it is important that Investigators submit any information requested by Sponsor as soon as it becomes available. As additional information becomes available, including but not limited to the outcome of the SAE and any medication or other therapeutic measures used to treat the event, it must be reported within 24 hours by using the study-specific SAE Form.

The Investigator should follow all SAEs until the events are resolved or have stabilized, or the patient is lost to follow-up. Resolution of SAEs (with dates) should be documented in the CRF, SAE Form, and in the patient's medical record.

Reporting of SAEs to the EC/IRB will be done in compliance with the standard operating procedures and policies of the EC/IRB, and with applicable regulatory requirements. Adequate documentation must be obtained by Sponsor showing that the EC/IRB was properly and promptly notified as required.

For the purposes of regulatory reporting, the Reference Safety Information for this study is in the Investigator's Brochure.

8.3. Adverse Event Reporting Period

After signing of the ICF but before the first dose of study drug (NX-5948), all SAEs regardless of causality, and all AEs caused by a protocol-mandated intervention, must be reported. After the first dose of study drug all AEs regardless of causality, must be reported through the end of

the protocol-defined AE reporting period (30 days after the last dose of study drug) or until initiation of another anti-cancer agent if less than 30 days. Any SAE assessed as related to study drug that is ongoing after 30 days from the last dose of study drug will be monitored and followed until stability, resolution, death, or study withdrawal, whichever comes first. After the end of the protocol-defined AE reporting period (as defined above), only SAEs assessed as related to study drug or study procedures are required to be reported.

8.4. Criteria for Permanent Discontinuation of Study Drug in Association with Liver Test Abnormalities

Study drug treatment should be discontinued permanently if all of the following 4 criteria are met (ie, potential severe drug-induced liver injury/Hy's law case):

- AST or ALT increases to ≥ 3 × ULN; AND
- Total bilirubin increases to > 2 × ULN or INR > 1.5 × ULN; AND
- Alkaline phosphatase value does not reach 2 × ULN; AND
- No alternative cause explains the combination of the above laboratory abnormalities; important alternative causes include, but are not limited to the following:
    - Cardiovascular causes, especially right heart failure, and hypotension;
    - Hepatobiliary tract disease;
    - Viral hepatitis (eg, hepatitis A/B/C/D/E, EBV);
    - Exposure to hepatotoxic agents/drugs or hepatotoxins, including herbal and dietary supplements, plants, and mushrooms;
    - Alcoholic hepatitis;
    - Nonalcoholic steatohepatitis; or
    - Autoimmune hepatitis.

If an alternative cause for hepatotoxicity is identified, then it should be determined (based on the severity of the hepatotoxicity or event) whether study drug treatment should be withheld or permanently discontinued as appropriate for the safety of the patient.

Any potential Hy's Law or Hy's Law case should be handled as a serious unexpected AE associated with the use of the drug and reported as an SAE with expedited reporting timelines (ie, within 24 hours of awareness), even before all other possible causes of liver injury have been excluded. Reporting should include all available information, especially that needed for evaluating the severity and likelihood that the drug caused the reaction and should initiate a close follow-up until complete resolution of the problem and completion of all attempts to obtain supplementary data. If unsure if there is an alternative cause for hepatotoxicity, the case

should be reported without delay, and updates to the case with more information should be made later, when available.

8.5. Overdose

In this clinical study, overdose is defined as the accidental or intentional use of study drug in an amount higher than the dose being studied. Overdose itself is not necessarily an AE, but overdose may be associated with 1 or more AEs or SAEs. Please refer to Section 6.6.5 for information regarding clinical management of overdose.

Any study drug overdose or incorrect administration of study drug should be recorded on the applicable Study Drug Administration CRF.

Any AE associated with an overdose or incorrect administration of study drug should be reported on the AE CRF and reported to Sponsor Pharmacovigilance, if it meets SAE criterion, via the SAE Form within 24 hours of awareness.

8.6. Pregnancy

Pregnancy in a patient or patient's partner should be reported within 24 hours of the site becoming aware of the pregnancy by faxing or emailing the Pregnancy Form to Sponsor Pharmacovigilance. In addition, pregnancy in a patient is also reported on the applicable CRFs if the pregnancy leads to discontinuation of study treatment or discontinuation from the study. In the event of pregnancy in a patient, the Investigator must make every effort to follow the patient through resolution of the pregnancy (delivery or termination) and to report the resolution on the Pregnancy Form. In the event of pregnancy in the female partner of a male study patient, the Investigator should make every reasonable attempt to obtain the woman's consent for release of protected health information. This consent for release of protected medical information will be documented on the Pregnancy Form and will be maintained in the patient's research record. The reporting time frame for all pregnancies begins at the time the patient signs the consent through 6 months (pregnancy in a female study participant) or 3 months (pregnancy in a female partner of a male study participant) following the last dose of study drug.

Pregnancy itself is not an AE, unless there is suspicion that the study drug interfered with effectiveness of contraception, in which case the pregnancy itself should be reported as an AE in the CRF in addition to completing a Pregnancy Form. Should the pregnancy result in an abortion (excluding elective abortion without complications) or stillbirth, these events should be reported as SAEs (with seriousness criterion of important medical event). Should the pregnancy result in an infant born with a congenital anomaly, the event should be reported as an SAE (with seriousness criterion of congenital anomaly or birth defect) in addition to completing a Pregnancy Form and Infant Report Form.

8.7. Death

In the event of a patient death on study, the conventions for capturing the event are as follows:

- If the primary cause of death is progression of the underlying disease under study, then death is not due to AE and there should be no Grade 5 fatal AE reported.
- If the primary cause of death is not progression of the underlying disease under study, then the primary cause of death should be reported as a Grade 5 fatal AE; at most, there should be only one Grade 5 fatal AE per patient.
- Any other AEs that contributed to death but were not the primary cause of death should be reported as AEs, but not Grade 5 or fatal.

8.8. Secondary Malignancy

Any event of secondary malignancy should be reported as SAE (with the seriousness criterion of Important Medical Event, if no other criteria apply) via the SAE form.

8.9. Sponsor Pharmacovigilance Contact Information

Contact information for Pharmacovigilance is as follows:

Email: Nurix_PhV@iqvia.com

The Investigator is encouraged to discuss with the Medical Monitor any AEs for which the issue of seriousness is unclear or questioned. Contact information for the 24/7 Medical Monitor is as follows:

Address: 1700 Owens Street, Suite 205, San Francisco, CA 94158

Email: NX-5948-MedicalMonitor@nurixtx.com

Additional contact information can be found on the Study Contact List.

9. APPROPRIATENESS OF MEASUREMENTS

The safety and PK endpoints in this study are standard and well understood endpoints in clinical oncology.

Tumor response, based on radiological measurement and B-cell malignancy indication-specific criteria, is a standard and well accepted efficacy endpoint in clinical oncology.

The PD assessments were selected as indicators of on-target inhibition of BTK signaling pathway and potential identification of patient populations that may benefit from NX-5948 in the clinical setting.

10. DATA QUALITY ASSURANCE

Sponsor personnel or designees will assess the study site prior to initiation of the study, and will review with the site personnel information about the investigational product, protocol, and other regulatory document requirements, any applicable enrollment procedures, source document requirements, CRFs, monitoring requirements, and procedures for reporting AEs and SAEs.

At visits during and after the study, a clinical research associate (CRA) will monitor the site for compliance with regulatory documentation, with a focus on accurate and complete recording of data on CRFs from source documents, adherence to protocol, SAE reporting and drug accountability records.

To ensure complete and accurate data, automated data validation checks programmed within the electronic data capture (EDC) system will flag missing and non-conformant data during data entry. Data review by the Sponsor's project team may result in additional questions. Items flagged by the automated data validation checks and by the project team will appear as electronic queries on the applicable CRF in the EDC system for a specified user role to resolve. All data entry and subsequent data changes are logged in an audit trail in the EDC system.

The PI is responsible for ensuring that the data entered into the CRFs are timely, complete, and accurate and will electronically sign the CRFs for each patient prior to database lock.

Following database lock, an electronic copy of the final patient casebook will be provided to the investigative site for archival.

Data quality control and analysis will be performed by Sponsor or a designee, based on a predefined analysis plan.

11. STATISTICAL METHODS AND DETERMINATION OF SAMPLE SIZE

Analyses of Phase 1a and Phase 1b (Part 1 and Part 2) portions of the study will be reported separately. The data will be summarized by dose level cohort (Phase 1a), safety and cohort expansion cohorts (Phase 1b, Part 1 and Part 2, respectively), and all patients. Safety expansion cohorts in Phase 1b will be summarized separately. The statistical analysis plan (SAP) will provide additional details on the planned statistical analyses. If discrepancies exist in the text of the planned statistical analysis between the protocol and the final SAP, the SAP will prevail. Changes from the protocol planned analyses will be described in the SAP and clinical study report as appropriate.

The Simon 2-stage design will be formally implemented as a rule-based futility stopping rule for CLL/SLL patients prior to initiating Phase 1b Part 2 enrollment. The decision criteria at the end of Phase 1b Part 1, Cohort 1 will adhere to the operating characteristics detailed in Section 4.2.

Data collected on demographics and baseline characteristics, anti-tumor activity, safety, PK/PD observations and measurements will be summarized. In Phase 1b, key safety, PK/PD, and efficacy information may be summarized by demographic subgroups to explore differences in characteristics, if any are observed. Summary statistics for continuous variables will include the number of patients (n), mean, standard deviation, median, minimum, and maximum values. Summary statistics for categorical variables will include the frequency and percentage of patients in each category. All assessments before the first dose of study drug will be considered as pre-dose assessments. If there are multiple pre-dose assessments, the most recent one will be used as "baseline assessment" for statistical analysis. No formal hypothesis testing is planned for efficacy, safety, PK, or PD parameters. All data collected as part of this study will be presented in data listings.

AE data will be descriptively evaluated. The incidence and duration of AEs will be tabulated by Medical Dictionary for Regulatory Activities (MedDRA) preferred term and system organ class, and by seriousness, severity, and relationship to study drug. All laboratory results and vital sign measurements will be summarized using appropriate descriptive statistics. The Schedules of Assessment (Section 15.3) describe the timing of required evaluations.

11.1. Analysis Set Definitions

11.1.1. Safety Analysis Set

The safety analysis set (SAS) consists of all patients who receive at least 1 dose of NX-5948. All safety and tolerability evaluations will be based on this analysis set.

11.1.2. DLT/MTD Evaluable Analysis Set

The DLT/MTD evaluable analysis set will include all patients in Phase 1a (Dose Escalation) who complete 1 cycle of treatment (receiving ≥ 21 doses of NX-5948) and assessment, or who discontinue before completing the first cycle because of toxicity. This will be the analysis set for MTD/recommended Phase 1b dose(s) evaluations.

11.1.3. PK Analysis Set

The PK analysis set (PKAS) includes all patients without protocol deviations affecting interpretability of PK. For noncompartmental assessments, patients must have sufficient plasma concentration data to allow for estimation of at least 1 PK parameter. PK analyses will be based on this analysis set.

11.1.4. Pharmacodynamic Analysis Set

The PD analysis set (PDAS) includes all patients without protocol deviations affecting the interpretability of the PD. All PD analyses are based on this analysis set.

11.1.5. Efficacy Analysis Set

The efficacy analysis set (EAS) includes all patients who received study treatment, have a baseline disease assessment, and at least 1 post-baseline indication-specific disease assessment, died, or discontinued due to clinical progression or toxicity. Unless otherwise specified, all efficacy analyses will be based on this analysis set.

11.1.6. Per Protocol Set

The per protocol set (PPS) is a subset of the EAS that may also be used to analyze select efficacy endpoints and will exclude patients with certain major protocol deviations. The criteria for inclusion in PPS will be finalized and documented prior to database lock.

11.2. Statistical Methods and Properties

11.2.1. Efficacy Analyses

All anti-tumor activity assessments will be calculated based on the following B-cell malignancy indication-specific criteria (see details in Section 15.4):

- iwCLL criteria (Hallek 2018) for CLL/SLL,
- AIHA response criteria (Barcellini 2012),
- Lugano Classification of Lymphoma response criteria (Cheson 2014) for DLBCL, FL, MCL, and MZL,
- WM response criteria (Owen 2013) for WM, and
- International PCNSL Collaboration Group criteria (Abrey 2005) for PCNSL and SCNSL.

In Phase 1b, 2-sided 95% exact confidence intervals (CIs) will be presented for the ORR for EAS. Analyses of ORR in Phase 1b may be repeated and presented for the PPS. CR rate (and CR/CRi rate for patients with CLL) will be analyzed using a similar method. Best overall response will be summarized with frequencies and percentages.

PFS will also be calculated using a Kaplan-Meier method overall and for appropriate subgroups and cohorts, using B-cell malignancy indication-specific criteria, as appropriate. The 6-, 12-, etc., month time points for PFS may also be reported. All time-to-event type endpoints (such as time to first response, DOR, time to next therapy) will be analyzed using a similar statistical method.

Efficacy assessments will be summarized separately for Phase 1a and Phase 1b. Missing values in the efficacy data will be treated as missing; no efficacy data will be imputed.

If feasible, OS will be calculated using a Kaplan-Meier method for appropriate subgroups and cohorts. The 6-, 12-, etc., month timepoints for OS may also be reported.

11.2.2. Safety Analysis

11.2.2.1. Adverse Events

AEs will be graded by Investigators according to the NCI CTCAE v5.0 and coded using the MedDRA. AE data will be analyzed using the principle of treatment emergence, where the treatment emergence period is defined as from the period from the first study drug administration to 28 days after the last study drug administration, death, or initiation of subsequent anti-cancer therapy (whichever occurs first). AE data will be descriptively evaluated. The incidence and duration of AEs will be tabulated by MedDRA preferred term and system organ class, and by seriousness, severity, and relationship to study drug. The number and percentage of patients who experienced an AE, SAE, treatment-related AE/SAE, or AEs leading to treatment discontinuation, will be reported according to the worst toxicity grade, unless otherwise stated.

11.2.2.2. Laboratory Test Abnormalities

All laboratory results and vital sign measurements will be summarized using appropriate descriptive statistics. Laboratory results with a corresponding term in CTCAE will be graded by NCI CTCAE v5.0. Laboratory results will be tabulated in listings.

11.2.2.3. Other Safety Observations

All other relevant safety measurements (including ECGs, vital sign measurement, and concomitant medications) will be summarized. Further details on other safety assessments will be provided in the SAP.

11.2.3. Pharmacokinetic and Pharmacodynamic Analyses

See Table 17 for PK/PD sampling timepoints. Concentration-time data for NX-5948 PK will be summarized and presented. PK parameters (including C_{max}, T_{max}, half-life, $AUC_{0\text{-last}}$, $AUC_{0\text{-tau}}$, C_{min}, accumulation ratio, CSF:plasma concentration ratio, if applicable) will be listed by patient and summarized in tables using descriptive statistics, and stratified by dose group. Mean concentration-time plots and other graphical displays will be generated. In addition, the concentration time data may be evaluated using nonlinear mixed effects modeling and reported outside of the clinical study report. Raw and change-from-baseline BTK levels in B-cells will be analyzed by summary statistics. Further details on other PK and PD analyses will be provided in PK/PD SAP.

11.2.4. Patient Reported Outcome Analyses

The FACIT-GP5 is a component of the FACIT measurement tool and is used to assess the overall impact of treatment toxicity and to monitor changes over time. It uses a single item that has demonstrated a significant relationship to overall quality of life. The item in FACIT-GP5 is scored on a scale ranging from 0 to 4, with higher scores indicating better quality of life. Details of the serial analysis of the PRO will be outlined in the SAP.

11.2.5. Randomization

Patients enrolled into Phase 1b Part 1 will be assigned using block randomization to reduce the likelihood of systematic differences between treatment groups. In this process, patients are divided into blocks, and within each block, patients are randomly assigned to different treatment groups. The size of the blocks can vary but is typically kept small to maintain balance. As an example, for 2 treatment groups, the block size of 4 will be used and block size of 6 for 3 treatment groups. Patients in Cohort 1 will be stratified according to whether they have previously received an ncBTKi. No stratification will be employed for the other cohorts. Additional criteria, such as replacement of drop-off patients and mid-trial inclusion of other treatment groups will be defined in the statistical analysis plan. Since this is an open-label randomization without use of control; treatment assignment isn't blinded.

11.2.6. Interim Analysis

An interim analysis based on Simon 2-stage analysis will be implemented for CLL/SLL patients in Phase 1b Cohort 1. Stopping rules based on Simon 2-stage design are described in Section 4.2.

11.3. Sample Size Determination

A total of up to approximately 572 patients are planned to be enrolled.

11.3.1. Phase 1a

Approximately 30 to 66 patients in each of the parallel dose escalations for CLL/SLL and NHL/WM are planned to be enrolled in Phase 1a Dose Escalation. The Phase 1a Dose Escalation stage of the study is a traditional 3+3 dose escalation design with no formal statistical hypothesis. During dose escalation, up to a maximum of 12 patients per previously cleared dose levels may be enrolled in order to further explore safety and tolerability, with approval of the Sponsor. Under this design, there is a 71% probability of escalation (ie, observing 0 DLTs in 3 patients or 1 DLT in 6 patients) if the true but unknown rate of DLT is 20%.

Given the exploratory nature of the Phase 1a Dose Escalation stage, the sample size is not based on power calculations. The number of patients to be included is typical for this kind of study, and the results give sufficient safety and tolerability data as well as for the description of PK and PD parameters without exposing too many patients.

11.3.2. Phase 1b Part 1 (Safety Expansion)

A total of approximately 440 patients are planned to be enrolled across the 17 cohorts in CLL and NHL indications in Phase 1b Part 1 and across Part 2, as shown in Table 14.

Table 14 Sample Size Planned for Phase 1b

Number of patients planned for Phase 1b			Phase 1b Part 1 (Stage 1)		Phase 1b Part 2 (Stage 2)
Dosing	**Indication**		**Dose 1**	**Dose 2**	
QD dosing	CLL	Cohort 1[a]	20	20	80
		Cohort 2	10	10	
		Cohort 3	10	10	
		Cohort 4	10	10	
		Cohort 5	10	10	
		Cohort 15	10	10	
		Cohort 16	10	10	
		Cohort 17	10	10	
	MCL	Cohort 6		20	
	MZL	Cohort 7		20	
	WM 3L	Cohort 8		20	
	WM 2L	Cohort 9		20	
	DLBCL	Cohort 10		20	
	FL	Cohort 11		20	
	PCNSL/SCNSL	Cohort 12		20	
	PCNSL 2L	Cohort 13		20	
	WM 1L/CIT unfit	Cohort 14		20	

[a] Simon 2-stage design (20+80) for the optimized dose level in Cohort 1.
Abbreviations: 1L, first line; 2L, second line; CIT, chemoimmunotherapy; CLL, chronic lymphocytic leukemia; DLBCL, diffuse large B-cell lymphoma; FL, follicular lymphoma; MCL, mantle cell lymphoma; MZL, marginal zone lymphoma; NHL, non-Hodgkin lymphoma; PCNSL, primary central nervous system lymphoma; QD, once daily; SCNSL, secondary central nervous system lymphoma; WM, Waldenström macroglobulinemia.

For the Phase 1b Part 1 portion of the study, up to approximately 360 additional response-evaluable patients (total) in 17 B-cell malignancy safety expansion cohorts (described in Section 4) are planned to be enrolled.

A Simon 2-stage approach will be used (Simon 1989) for each dose level in Cohort 1, with a 1-sided type I error rate of 5% and power of 80%. The null hypothesis for this cohort states that the true response rate is 35%, and the alternative hypothesis is 55%. The trial is carried out in 2 stages: Stage 1 (Phase 1b Part 1) and Stage 2 (Phase 1b Part 2).

In Phase 1b Part 1 (Stage 1), futility will be assessed once a total of 20 patients in a given dose level are accrued. If 8 (rl) or fewer responses are observed among these 20 patients, the dose from Cohort 1 will be stopped and will not be carried forward to Phase 1b Part 2 (Stage 2).

For all remaining cohorts in Phase 1b Part 1, analyses will be descriptive in nature, and the sample size is determined by practical, rather than power, considerations. With up to 20 evaluable patients anticipated per cohort, the precision of the point estimate for the ORR, based

on the width of its 95% Clopper-Pearson CI, ranges from a width of 0.247 (95% CI = [0.001, 0.249]) for a response rate of 1/20 to a maximum width of 0.456 (95% CI = [0.272, 0.728]) when the response rate is 10/20. Table 15 provides several examples for the 90% CI calculations based on a 20-patient expansion cohort. All calculations are based on Clopper-Pearson exact 90% CIs.

Table 15 Observed ORR and 90% Confidence Interval for Evaluable Patients

Evaluable Patients per Cohort (n=20)		
Observed ORR	**90% CI[a]**	**Width of 90% CI**
5% (1/20)	(0.3%, 21.6%)	0.214
25% (5/20)	(10.4%, 45.6%)	0.352
50% (10/20)	(30.2%, 69.8%)	0.396
75% (15/20)	(54.4%, 89.6%)	0.352

[a] CIs are calculated using Pearson-Koller exact method (Clopper 1934).
Abbreviations: CI, confidence interval; ORR, overall response rate.

Based on the table above, if 10 out of 20 evaluable patients have an overall response (ie, PR-L or better), we will have evidence that the ORR following NX-5948 treatment will be above 30%.

11.3.3. Phase 1b Part 2 (Cohort Expansion)

Stage 2 of the Simon 2-stage design for Cohort 1 will be initiated in the dose(s) established in Phase 1b Part 1 per the Sponsor's discretion upon establishing lack of futility. An additional 80 response evaluable patients with CLL/SLL (post-BTKi and post-BCL-2 inhibitor) may be enrolled in Phase 1b Part 2 (Stage 2), resulting in a total sample size of 100 for the optimized dose level of Cohort 1. If there are 46 (r) or more responses among these cumulative 100 (n) patients, we reject the null hypothesis and claim proof of concept is achieved.

11.4. Statistical Criteria for Termination of the Study

There are no statistical criteria for the termination of this study during Phase 1a (Dose Escalation). In Phase 1b, the IDMC will formally implement the futility stopping rules based on decision criteria described in Section 4.2.

12. CLINICAL TRIAL DATA MONITORING AND GOVERNANCE

12.1. Safety Review Committee (Phase 1a)

Decisions on dose escalation will be made by an SRC.

The SRC will be comprised of representatives of the Sponsor, the Sponsor's Medical Monitor, the CRO's Medical Monitor (if applicable), and at a minimum, a quorum of Investigators with patients under review. One or more relevant external medical experts may be added at the committee's discretion.

The SRC's responsibilities will be detailed in a separate Charter, and at a minimum will entail:

- Meet to evaluate the DLT period for each dose-escalation cohort to review any significant safety event (such as drug-related death or the equivalent of ≥ 33% of patients experiencing DLT events) as they occur.
- Review all relevant safety findings and confirm all DLTs.
- Determine all cohort dose escalation and de-escalation decisions during Phase 1a (Dose Escalation), including any cohort dose modifications based on toxicity.
- Recommend the dose(s) or dosing schedule(s), to extend into Phase 1b (Safety Expansion; ie, the recommended Phase 1b dose[s]).

12.2. Independent Data Monitoring Committee (Phase 1b)

An IDMC will be established during dose expansion (Phase 1b) to monitor trial progress and reduce risks to patients in the trial. The IDMC will comprise of independent experts in relevant fields such as clinical medicine (specifically hematologist-oncologists with expertise in B-cell malignancies) and biostatistics. Operating autonomously from the sponsor and investigators, the IDMC's primary responsibility will be to conduct ongoing review and analysis of accumulating trial data. IDMC's role includes:

- Analysis of expedited safety reports,
- Review of cumulative summaries of all adverse events,
- Assessing protocol adherence, and
- Evaluating clinical data periodically to determine if modifications or termination are warranted.
- Evaluating safety and futility stopping rules as defined in Section 4.2

The IDMC may make recommendations to the Sponsor regarding protocol modifications to reduce risks to patients enrolled in the trial. By providing impartial oversight and recommendations based on rigorous data analysis, the IDMC intends to enhance the credibility and reliability of trial findings. The IDMC's objectives, roles, and responsibilities will be detailed in a separate charter.

13. ADMINISTRATIVE ITEMS

13.1. Ethics

13.1.1. Ethics Committee/Institutional Review Board

Prior to initiating the study, the Investigator will obtain written confirmation that the EC/IRB is properly constituted and compliant with ICH and GCP requirements, applicable laws, and local regulations. A copy of the confirmation from the EC/IRB will be provided to Nurix Therapeutics, Inc. (Sponsor) or its designee. The Investigator will provide the EC/IRB with all appropriate material, including the protocol, Investigator's Brochure, the ICF including compensation procedures, and any other written information provided to the patients, including all ICFs translated to a language other than the native language of the clinical site. The study will not be initiated, and investigational product supplies will not be shipped to the site until appropriate documents from the EC/IRB confirming unconditional approval of the protocol, the ICF, and all patient recruitment materials are obtained in writing by the Investigator and copies are received at Sponsor or its designee. The approval document should refer to the study by protocol title and Sponsor protocol number (if possible), identify the documents reviewed, and include the date of the review and approval. Sponsor will ensure that the appropriate reports on the progress of the study are made to the EC/IRB and Sponsor by the Investigator in accordance with applicable guidance documents and governmental regulations.

13.1.2. Ethical Conduct of Study

This study will be conducted in accordance with the following:

- US Code of Federal Regulations (CFR) sections that address clinical research studies, GCP as outlined in ICH E6, Declaration of Helsinki, and/or other national and local regulations that address clinical research studies, as applicable.

Specifically, this study is based on adequately performed laboratory and animal experimentation. The study will be conducted under a protocol reviewed and approved by an EC/IRB and will be conducted by scientifically and medically qualified persons. The benefits of the study are in proportion to the risks. The rights and welfare of the patients will be respected, and patient participation continued as long as the Investigators conducting the study do not find the hazards to outweigh the potential benefits.

Each patient, or his/her legally authorized representative, will provide written, informed consent before any study-related tests or evaluations are performed.

The Sponsor has a liability insurance policy and has also taken the necessary measures to obtain insurance policies that are in accordance with legal requirements in participating regions/countries to provide coverage for damage to research patients through injury or death caused by the study.

13.1.3. Patient Information and Informed Consent

A properly written and executed ICF, in compliance with US CFR, 21 CFR §50, the Declaration of Helsinki, and/or applicable local regulations will be obtained for each patient prior to entering the patient into the study. The Investigator will prepare the ICF and provide the documents to Sponsor for approval prior to submission to the EC/IRB approval. Sponsor and the EC/IRB must approve the documents before they are implemented. A copy of the approved ICF from each investigational site and, if applicable, a copy of the approved patient information sheet and all ICFs translated to a language other than the native language of the clinical site, must also be received by Sponsor or designee prior to any study-specific procedures being performed.

13.2. Study Administrative Structure

The study will be administered and monitored by employees or representatives of Sponsor, Nurix. CRAs or trained designees will monitor each site on a periodic basis and perform verification of source documentation for each patient as well as other required review processes. The Sponsor will be responsible for the timely reporting of SAEs to appropriate regulatory authorities as required.

Clinical safety laboratory evaluations will be performed at the local laboratories associated with the study sites. The specific laboratories and instructions for biomarker and PK sample collection, processing, and shipment are provided in the laboratory manual.

13.3. Investigator Responsibilities

Prior to beginning the study, the Investigator at each site must provide to Sponsor or designee a fully executed and signed US FDA Form FDA 1572 (or equivalent if applicable) and a Financial Disclosure Form. Financial Disclosure Forms must also be completed for all sub-Investigators who will be directly involved in the treatment or evaluation of patients in this study. Each Investigator will notify Nurix, or its authorized representative, of any relevant changes to the information included on such Financial Disclosure Form during the conduct of the study and for one (1) year after the study has completed.

The Investigator is responsible for ensuring that all study site personnel, including sub-Investigators and other responsible study staff members, conduct the study in compliance with the Declaration of Helsinki and the ICH E6 Guideline for GCP, including the archiving of essential documents.

Study completion at a site is defined as the date when the study database is locked.

If it is necessary to amend either the protocol or the study ICF, the Investigator will be responsible for ensuring that the EC/IRB reviews and approves the amended documents, and that patients are informed of applicable changes, and updates.

The Investigator will sign and return to Nurix Therapeutics, Inc. (or designee) the “Protocol Signature Page” of the original protocol and any protocol amendment, provide current

medical licenses, and curriculum vitae. All forms must be updated as applicable throughout the study.

13.4. Protection of Human Patients

13.4.1. Compliance with Informed Consent Regulations and Relevant Country Regulations

Written informed consent is to be obtained from each patient prior to enrollment into the study, and/or from the patient's legally authorized representative. Sufficient time will be allowed to discuss any questions raised by the patient. The Investigator or designated staff will document this process in the study records. The Investigator must use the current EC/IRB-approved consent form for documenting written informed consent. The signed consent form(s) will be retained with the study center's records. Each patient will also be given a copy of his or her signed consent form(s).

13.4.2. Compliance with EC/IRB Regulations

This study is to be conducted in accordance with EC/IRB regulations. The Investigator must obtain approval from a properly constituted EC/IRB prior to initiating the study and re-approval or review at least annually. Sponsor is to be notified immediately if the responsible EC/IRB has been disqualified or if proceedings leading to disqualification have begun. Copies of all EC/IRB correspondence with the Investigator should be provided to Sponsor.

13.4.3. Compliance with Good Clinical Practice

This study will be conducted in accordance with Protocol NX-5948-301, relevant Sponsor/Sponsor designee's policies and procedures, the ICH, GCP guidelines, and the applicable country and regional regulatory requirements.

13.4.4. Patient Confidentiality

A report of the results of this study may be published or sent to the appropriate health authorities in any country in which the study drug may ultimately be marketed, but the patient's name will not be disclosed in these documents. The patient's name may be disclosed to the Sponsor of the study, Sponsor's representatives, or the governing health authorities if they inspect the study records. Appropriate precautions will be taken to maintain confidentiality of medical records and personal information.

13.4.5. Patient Privacy

Written authorization is to be obtained from each patient prior to enrollment into the study, and/or from the patient's legally authorized representative in accordance with the applicable privacy requirements (eg, GDPR, HIPAA).

In the EU, UK, and Switzerland, according to GDPR (and equivalent laws in Switzerland), the patient specifically provides written informed consent to:

- Processing of special categories of personal data (as defined in GDPR; hereinafter described as personal data), for the purposes explained to them. Such personal data will be treated in accordance with the principles of Data Protection Regulations and Laws in EU/UK/Switzerland and in countries where the Sponsor will apply for a marketing license.
- The data collected for the study can be transferred to other countries, which do not have the same data protection laws. The patient should be informed that their identity will not be revealed in any such data.

In the EU, UK, Switzerland and US, according to GDPR (and equivalent laws in Switzerland), and in accordance with HIPAA requirements, additional purposes of this study include the following:

- To publish anonymous patient data from the study; and
- To create and maintain a data repository.

The Sponsor is incorporated in the United States and is required to comply with a European data protection law known as the GDPR no. 2016/679 to ensure the appropriate transfer of personal data. The Sponsor will implement reasonable and appropriate legal, technical, and organizational security measures as required under law to protect the security of the processing of personal data and to protect personal data from unauthorized access, use or disclosure. Personal data will only be transferred as permitted by the Competent Supervisory Authorities (which oversees the data privacy laws in the EU) or other data protection authority, and that may involve reliance on the EU Standard Contractual Clauses which are terms and conditions adopted by the European Commission and which set out how personal data can be transferred from the EU to a country like the United States which is outside of the EU. These Standard Contractual Clauses s provide specific contractual obligations to help safeguard personal data, keep it secure, and ensure that it is not used for any purpose other than those the Sponsor instructs to its partners or suppliers.

13.5. Changes to the Protocol

Protocol modifications, except those intended to reduce immediate risk to study patients, may be initiated only by Nurix. All protocol modifications must be submitted to competent authorities according to local requirements and IRB/EC together with, if applicable, a revised ICF in accordance with local requirements. Written documentation from competent authorities (according to local requirements) and from the IRB/EC and any required site approval must be obtained by the sponsor before the change can be implemented.

Information on any change in risk and or change in scope must be provided to patients already actively participating in the study, and they must read, understand and sign each revised ICF confirming willingness to remain in the study.

13.6. Documentation

13.6.1. Source Documents

Source documents may include a patient's medical records, hospital charts, clinic charts, the Investigator's patient study files, as well as the results of diagnostic tests such as X-rays, laboratory tests, and ECGs. The Investigator's copy of the CRFs serves as part of the Investigator's record of a patient's study-related data. All data entered into the CRF must be substantiated by a source document.

Source documents should be completed in accordance with Attributable, Legible, Contemporaneous, Original, and Accurate (ALCOA) guidelines.

13.6.2. Case Report Form Completion

The Investigator is responsible for ensuring that data are properly recorded on each patient's CRFs and related documents. An Investigator who has signed the protocol signature page should personally sign for the CRFs (as indicated in the CRFs) to ensure that the observations and findings are recorded on the CRFs correctly and completely. The CRFs are to be completed promptly and submitted to Sponsor or designee in a timely manner at the completion of the study, or as otherwise specified by Sponsor.

Sponsor or designee will conduct data processing and review all completed CRFs for completeness. The Investigator will be contacted for corrections and/or clarifications.

13.6.3. Retention of Documentation

All study-related correspondence, patient records, consent forms, patient privacy documentation, records of the distribution and use of all investigational products, and copies of CRFs should be maintained on file.

For countries falling within the scope of the ICH guidelines, the Sponsor-specific essential documents should be retained until: at least 2 years after the last approval of a marketing application in an ICH region and until there are no pending or contemplated marketing applications in an ICH region; or at least 2 years have elapsed since the formal discontinuation of clinical development of the investigational product. However, these documents should be retained for a longer period if required by the applicable regulatory requirement(s) or if needed by the Sponsor.

Sponsor requires that it be notified in writing if the Investigator wishes to relinquish ownership of the data so that mutually agreed-upon arrangements can be made for transfer of ownership to a suitably qualified, responsible person.

13.7. Inventory, Accountability, and Destruction or Return of Study Drug

All study drug and related materials should be inventoried, reconciled, and destroyed or returned according to applicable local and national regulations and study procedures.

13.7.1. Study Drug Inventory Records

A detailed inventory must be kept for all study drug. The medication is to be dispensed in accordance with the protocol to patients who are under the direct supervision of an Investigator. Discrepancies in study drug inventory must be documented.

13.7.2. Study Drug Accountability

A detailed accountability must be kept for all study drug. Unused study drug should be kept in a secure location for accountability and reconciliation by a monitor. Sites with limited capacity or processes that do not allow for unused study drug to be stored for accountability and reconciliation by a monitor should provide documentation of these constraints to the Sponsor for approval prior to receiving their first shipment of study drug. Discrepancies in study drug accountability must be documented.

13.7.3. Destruction or Return of Study Drug

Unused study drug may be destroyed on site, per the site's standard operating procedures, after Sponsor has granted approval for drug destruction. All study drug destroyed on site must be documented. If a site is unable to destroy study drug appropriately, the site may return unused study drug upon request as detailed in the Pharmacy Manual. The return of study drug or study drug materials must be documented. Documentation must be provided to Sponsor and retained in the Investigator study files.

13.8. Monitoring by the Sponsor

A representative of the Sponsor will monitor the study on a periodic basis. The determination of the extent and nature of monitoring will be based on considerations such as the objective, purpose, design, complexity, size, and endpoints of the study.

Monitoring will be conducted according to GCP and standard operating procedures for compliance with applicable government regulations. The Investigators must agree to allow the monitor access to the clinical supplies, dispensing, and storage areas and to the clinical files of the study patients, and, if requested, agrees to assist the monitors in order to verify the accuracy of the data recorded in the CRF.

13.9. Site Audits and Regulatory Inspections

Representatives of regulatory authorities, Nurix or designee, or EC/IRBs may conduct inspections or audits of the clinical study. If the Investigator is notified of an inspection by a regulatory authority, the Investigator agrees to notify the Sponsor and CRA immediately. The Investigator agrees to provide representatives of a regulatory agency or Nurix access to source documents, records, facilities, and personnel for the effective conduct of any inspection or audit.

13.10. Handling of Biological Specimens

Samples of blood and urine for evaluation of hematology, coagulation, chemistry, pregnancy, serology, thyroid function, and urinalysis will be analyzed locally at appropriately accredited and certified local laboratories and will be destroyed after analysis in accordance with institutional policies.

Blood and CSF samples for biomarker analysis and determination of concentrations of study drug/metabolites will be handled and processed according to the laboratory manual provided to clinical sites and will be analyzed at contract laboratories designated by Sponsor.

Lymph node biopsy samples and bone marrow biopsy and aspirate samples will be handled and processed according to the laboratory manual and will be analyzed at a laboratory designated by Sponsor.

Biological samples present at the conclusion of this study may be retained in storage by the Sponsor for a period up to 10 years or longer if required under local law and analyzed at a later date. Patients who participate in this study are consenting to the biological sample retention duration. A patient may request the destruction of their samples at any time. The Sponsor may destroy the samples sooner than 10 years after the study completes. Other analyses may include additional biomarkers for NX-5948 activity, disease-specific biomarkers, and/or biomarkers of safety. This may also include the development of ways to detect, monitor, or treat cancer. A decision to perform such exploratory research studies would be based on outcome data from this study or from new scientific findings related to the drug class or disease, as well as reagent and assay availability.

Clinical samples shall not be retained or used by the Investigator or institution except as expressly permitted by Sponsor in writing.

13.11. Reporting Requirements

In accordance with applicable regulatory requirements, the Investigator is solely obligated to inform the EC/IRB of progress of the study and notify the EC/IRB of study closure. The Investigator must also provide Nurix (or designee) with copies of all EC/IRB correspondence that relate to study approvals, updates, or changes. The Investigator must also forward all EC/IRB renewals to Nurix (or designee).

Protocol deviations must be sent to the Sponsor and to the site's EC/IRB per their policies. The Investigator is responsible for knowing and adhering to the institution's EC/IRB requirements.

13.12. Publications

Nurix, as the Sponsor, has proprietary interest in this study. Authorship and manuscript composition will reflect cooperation between multiple Investigators, sites, and Sponsor personnel. Authorship will be established prior to the writing of the manuscript. As this study involves multiple centers, no individual publications will be allowed prior to completion of the final report of the multicenter study except as agreed with Sponsor.

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15. APPENDICES

15.1. ECOG Performance Status

0	Fully active, able to carry on all pre-disease activities without restriction.
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, eg, light housework or office work.
2	Ambulatory and capable of all selfcare but unable to carry out any work activities. Up and about more than 50% of waking hours.
3	Capable of only limited selfcare, confined to bed or chair more than 50% of waking hours.
4	Completely disabled. Cannot carry on any selfcare. Totally confined to bed or chair.
5	Dead.

15.2. Common Terminology Criteria for Adverse Events

Please refer to: https://ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm

15.3. Study Assessments

Table 16 Schedule of Assessments for Phase 1a and Phase 1b

	Screening	Cycle 1					Cycle 2					Cycle 3 & 4		Cycle 5+	SFU			LTFU
Procedures/ Examinations[a,b]	**Days -28 to -1**	**Day 1**	**Day 2**	**Day 8 (±2)**	**Day 15 (±2)**	**Day 22 (±2)**	**Day 1 (±2)**	**Day 2**	**Day 8 (±2)**	**Day 15 (±2)**	**Day 22 (±2)**	**Day 1 (±7)**	**Day 15 (±7)**	**Day 1 (±7)**	**EOT (+3)**	**15-day Post Tx (±5)**	**30-day Post Tx (±7)**	**Q12 weeks (±7)**
Informed consent[c]	X																	
Inclusion/exclusion criteria[d]	X	X																
Medical/oncologic history[e]	X																	
Height, weight, & physical examination[f]	X	X		X	X	X	X					X		X	X		X	
Vital signs, pulse oximetry, and ECOG performance status[g]	X	X		X	X	X	X					X		X	X		X	
12-lead ECG[h]	X	X		X			X					X		X				
Echocardiogram or MUGA[h]	X																	
Pregnancy test[i]	X	X					X					X		X	X		X	
Serum IgM level (WM only)[j]	X	X					X					X		X				
Serum IgA and IgG levels (WM patients)	X											q8wks		q8wks	X			
Serum IgM monoclonal protein by immunofixation (WM only)	X										at Weeks 8, 16, 24, and q12wks (± 7 days) in conjunction with disease assessments							

	Screening	Cycle 1					Cycle 2					Cycle 3 & 4		Cycle 5+	SFU			LTFU
Procedures/ Examinations[a,b]	**Days -28 to -1**	**Day 1**	**Day 2**	**Day 8 (±2)**	**Day 15 (±2)**	**Day 22 (±2)**	**Day 1 (±2)**	**Day 2**	**Day 8 (±2)**	**Day 15 (±2)**	**Day 22 (±2)**	**Day 1 (±7)**	**Day 15 (±7)**	**Day 1 (±7)**	**EOT (+3)**	**15-day Post Tx (±5)**	**30-day Post Tx (±7)**	**Q12 weeks (±7)**
Serum immunoglobulins (IgG, IgA, and IgM) (non-WM patients)	X											q8wks		q8wks	X			
Serology (HBV, HCV, HIV, CMV, and EBV)[k]	X																	
Hematology (including haptoglobin)[l]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X			
Chemistry, LFTs, uric acid, LDH, [m]	X	X	X	X	X	X	X	X	X	X	X	X		X	X			
Direct antiglobulin test (DAT) (WM/CLL/SLL patients)[n]	X	X										q8wks		q8wks	X			
Coagulation[o]	X	X										q8wks		q8wks				
Thyroid function tests, serum amylase/lipase, C-reactive protein[p]	X											q8wks		q8wks	X			
Urinalysis (dipstick)[q]	X	X										q8wks		q8wks	X			
β-2-microglobulin (for CLL patients only)	X																	
Lymph node biopsy (all non-WM indications) or bone marrow aspirate and biopsy (WM patients)[r]	X			X														
Bone marrow biopsy and/or aspirate[s]			As clinically indicated, and to confirm CR, as required per disease-specific response criteria															

	Screening	Cycle 1					Cycle 2					Cycle 3 & 4		Cycle 5+	SFU			LTFU
Procedures/ Examinations[a,b]	**Days -28 to -1**	**Day 1**	**Day 2**	**Day 8 (±2)**	**Day 15 (±2)**	**Day 22 (±2)**	**Day 1 (±2)**	**Day 2**	**Day 8 (±2)**	**Day 15 (±2)**	**Day 22 (±2)**	**Day 1 (±7)**	**Day 15 (±7)**	**Day 1 (±7)**	**EOT (+3)**	**15-day Post Tx (±5)**	**30-day Post Tx (±7)**	**Q12 weeks (±7)**
PK blood[t,u]		X	X	X	X	X	X	X				C3D1		C5D1, C7D1, and q12wks (±7d)	X	X	X	
Ophthalmologic examination (CNSL patients)[v, w]	X																	
Lumbar puncture (or Ommaya reservoir if available) with CSF cytology and protein (CNSL patients)[x]	X						X								X			
Karnofsky performance score (PCNSL patients)[v]	X																	
PD flow cytometry[u, y]		X	X	X	X	X	X					C3D1		C5D1, C7D1 and q12wks (±7d)	X	X	X	
Exploratory PD cytokines[y]		X		X			X					C3D1			X			
Exploratory PD NGS[z]		X												C7D1 and C13D1 (±7d)	X			
Exploratory PD PBMC[y]	X	X	X	X			X					C3D1		C5D1, C7D1 and q12wks (±7d)	X			
PRO (FACIT-GP5)[aa]	X	X		X	X	X	X					X		X[aa]	X			

	Screening	Cycle 1					Cycle 2					Cycle 3 & 4		Cycle 5+	SFU			LTFU
Procedures/ Examinations[a,b]	**Days -28 to -1**	**Day 1**	**Day 2**	**Day 8 (±2)**	**Day 15 (±2)**	**Day 22 (±2)**	**Day 1 (±2)**	**Day 2**	**Day 8 (±2)**	**Day 15 (±2)**	**Day 22 (±2)**	**Day 1 (±7)**	**Day 15 (±7)**	**Day 1 (±7)**	**EOT (+3)**	**15-day Post Tx (±5)**	**30-day Post Tx (±7)**	**Q12 weeks (±7)**
Disease assessment (Excluding patients with CNSL)[w]	X										at Weeks 8, 16, 24, and then q12wks up to 7 days prior to the scheduled assessment							
Disease assessment (patients with CNSL only)[w]	X	between Cycle 1 Day 22 and Cycle 1 Day 28					at Weeks 8, 16, 24, and q12wks up to 7 days prior to the scheduled assessment											
AIHA response assessment (Cohort 16 only)	X										at Weeks 8, 16, 24, and then q12wks up to 7 days prior to the scheduled assessment							
NX-5948 study drug[bb, l]		X	→	→	→	→	→	→	→	→	→	→	→	→				
TLS prophylaxis (DLBCL, MCL, CLL/SLL only)[cc]		X	→	→	→	→												
Adverse events[u, dd]	X	→	→	→	→	→	→	→	→	→	→	→	→	→	→	→	→	→
Concomitant medications[ee]	X	→	→	→	→	→	→	→	→	→	→	→	→	→	→	→	→	
Survival follow-up contact																		X
PK CSF (for patients with CNSL only)[ff]							X								X			
Exploratory PD ctDNA CSF (for patients with CNSL only)[gg]	X						X								X			
Exploratory PD (buccal swab) ctDNA germline sample collection (for patients with CNSL only)[hh]	X																	
Exploratory PD CSF Omic (For CNSL only)	X						X								X			

	Screening	Cycle 1					Cycle 2					Cycle 3 & 4		Cycle 5+	SFU			LTFU
Procedures/ Examinations[a,b]	**Days -28 to -1**	**Day 1**	**Day 2**	**Day 8 (±2)**	**Day 15 (±2)**	**Day 22 (±2)**	**Day 1 (±2)**	**Day 2**	**Day 8 (±2)**	**Day 15 (±2)**	**Day 22 (±2)**	**Day 1 (±7)**	**Day 15 (±7)**	**Day 1 (±7)**	**EOT (+3)**	**15-day Post Tx (±5)**	**30-day Post Tx (±7)**	**Q12 weeks (±7)**
Baseline biomarker plasma sample	X																	
Fitzpatrick skin color assessment	X																	
MMSE-2: BV[ii]	X										at Weeks 8, 16, 24, and q12wks (± 7 days) in conjunction with disease assessments							
Exploratory PD MRD (for patients with CLL/SLL only)[ij]		at Cycle 1 Day 1 (Pre-dose), Weeks 24, 48, and yearly (±7 days) in conjunction with disease assessments																

Abbreviations: AIHA, autoimmune hemolytic anemia; C, cycle; CLL, chronic lymphocytic leukemia; CMV, cytomegalovirus; CNS, central nervous system; CNSL, central nervous system lymphoma (includes both primary and secondary CNSL); CSF, cerebrospinal fluid; CT, computed tomography; ctDNA, circulating tumor deoxyribonucleic acid; D, day; DAT, direct antiglobin test; DLBCL, diffuse large B-cell lymphoma; EBV, Epstein-Barr virus; ECG, electrocardiogram; ECOG, Eastern Cooperative Oncology Group; EOT, End of Treatment; GCB, germinal center B-cell subtype; HBV, hepatitis B virus; HCV, hepatitis C virus; HIV, human immunodeficiency virus; Ig, immunoglobulin; LDH, lactate dehydrogenase; LFT, liver function test; LTFU, Long-term Follow-up; MCL, mantle cell lymphoma; MMSE-2: BV, Mini Mental State Exam, 2nd Edition: Brief Version; MUGA, multiple gated acquisition scan; NGS, next-generation sequencing; PBMC, peripheral blood mononuclear cell; PCNSL, primary central nervous system lymphoma; PD, pharmacodynamics; PK, pharmacokinetics; q, every; SFU, Safety Follow-up; SLL, small lymphocytic lymphoma; SoC, standard of care; TLS, tumor lysis syndrome; Tx, treatment; wAIHA, warm autoimmune hemolytic anemia; wks, weeks; WM, Waldenström macroglobulinemia

[a] All procedures and examinations should be performed before the administration of NX-5948, except as indicated below. There will not be a pause in the study calendar in the event of a treatment interruption.

[b] Patients should be assessed for eligibility prior to dosing on Cycle 1 Day 1. To accommodate logistics, safety labs may be obtained within 96 hours of first dose on Cycle 1 Day 1 and assessed for eligibility and do not need to be repeated on Cycle 1 Day 1 with the exception of hematology. Cycle 1 Day 1 hematology should be redrawn but does not need to be evaluated for eligibility purposes if previously collected within 96 hours of first dose. Serologies may be obtained at any time during the Screening period and assessed for eligibility purposes (EBV, CMV, HIV, Hep B/C). Serum amylase and lipase may be obtained at any time during the Screening period and assessed for eligibility purposes. Safety assessments may be obtained up to 96 hours prior to dosing before Day 1 of each dosing cycle beginning with Cycle 2.

[c] Ethics committee (EC)/Institutional Review Board (IRB)-approved informed consent form (ICF) must be obtained before any study-specific procedures or examinations are performed.

[d] Patient must meet eligibility criteria described in Section 5.1. Patient demographic information (age, sex, race, and ethnicity) to be collected for eligibility review at Screening. For Phase 1b Part 1 CLL Cohort 2, documentation of BTK mutations from local lab is required.

[e] Including prior cancer treatments and cytogenetic data.

[f] Height recorded at Screening only. Height and weight measured while wearing indoor clothing and with shoes removed. Full physical examination to be performed at Screening. Symptom-directed physical examination should be performed at all other visits, if clinically indicated.

[g] On the Cycle 1 Day 1 visit, vital signs (pulse, systolic and diastolic blood pressure, temperature, pulse oximetry, and respiration rate) should be collected pre-dose (within 5 minutes of NX-5948 administration) then 15, 30, and 60 minutes (± 10 minutes) after NX-5948 administration. Vital signs on subsequent visits should be taken before NX-5948 administration. When post-dose vital sign measurements are scheduled at the same time as a blood draw, the vital signs will be obtained prior to the blood draw and, if applicable, the CSF collection. Eastern Cooperative Oncology Group (ECOG): if a patient's ECOG performance status declines to 2 or worse (or worse than 2 for patients with PCNSL and secondary CNS involvement) between Screening/baseline and Cycle 1 Day 1, patient will be screen failed from the study without treatment as they are no longer eligible.

[h] 12-lead triplicate ECGs will be performed locally for all patients at Screening. ECGs should be recorded digitally and in triplicate (approximately 10 seconds per ECG over a 5-minute period) on Cycle 1 Day 1, Cycle 1 Day 8, and Cycle 2 Day 1 pre-dose (within 30 minutes prior to dosing) and 3 and 4 hours post-dose (± 15 minutes). On subsequent cycles, a pre-dose (within 30 minutes prior to dosing), single-tracing ECG will be performed. When a post-dose ECG is scheduled at the same time as a blood draw, the ECG should be obtained prior to the blood draw and, if applicable, the CSF collection. Frequency of ECGs should be increased if clinically indicated. Prior to performing ECGs, patients should rest in the supine or semi-recumbent position for at least 5 minutes. Patients who are taking concomitant medications that are known to prolong the corrected QT interval (QTc) should have a repeat ECG after stopping the medication. Echocardiogram or MUGA done within 6 weeks of Cycle 1 Day 1 can be used for Screening.

[i] For women of childbearing potential (WOCBP), serum pregnancy test is required at Screening. Serum or urine pregnancy test to be performed within 24 hours of first dose on Cycle 1 Day 1 then on Day 1 of every cycle thereafter. If treatment with study drug is stopped for ≥ 14 days, WOCBP should have a negative pregnancy test prior to restarting study drug. For WOCBP, pregnancy tests are also required at the End of Treatment (EOT) and 30-day Post Treatment Safety Follow-up visits.

[j] Blood sample will be collected at Screening. Sample collection should be repeated prior to first dose of study drug on Cycle 1 Day 1, and then on Day 1 of every cycle thereafter.

[k] Blood sample will be collected at Screening and will be tested for presence of human immunodeficiency virus (HIV; Type 1 and Type 2), hepatitis B and C (hepatitis B core antibody [HBcAb], hepatitis B surface antibody [HBsAb], hepatitis B surface antigen [HbsAg], hepatitis C antibody), cytomegalovirus (CMV), and Epstein-Barr virus (EBV) serologies (and/or polymerase chain

reaction [PCR]) may include viral capsid antigen-immunoglobulin (Ig)M, viral capsid antigen-IgG, Epstein-Barr virus nuclear antigen IgG. Patients with hepatitis B virus (HBV) who are negative for HBV deoxyribonucleic acid (DNA) by quantitative PCR are eligible irrespective of serology findings.

[l] Hematology performed at Screening then on Days 1, 2, 8, 15, and 22 by local laboratory for the first 2 cycles, every 2 weeks in Cycles 3 and 4, and on Day 1 of each subsequent cycle: red blood cell (RBC) count, hemoglobin, hematocrit, mean corpuscular volume (MCV), reticulocyte count, haptoglobin, platelet count, white blood cell (WBC) count, and WBC differential (absolute neutrophils, lymphocytes, monocytes, eosinophils, and basophils), CD4 counts if a patient has HIV. If hematology demonstrates Common Terminology Criteria for Adverse Events (CTCAE) Grade 3/4 cytopenias, may increase frequency of monitoring as clinically appropriate. Where possible, Day 1 draws for each cycle should be performed at the study clinic. Hematology assessments may be performed at a local lab outside of the investigative site lab as long as laboratory reports are provided to the Investigator prior to dosing on Day 1. Hematology assessments scheduled for Day 1 of each cycle must be available and assessed for toxicity before dosing. Cycle 1 Day 1 hematology may be obtained and assessed for eligibility up to 96 hours prior to dosing. Repeat testing on Cycle 1 Day 1 is required but does not need to be evaluated for eligibility prior to dosing if previously drawn within 96 hours of first dose. Samples for Cycles 2 and beyond may be obtained up to 96 hours prior to dosing.

[m] Serum chemistry, LFTs (AST, ALT, total bilirubin, indirect bilirubin, total protein, albumin and alkaline phosphatase), uric acid, and LDH performed at Screening then on Days 1, 2, 8, 15, and 22 by local laboratory for the first 2 cycles, and on Day 1 of each cycle from Cycle 3 and beyond. Serum chemistry includes: sodium, potassium, calcium, chloride, bicarbonate, phosphorus or phosphate, magnesium, creatinine with estimated glomerular filtration rate (eGFR) calculated by Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) creatinine equation (Section 15.5), glucose (fasting required for Screening visit or Cycle 1 Day 1 [pre-dose]; fasting status is optional for all other visits where glucose is collected), and blood urea nitrogen (BUN) or urea. Where possible, Day 1 draws for each cycle should be performed at the study clinic. Serum chemistry may be performed at a local lab outside of the investigative site lab as long as laboratory reports are provided to the Investigator prior to dosing on Day 1. Cycle 1 Day 1 chemistry may be obtained and assessed for eligibility up to 96 hours prior to dosing. Repeat testing on Cycle 1 Day 1 is not required if collected within 96 hours prior to first dose. Patients who have an increase not only in transaminases, but also elevated bilirubin will be required to return to the clinic for repeat testing within 3 days. Samples for Cycles 2 and beyond may be obtained up to 96 hours prior to dosing.

[n] Direct antiglobulin test (DAT) at Screening only for all patients with CLL/SLL or WM. Additional DAT at all listed timepoints only for Cohort 16 (CLL with secondary wAIHA).

[o] Coagulation parameters (international normalized ratio [INR] and either partial thromboplastin time [PTT] or activated partial thromboplastin time [aPTT]) are required at Screening and Cycle 1 Day 1, and to be monitored at least every 8 weeks (± 7 days). Therapeutic anticoagulation with warfarin or equivalent vitamin K antagonist is prohibited. Where possible, draws should be performed at the study clinic. Cycle 1 Day 1 coagulation may be obtained and assessed for eligibility up to 96 hours prior to

dosing. Repeat testing on Cycle 1 Day 1 is not required if collected within 96 hours prior to first dose. Samples for Cycles 2 and beyond may be obtained up to 96 hours prior to dosing.

[p] Thyroid stimulating hormone (TSH), and, if abnormal, obtain free or total T3 or T4. TSH, serum amylase/lipase should be measured at Screening and every 8 weeks (± 7 days) and at EOT visit. Where possible, draws should be performed at the study clinic. See footnote b above for Cycle 1 Day 1 samples. Samples for Cycles 2 and beyond may be obtained up to 96 hours prior to dosing.

[q] Urinalysis is required at Screening and Cycle 1 Day 1. On-study urinalysis is required every 8 weeks (± 7 days) and at EOT visit. All urinalysis may be point of care (dipstick). Urine microscopy is required only to follow-up on clinically significant urine dipstick findings Where possible, should be performed at the study clinic. See footnote b above for Cycle 1 Day 1 samples. Samples for Cycles 2 and beyond may be obtained up to 96 hours prior to dosing.

[r] Lymph node biopsies are optional during Phase 1a Dose Escalation. In Phase 1b, lymph node biopsies will be mandatory, if feasible, for up to 10 patients per safety expansion cohort (excluding WM cohorts), if deemed appropriate by the Sponsor. For patients with WM, bone marrow aspirate and biopsy should be utilized for these assays due to limited nodal involvement. Therefore, samples consisting of both bone marrow aspirate and biopsy will be mandatory, if feasible, for up to 10 patients enrolled in WM cohorts in Phase 1b safety expansion cohorts.

[s] Bone marrow biopsy and/or aspirate should be performed when clinically indicated, and to assess response as required per disease-specific response criteria.

[t] Blood plasma samples for pharmacokinetic (PK) analyses will be collected at the timepoints shown in Table 17. Pre-dose samples to be drawn within 2 hours prior to dosing. Samples will also be taken at EOT, 15-day Post Treatment and 30-day Post Treatment Safety Follow-up visits. In the event of a treatment interruption, PK samples should still be collected.

[u] In the event of a dose modification, PK samples for trough levels of NX-5948 should be collected. A single pre-dose PK sample and pharmacodynamic (PD) flow cytometry sample should be collected within 2 hours prior to dosing on the day of the dose modification. Single PK/PD Flow should be collected per the Schedule of Assessments for subsequent visits. See Table 17.

[v] Ophthalmologic exam is required for central nervous system lymphoma (CNSL) patients and Karnofsky performance score (Section 15.6) is required for PCNSL patients. Detailed ophthalmologic exam should include a dilated fundoscopic examination to evaluate vitreous, retinal, or optic nerve involvement. For patients with disease restricted to orbits vitrectomy (eg, primary intraocular lymphoma) pathology should be submitted in lieu of tumor biopsy. Ophthalmologic exams performed within 6 weeks of Cycle 1 Day 1 that were completed as SoC may be used for baseline.

[w] Computed tomography (CT), CT/positron emission tomography (PET), PET, magnetic resonance imaging (MRI) scans and ophthalmologic exam obtained within 6 weeks of Cycle 1 Day 1 that were completed as SoC may be used for baseline. Disease assessments will take place at Screening, end of Cycle 2 (Week 8), and then at Weeks 16, 24, and every 12 weeks (–7 day

window) thereafter and at time of clinical suspicion of disease progression. PCNSL/SCNSL patients will undergo an additional disease assessment between Cycle 1 Day 22 and Cycle 1 Day 28. Disease assessments should be obtained prior to Day 1 of the next cycle. Response criteria specific to B-cell malignancy type will be used to determine response. For patients with PCNSL, disease assessment should include ophthalmologic examination only if there was evidence of ocular lymphoma at Screening or due to interim development of ocular symptoms. In the case of patients with positive CSF cytology at study entry, lumbar puncture (or Ommaya reservoir, if available) with CSF cytology should be performed as part of each disease assessment until there is no evidence of disease in the CSF.

[x] Cerebrospinal fluid (CSF) cytology and protein information should be submitted for all patients with primary or secondary CNSL at Screening. CSF may be collected by lumbar puncture or Ommaya reservoir (if available). CSF cytology should be included as part of subsequent disease assessments only if positive at Screening. CSF collection at Cycle 2 Day 1 and EOT is also mandatory if feasible. CSF collection does not need to be repeated at EOT if a collection is done within the past 7 days of EOT as part of a disease assessment.

[y] Pharmacodynamic (PD) flow cytometry samples, cytokine samples, and peripheral blood mononuclear cell (PBMC) samples will be collected at the timepoints shown in Table 17. Pre-dose samples to be drawn within 2 hours prior to dosing. In the event of a treatment interruption, PD samples should still be collected.

[z] Blood samples for exploratory next generation sequencing (NGS) analysis will be collected at the timepoints shown in Table 17. Pre-dose sample to be drawn within 2 hours prior to dosing. In the event of a treatment interruption, PD samples should still be collected. For Cycle 7 Day 1 and Cycle 13 Day 1 collections, these samples should be collected from patients with CLL/SLL only.

[aa] Single item PRO (FACIT-GP5) will be provided in paper format to the patient by study site personnel (eg, study coordinator or research nurse) prior to dosing and prior to conducting any other study tests or procedures on that visit, whenever feasible or possible. Patients will address the statement “I am bothered by the side effects of treatment” within the past 7 days by using the 5-point scale (0=not at all, 1=a little bit, 2=somewhat, 3=quite a bit, 4=very much). Results will be recorded in the eCRF. FACIT-GP5 will be collected on the following days: Screening, Cycle 1 Day 1, Cycle 1 Day 8, Cycle 1 Day 15, Cycle 1 Day 22, Cycle 2 Day 1, Cycle 3 Day 1, Cycle 4 Day 1, Cycle 5 Day 1, Cycle 6 Day 1, Cycle 9 Day 1, Cycle 12 Day 1, and at EOT.

[bb] NX-5948 will be dispensed at the start of each 28-day treatment cycle as bottles of capsules to be taken daily with water. Patients should fast for at least 2 hours before and 2 hours after taking NX-5948, except for intensive PK collection days (Cycle 1 Day 1 and Cycle 2 Day 1) when patients should fast for a minimum of 8 hours before and 2 hours after taking NX-5948.

[cc] For patients with diffuse large B-cell lymphoma (DLBCL), chronic lymphocytic leukemia (CLL)/small lymphocytic lymphoma (SLL), and mantle cell lymphoma (MCL), tumor lysis syndrome (TLS) prophylaxis is required as per institutional standards.

[dd] After signing of the ICF, all serious adverse events (SAEs) regardless of causality, and all adverse events (AEs) caused by a protocol-mandated intervention, must be reported. After the first dose of study drug (NX-5948), all AEs/SAEs regardless of causality, must be reported through the end of the protocol-defined AE reporting period (30 days after the last dose of study drug)

or until initiation of another anti-cancer agent if less than 30 days. After the end of the protocol-defined AE reporting period, only SAEs assessed as related to study drug or study procedures are required to be reported. Refer to Section 8.2.

[ee] All concomitant medications (including over-the-counter and herbal treatments) should be recorded from 28 days before the first dose until 30 days after the last dose of NX-5948 or until initiation of another anti-cancer agent if less than 30 days. If a related SAE is reported beyond the end of the protocol-defined AE reporting period, concomitant medications should also be recorded at the time of the related SAE report.

[ff] CSF samples for PK analyses will be collected for patients with primary or secondary CNSL at the timepoints shown in Table 17. Note: CSF protein should be measured at any time point at which CSF is collected for PK.

[gg] CSF samples for exploratory ctDNA NGS analysis will be collected for patients with primary or secondary CNSL at the timepoints shown in Table 17.

[hh] Exploratory ctDNA analysis in CSF requires a buccal swab for germline analysis for patients with primary or secondary CNSL. The buccal swab will be collected at Screening only as shown in Table 17.

[ii] Mini Mental State Examination, 2nd Edition: Brief Version (MMSE-2: BV) will be provided in paper format to sites along with an MMSE-2 User's Manual. The person responsible for administering the exam should alternate between the red and blue forms each time the test is administered (at Screening and in conjunction with disease assessments).

[jj] Blood samples for exploratory PD Minimal Residual Disease (MRD) will be collected from patients with CLL/SLL at baseline and at the specific timepoints shown in Table 17. In Phase 1b Part 1 blood samples will be collected from patients in Cohort 1 and Cohort 15 at the specified timepoints regardless of their response to treatment; this will involve up to 10-15 patients from each dose level in Cohort 1 and all patients in Cohort 15. In Phase 1b Part 2, MRD blood samples will also be collected from all patients with CLL/SLL at the indicated timepoints, regardless of their response to treatment.

Table 17 PK, PD, and Exploratory Investigation Sampling Times for Phase 1a and Phase 1b

Cycle/Day	Planned Time	Baseline Biomarker Plasma Sample	PK		Exploratory PD Samples										Window
			Plasma	CSF[a]	Lymph Node Biopsy (all non-WM patients)	Bone Marrow Aspirate and Biopsy (only WM patients)	Flow Cytometry	PBMC	Cytokines	NGS	ctDNA CSF[a]	Omic CSF[a]	Buccal swab[a]	MRD (for patients with CLL/ SLL only)[d]	
Screening		X			X	X		X			X	X	X		
C1 D1	Pre-dose		X				X	X	X	X				X	<2 hr prior to study drug
	0.5 hr		X				X								± 5 min
	1 hr		X				X								± 5 min
	1.5 hr		X												± 5 min
	2 hr		X				X								± 5 min
	3 hr		X												± 10 min
	4 hr		X				X	X							± 10 min
	6 hr		X												± 30 min
	8 hr		X				X	X	X						± 30 min
C1 D2	24 hr (Pre-dose)		X				X	X							± 60 min <2 hr prior to study drug
C1 D8	Pre-dose		X		X	X	X	X	X						<2 hr prior to study drug
C1 D15	Pre-dose		X				X								<2 hr prior to study drug
C1 D22	Pre-dose		X				X								<2 hr prior to study drug
C2 D1 (Day 29)	Pre-dose		X				X	X	X						<2 hr prior to study drug
	0.5 hr		X												± 5 min
	1 hr		X												± 5 min
	1.5 hr		X												± 5 min
	2 hr		X												± 5 min

Cycle/Day	Planned Time	Baseline Biomarker Plasma Sample	PK		Exploratory PD Samples										Window
			Plasma	CSF[a]	Lymph Node Biopsy (all non-WM patients)	Bone Marrow Aspirate and Biopsy (only WM patients)	Flow Cytometry	PBMC	Cytokines	NGS	ctDNA CSF[a]	Omic CSF[a]	Buccal swab[a]	MRD (for patients with CLL/ SLL only)[d]	
	3 hr		X												± 10 min
	4 hr		X	X[b]							X[b]	X[b]			± 10 min
	6 hr		X												± 30 min
	8 hr		X												± 30 min
C2 D2	24 hr (Pre-dose)		X												± 60 min <2 hr prior to study drug
C3 D1	Pre-dose		X				X	X	X						<2 hr prior to study drug
C5 D1	Pre-dose		X				X	X							<2 hr prior to study drug
C7 D1, and q12wks after Cycle 7 (± 7 days) and as clinically indicated	Pre-dose		X				X	X		C7D1, C13D1[c]				C7D1, C13D1, then yearly[d]	<2 hr prior to study drug
EOT Safety Follow-up			X	X			X	X	X	X	X	X			
15-day Post Tx Safety Follow-up			X				X								
30-day Post Tx Safety Follow-up			X				X								
Unscheduled for significant safety event (eg, DLT, related Grade ≥ 3 AE, SAE)			X				X								As close as possible when aware of AE

Abbreviations: AE, adverse event; C, cycle; CSF, cerebrospinal fluid; ctDNA, circulating tumor deoxyribonucleic acid; D, day; hr, hour; min, minute; DLT, dose limiting toxicity; EOT, End of Treatment; MRD, minimal residual disease; NGS, next generation sequencing; PBMC, peripheral blood mononuclear cell; PD, pharmacodynamic; PK, pharmacokinetic; Q, every; SAE, serious adverse event; Tx, treatment; wks, weeks.

[a] Cerebrospinal fluid (CSF) samples and buccal swabs will be collected for patients with primary and secondary CNSL. CSF sample collection time points may be adjusted based on emerging human pharmacokinetic (PK) data.

[b] PK, ctDNA, and Omic CSF samples should be collected as close as possible to the 4-hour post-dose timepoint, up to 6 hours post-dose.

[c] PD NGS samples at Cycle 7 Day 1 and Cycle 13 Day 1 will be collected for patients with CLL/SLL in Phase 1b (Part 1 and Part 2).

[d] PD MRD blood samples will be collected for patients with CLL/SLL at baseline and at the indicated timepoints. In Phase 1b Part 1, blood samples will be collected from patients in Cohort 1 and Cohort 15 at the specified timepoints regardless of their response to treatment; this will involve up to 10-15 patients from each dose level in Cohort 1 and all patients in Cohort 15. In Phase 1b Part 2, MRD blood samples will also be collected from all patients with CLL/SLL at the specified timepoints regardless of their response to treatment.

15.4. Response Assessment

15.4.1. Modified iwCLL Response Criteria: Hallek 2018

Group	Parameter	CR	PR	PD	SD
A	Lymph nodes	None ≥1.5 cm	Decrease ≥50% (from baseline)*	Increase ≥50% from baseline or from response	Change of –49% to +49%
	Liver and/or spleen size†	Spleen size <13cm; liver size normal	Decrease ≥50% (from baseline)	Increase ≥50% from baseline or from response	Change of –49% to +49%
	Constitutional symptoms	None	Any	Any	Any
	Circulating lymphocyte count	Normal	Decrease ≥50% from baseline	Increase ≥50% over baseline	Change of –49% to +49%
B	Platelet count	≥100 × 10^9/L	≥100 × 10^9/L or increase ≥50% over baseline	Decrease of ≥50% from baseline secondary to CLL	Change of –49 to +49%
	Hemoglobin	≥11.0 g/dL (untransfused and without erythropoietin)	≥11 g/dL or increase ≥50% over baseline	Decrease of ≥2 g/dL from baseline secondary to CLL	Increase <11.0 g/dL or <50% over baseline, or decrease <2 g/dL
	Marrow	Normocellular, no CLL cells, no B-lymphoid nodules	Presence of CLL cells, or of B-lymphoid nodules, or not done	Increase of CLL cells by ≥50% on successive biopsies	No change in marrow infiltrate

For a detailed description of the response parameters, see section 5 of Hallek 2018.
*Sum of the products of 6 or fewer lymph nodes (as evaluated by CT scans and physical examination in clinical trials or by physical examination in general practice).
†Spleen size is considered normal if < 13 cm. There is not firmly established international consensus of the size of a normal liver; therefore, liver size should be evaluated by imaging and manual palpation in clinical trials and be recorded according to the definition used in a study protocol.
Abbreviations: CLL, chronic lymphocytic leukemia; CR, complete remission (all of the criteria have to be met); CT, computed tomography; PD, progressive disease (at least 1 of the criteria of group A or group B has to be met); PR, partial remission (for a PR, at least 2 of the parameters of group A and 1 parameter of group B need to improve if previously abnormal; if only 1 parameter of both groups A and B is abnormal before therapy, only 1 needs to improve); SD, stable disease (all of the criteria have to be met; constitutional symptoms alone do not define PD).

Additional Response Assessments for CLL Patients Enrolled in Phase 1b:

PR with lymphocytosis (PR-L): Only 1 abnormal Group A criteria met in the setting of lymphocytosis (blood lymphocytes decreased < 50% from baseline or are increased from baseline) not meeting PR criteria.

Nodular PR (nPR): All the criteria met for CR except for the presence of lymphoid nodules in the bone marrow.

CRi: All the criteria met for CR except that patient has at least 1 persistent peripheral blood cytopenia (anemia, thrombocytopenia, and/or neutropenia) and the investigator attributes the cytopenia(s) as related to investigational drug and not due to underlying disease.

Reference:

Hallek M, Cheson BD, Catovsky D, et al. iwCLL guidelines for diagnosis, indications for treatment, response assessment, and supportive management of CLL. *Blood*. 2018;131(25):2745-2760.

15.4.2. WM Response Criteria: VIth International Workshop

Response category	Definition
Complete response (CR)	Absence of serum monoclonal IgM protein by immunofixation Normal serum IgM level Complete resolution of extramedullary disease, i.e., lymphadenopathy and splenomegaly if present at baseline Morphologically normal bone marrow aspirate and trephine biopsy
Very good partial response (VGPR)	Monoclonal IgM protein is detectable ≥90% reduction in serum IgM level from baseline* Complete resolution of extramedullary disease, i.e., lymphadenopathy/splenomegaly if present at baseline No new signs or symptoms of active disease
Partial response (PR)	Monoclonal IgM protein is detectable ≥50% but <90% reduction in serum IgM level from baseline* Reduction in extramedullary disease, i.e., lymphadenopathy/splenomegaly if present at baseline No new signs or symptoms of active disease
Minor response (MR)	Monoclonal IgM protein is detectable ≥25% but<50% reduction in serum IgM level from baseline* No new signs or symptoms of active disease
Stable disease (SD)	Monoclonal IgM protein is detectable <25% reduction and <25% increase in serum IgM level from baseline* No progression in extramedullary disease, i.e., lymphadenopathy/splenomegaly No new signs or symptoms of active disease
Progressive disease (PD)	≥25% increase in serum IgM level* from lowest nadir (requires confirmation) and/or progression in clinical features attributable the disease

*Sequential changes in IgM levels may be determined either by M protein quantitation by densitometry or total serum IgM quantitation by nephelometry.

Reference:

Owen RG, Kyle RA, Stone MJ, et al. VIth International Workshop on Waldenström macroglobulinaemia. Response assessment in Waldenström macroglobulinaemia: update from the VIth International Workshop. *Br J Haematol.* 2013;160(2):171-176.

15.4.3. Lugano Classification of Lymphoma Response Criteria: Cheson 2014

Table 3. Revised Criteria for Response Assessment

Response and Site	PET-CT–Based Response	CT-Based Response
Complete	Complete metabolic response	Complete radiologic response (all of the following)
Lymph nodes and extralymphatic sites	Score 1, 2, or 3* with or without a residual mass on 5PS†	Target nodes/nodal masses must regress to < 1.5 cm in LDi
	It is recognized that in Waldeyer's ring or extranodal sites with high physiologic uptake or with activation within spleen or marrow (eg, with chemotherapy or myeloid colony-stimulating factors), uptake may be greater than normal mediastinum and/or liver. In this circumstance, complete metabolic response may be inferred if uptake at sites of initial involvement is no greater than surrounding normal tissue even if the tissue has high physiologic uptake	No extralymphatic sites of disease
Nonmeasured lesion	Not applicable	Absent
Organ enlargement	Not applicable	Regress to normal
New lesions	None	None
Bone marrow	No evidence of FDG-avid disease in marrow	Normal by morphology; if indeterminate, IHC negative
Partial	Partial metabolic response	Partial remission (all of the following)
Lymph nodes and extralymphatic sites	Score 4 or 5† with reduced uptake compared with baseline and residual mass(es) of any size	> 50% decrease in SPD of up to 6 target measurable nodes and extranodal sites
	At interim, these findings suggest responding disease	When a lesion is too small to measure on CT, assign 5 mm x 5 mm as the default value
	At end of treatment, these findings indicate residual disease	When no longer visible, 0 x 0 mm
		For a node > 5 mm x 5 mm, but smaller than normal, use actual measurement for calculation
Nonmeasured lesions	Not applicable	Absent/normal, regressed, but no increase
Organ enlargement	Not applicable	Spleen must have regressed by > 50% in length beyond normal
New lesions	None	None
Bone marrow	Residual uptake higher than uptake in normal marrow but reduced compared with baseline (diffuse uptake compatible with reactive changes from chemotherapy allowed). If there are persistent focal changes in the marrow in the context of a nodal response, consideration should be given to further evaluation with MRI or biopsy or an interval scan	Not applicable
No response or stable disease	No metabolic response	Stable disease
Target nodes/nodal masses, extranodal lesions	Score 4 or 5 with no significant change in FDG uptake from baseline at interim or end of treatment	< 50% decrease from baseline in SPD of up to 6 dominant, measurable nodes and extranodal sites; no criteria for progressive disease are met
Nonmeasured lesions	Not applicable	No increase consistent with progression
Organ enlargement	Not applicable	No increase consistent with progression
New lesions	None	None
Bone marrow	No change from baseline	Not applicable
Progressive disease	Progressive metabolic disease	Progressive disease requires at least 1 of the following
Individual target nodes/nodal masses	Score 4 or 5 with an increase in intensity of uptake from baseline and/or	PPD progression:
Extranodal lesions	New FDG-avid foci consistent with lymphoma at interim or end-of-treatment assessment	An individual node/lesion must be abnormal with: LDi > 1.5 cm and Increase by ≥ 50% from PPD nadir and An increase in LDi or SDi from nadir 0.5 cm for lesions ≤ 2 cm 1.0 cm for lesions > 2 cm In the setting of splenomegaly, the splenic length must increase by > 50% of the extent of its prior increase beyond baseline (eg, a 15-cm spleen must increase to > 16 cm). If no prior splenomegaly, must increase by at least 2 cm from baseline New or recurrent splenomegaly
Nonmeasured lesions	None	New or clear progression of preexisting nonmeasured lesions

(continued on following page)

Table 3. Revised Criteria for Response Assessment (continued)

Response and Site	PET-CT–Based Response	CT-Based Response
New lesions	New FDG-avid foci consistent with lymphoma rather than another etiology (eg, infection, inflammation). If uncertain regarding etiology of new lesions, biopsy or interval scan may be considered	Regrowth of previously resolved lesions A new node > 1.5 cm in any axis A new extranodal site > 1.0 cm in any axis; if < 1.0 cm in any axis, its presence must be unequivocal and must be attributable to lymphoma Assessable disease of any size unequivocally attributable to lymphoma
Bone marrow	New or recurrent FDG-avid foci	New or recurrent involvement

Abbreviations: 5PS, 5-point scale; CT, computed tomography; FDG, fluorodeoxyglucose; IHC, immunohistochemistry; LDi, longest transverse diameter of a lesion; MRI, magnetic resonance imaging; PET, positron emission tomography; PPD, cross product of the LDi and perpendicular diameter; SDi, shortest axis perpendicular to the LDi; SPD, sum of the product of the perpendicular diameters for multiple lesions.

*A score of 3 in many patients indicates a good prognosis with standard treatment, especially if at the time of an interim scan. However, in trials involving PET where de-escalation is investigated, it may be preferable to consider a score of 3 as inadequate response (to avoid undertreatment). Measured dominant lesions: Up to six of the largest dominant nodes, nodal masses, and extranodal lesions selected to be clearly measurable in two diameters. Nodes should preferably be from disparate regions of the body and should include, where applicable, mediastinal and retroperitoneal areas. Non-nodal lesions include those in solid organs (eg, liver, spleen, kidneys, lungs), GI involvement, cutaneous lesions, or those noted on palpation. Nonmeasured lesions: Any disease not selected as measured, dominant disease and truly assessable disease should be considered not measured. These sites include any nodes, nodal masses, and extranodal sites not selected as dominant or measurable or that do not meet the requirements for measurability but are still considered abnormal, as well as truly assessable disease, which is any site of suspected disease that would be difficult to follow quantitatively with measurement, including pleural effusions, ascites, bone lesions, leptomeningeal disease, abdominal masses, and other lesions that cannot be confirmed and followed by imaging. In Waldeyer's ring or in extranodal sites (eg, GI tract, liver, bone marrow), FDG uptake may be greater than in the mediastinum with complete metabolic response, but should be no higher than surrounding normal physiologic uptake (eg, with marrow activation as a result of chemotherapy or myeloid growth factors).

†PET 5PS: 1, no uptake above background; 2, uptake < mediastinum; 3, uptake > mediastinum but < liver; 4, uptake moderately > liver; 5, uptake markedly higher than liver and/or new lesions; X, new areas of uptake unlikely to be related to lymphoma.

Reference:

Cheson BD, Fisher RI, Barrington SF, et al. Recommendations for Initial evaluation, staging, and response assessment of Hodgkin and non-Hodgkin lymphoma: the Lugano classification. *J Clin Oncol.* 2014;32(27):3059-3068.

15.4.4. International PCNSL Collaboration Group Criteria: Abrey 2005

Table 4. Response Criteria for Primary Central Nervous System Lymphoma

Response	Brain Imaging	Corticosteroid Dose	Eye Examination	CSF Cytology
CR	No contrast enhancement	None	Normal	Negative
CRu	No contrast enhancement	Any	Normal	Negative
	Minimal abnormality	Any	Minor RPE abnormality	Negative
PR	50% decrease in enhancing tumor	Irrelevant	Minor RPE abnormality or normal	Negative
	No contrast enhancement	Irrelevant	Decrease in vitreous cells or retinal infiltrate	Persistent or suspicious
PD	25% increase in lesion	Irrelevant	Recurrent or new ocular disease	Recurrent or positive
	Any new site of disease: CNS or systemic			

Abbreviations: CR, complete response; CRu, unconfirmed complete response; RPE, retinal pigment epithelium; PR, partial response; PD, progressive disease.

Reference:

Abrey LE, Batchelor TT, Ferreri AJM, et al. Report of an international workshop to standardize baseline evaluation and response criteria for primary CNS lymphoma. *J Clin Oncol.* 2005;23(22):5034-5043.

15.4.5. AIHA Response Criteria: Barcellini 2012

AIHA Complete Response (CR^{AIHA})	Hemoglobin > 12 g/dL, AND no transfusion requirements No laboratory evidence of ongoing hemolysis (haptoglobin, indirect bilirubin within normal limit, reticulocyte count < 1.5 ULN)
AIHA Partial Response (PR^{AIHA})	Hemoglobin level 10-12 g/dL (without transfusions), OR hemoglobin increase of at least 2 g/dL above pretreatment level Presence of laboratory evidence of ongoing hemolysis

Reference:

Barcellini W, Zaja F, Zaninoni A, et al. Low-dose rituximab in adult patients with idiopathic autoimmune hemolytic anemia: clinical efficacy and biologic studies. Blood. 2012;119(16):3691-3697.

- Percentage of patients with autoimmune hemolytic anemia (AIHA) complete response (CR^{AIHA}), AIHA partial response (PR^{AIHA}), and overall AIHA response as assessed by Investigator
- Sustained response: Time from CR^{AIHA} or PR^{AIHA} until death or relapse
- Relapse-free survival: Time from start of treatment until death or relapse of AIHA
- Durable response: Percentage of patients achieving a CR^{AIHA} or PR^{AIHA} that is maintained for 8 consecutive weeks in the absence of rescue therapy

Frequency of packed red blood cell (PRBC) transfusion while receiving NX-5948

15.5. CKD-EPI Creatinine Equation for Estimating Creatinine Clearance or GFR

$$eGFR = 141 \times \min(SCr/\kappa, 1)^{\alpha} \times \max(Scr/\kappa, 1)^{-1.200} \times 0.9938^{Age} \times 1.012 \text{ [if female]}$$

Abbreviations / Units:

eGFR (estimated glomerular filtration rate) = mL/min/1.73 m^2

SCr (standardized serum creatinine) = mg/dL

κ = 0.7 (females) or 0.9 (males)

α = -0.241 (females) or -0.302 (males)

min = indicates the minimum of SCr/κ or 1

max = indicates the maximum of SCr/κ or 1

age = years

Reference:

Miller WG, Kaufman HW, Levey AS, et al. National Kidney Foundation Laboratory Engagement Working Group recommendations for implementing the CKD-EPI 2021 race-free equations for estimated glomerular filtration rate: practical guidance for clinical laboratories. *Clin Chem.* 2022;68(4):511-520.

15.6. Karnofsky Performance Status Scale Definitions Rating (%) Criteria

Able to carry on normal activity and to work; no special care needed.	100	Normal no complaints; no evidence of disease.
	90	Able to carry on normal activity; minor signs or symptoms of disease.
	80	Normal activity with effort; some signs or symptoms of disease.
Unable to work; able to live at home and care for most personal needs; varying amount of assistance needed.	70	Cares for self; unable to carry on normal activity or do active work.
	60	Requires occasional assistance, but is able to care for most of his or her personal needs.
	50	Requires considerable assistance and frequent medical care.
Unable to care for self; requires equivalent institutional or hospital care; disease may be progressing rapidly.	40	Disabled; requires special care and assistance.
	30	Severely disabled; hospital admission is indicated although death not imminent.
	20	Very sick; hospital admission necessary; active supportive treatment necessary.
	10	Moribund; fatal processes progressing rapidly.
	0	Dead.

Reference:

Schag CC, Heinrich RL, Ganz PA. Karnofsky performance status revisited: reliability, validity, and guidelines. *J Clin Oncol.* 1984;2(3):187-193.

15.7. Fitzpatrick Skin Color Assessment

The Fitzpatrick skin color assessment is a commonly used system to describe a person's skin type in response to ultraviolet radiation exposure. In this study, patients' skin color will be assessed according to the Fitzpatrick standardized descriptive color scale.

Type of skin	Description
1	Pale white skin
2	White skin
3	Light brown skin
4	Moderate brown skin
5	Dark brown skin
6	Deeply pigmented dark brown to black skin

Figure 6 Fitzpatrick Classification of Skin Types I through VI

Reference:

Fitzpatrick TB. The validity and practicality of sun-reactive skin types I through VI. *Arch Dermatol.* 1988;124(6):869-871.

15.8. GELF Criteria

Patients with FL who meet ≥ 1 of the following criteria are considered to have “high” tumor burden according to GELF:

- Involvement of >3 nodal sites, each with a diameter of >3 cm
- Any nodal or extranodal tumor mass with a diameter of 7 cm
- Presence of type B symptoms
- Risk of local compressive symptoms that may result in organ compromise
- Cytopenias (leukocytes <1.0×10^9/L and/or platelets <100×10^9/L)
- Leukemia (>5.0×10^9/L malignant cells)
- Splenomegaly (>16 cm on CT scan)
- Pleural effusion or peritoneal ascites

References:

Barraclough A, Yoo E, Cheah CY, et al. The 'real world' uptake and prognostic impact of GELF in newly diagnosed follicular lymphoma: an Australasian Alliance initiative. *Blood.* 2019;134(suppl_1):3986.

Chen Q, Ayer T, Nastoupil LJ, et al. Initial management strategies for follicular lymphoma. *Int J Hematol Oncol*. 2012;1(1):35-45.

15.9. Clinical Trials Facilitation and Coordination Group (CTFG) Recommendations Related to Contraception and Pregnancy Testing in Clinical Trials, Version 1.1

Definitions

A woman is considered to be a WOCBP following menarche and until becoming postmenopausal, unless permanently sterile; permanent sterilization methods include hysterectomy, bilateral salpingectomy, and bilateral oophorectomy. Postmenopausal is defined as >12 months with no menses without an alternative medical cause. A high follicle stimulating hormone (FSH) level in the postmenopausal range may be used to confirm a postmenopausal state in women not using hormonal contraception or hormonal replacement therapy, but in the absence of >12 months of amenorrhea, a single FSH measurement is insufficient.

Contraception methods are considered highly effective if they achieve a failure rate of <1% per year when used consistently and correctly. Such methods include, but are not limited to, the following: combined estrogen- and progestogen-containing hormonal contraception associated with inhibition of ovulation (oral, intravaginal, or transdermal), progestogen-only hormonal contraception associated with inhibition of ovulation (oral, injectable, or implantable), intrauterine device or intrauterine hormone-releasing system, bilateral tubal occlusion, or vasectomized partner with medical confirmation of vasectomy surgical success.

Note that caution should be taken when using hormonal contraception methods if there are possible drug-drug interactions with an investigational drug. In the absence of definitive drug interaction studies for NX-5948, hormonal contraception methods (including those listed above) for WOCBP must be supplemented with a barrier contraception method (ie, male condom; also see below).

Sexual abstinence may be considered a highly effective method of contraception only if defined as refraining from heterosexual intercourse during the entire period of risk. The reliability of sexual abstinence needs to be evaluated in relation to the duration of the clinical trial and the preferred and usual lifestyle of the patient.

Barrier contraception (required for men with WOCBP partners) is defined as male condom with or without spermicide.

If a contraception method is not permitted according to local regulations, it does not qualify as a highly effective method of contraception for patients participating at sites in the relevant country or region.

For additional detailed guidance and definitions, please refer to the 2020 Clinical Trials Facilitation and Coordination Group (CTFG) recommendations related to contraception and pregnancy testing in clinical trials (CTFG 2020).